Military Review

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September-October 2026

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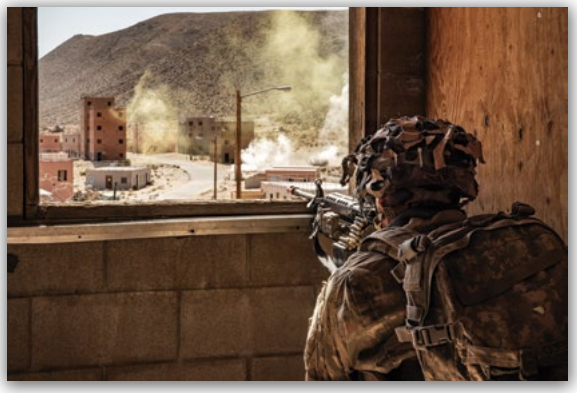
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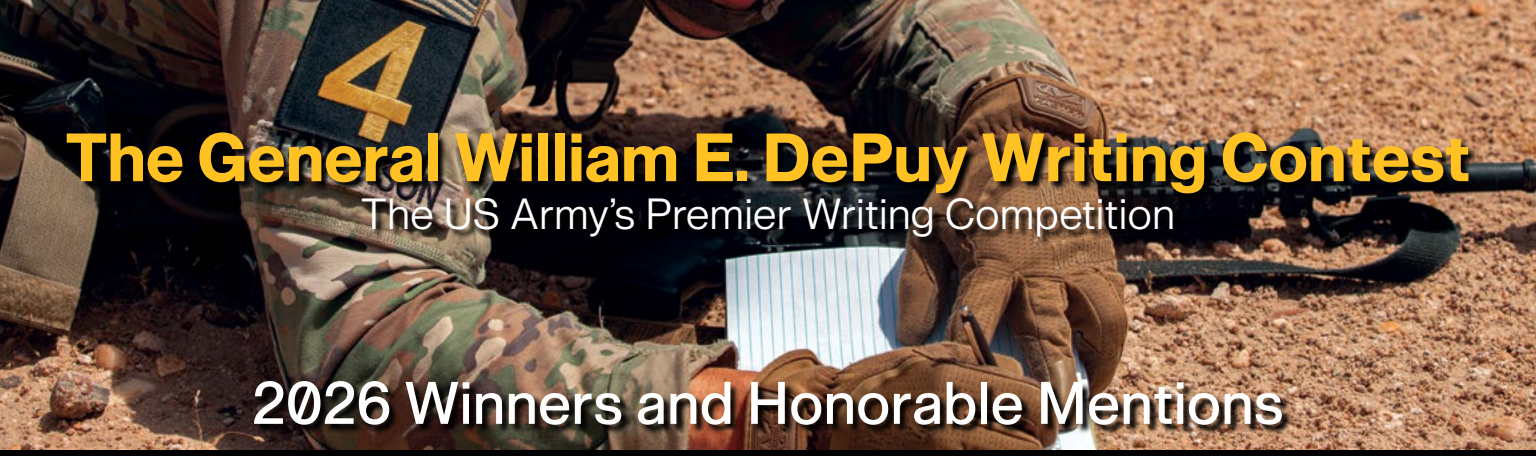
MATTHEW L. SANNITO
Administrative Assistant
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2623107



A soldier assigned to Bravo Company, 1st Battalion, 8th Infantry Regiment, 3rd Armored Brigade Combat Team, 4th Infantry Division, conducts security during Project Convergence Capstone 6 at Fort Irwin, California, 24 July 2026. Project Convergence is the premier Army-hosted experiment that allows the joint force and multinational allies and partners to integrate people, equipment, and technologies to support continuous transformation efforts aimed at shaping how future warfighters will operate in fast-paced, high-tech environments. (Photo by Spc. Jonathan Reyes, US Army)

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"The Benning Revolution 2.0—Upgrading Army PME to Close the Adaptation Gap"
Maj. Dalton T. Fuss, US Army

Second Place

"Sooner or Later, Someone Is Going to Need to Think: Modernizing PME for the Age of AI"
Prof. Kristan J. Wheaton, US Army War College

Third Place

"When Speed Outruns Judgment: Preserving Human Command in an AI-Accelerated Army"
Staff Sgt. Shantelle M. Davis, US Army

Honorable Mentions

"Learning at the Speed of Relevance: A Better Battle Rhythm for Adaptation"
Maj. Trevor Barton, US Army

"Learning at the Speed of War: Modernizing Army Professional Military Education for Joint All-Domain Operations, Artificial Intelligence-Enabled Decision Making, Electromagnetic Warfare, and Moral Durability"
Capt. Matthew Juntz, US Army Reserve

"Invisibility Before Lethality: Modernizing PME for the Transparent Battlefield"
Capt. Austin Strobel, US Army

"From Death by PowerPoint to Dual Instruction: Governing AI in Professional Military Education"
1st Lt. Ji Hyun An, US Air Force

Military Review thanks our esteemed panel of judges for their careful, insightful, and timely evaluation of the manuscript submissions.

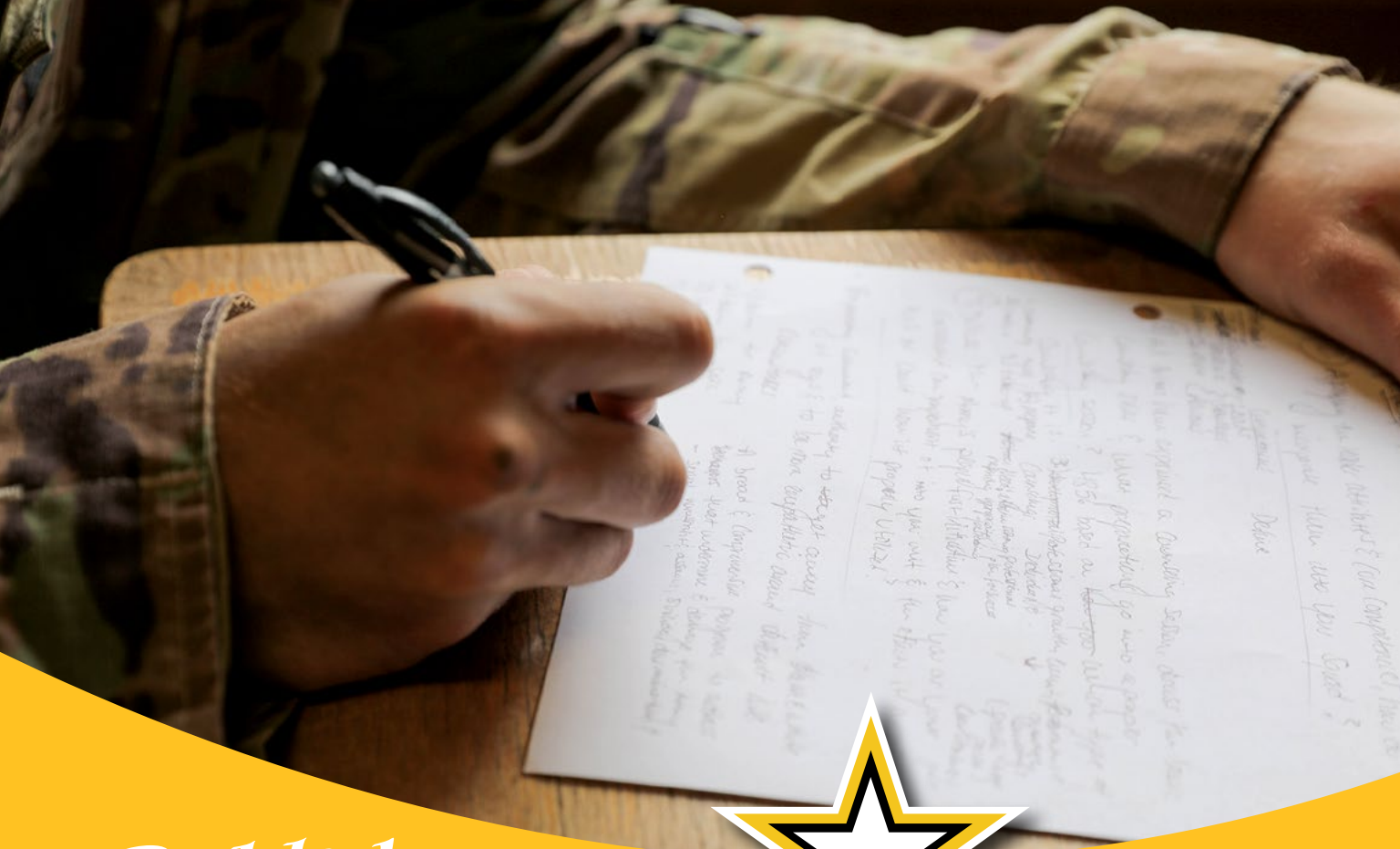
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Write for *Military Review*
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- Using a case study of any Ukrainian campaign in its ongoing war with Russia, discuss and analyze Ukraine's use of multidimensional warfare to achieve specified military and political objectives from 2022 to the present. Examples could include Ukraine's defense of the city of Pokrovsk, denial of full occupation of the Donbas region, and ongoing campaign against the Russian occupation of Crimea.
- Discuss and analyze the goals of the People's Republic of China for territorial expansion and establishing territorial influence. Compare and contrast Imperial Japanese justifications and actions used for seizure of terrain leading up to World War II to current claims asserted by Communist China for justifying seizure of terrain also claimed by its neighbors. Nations who have territory claimed by China include Russia, Japan, India, Vietnam, and the Philippines. How are the Chinese planning for multidimensional global campaigns to support territorial expansion and territorial influence?
- From a US military perspective, what are the greatest security threats to the United States? How specifically is the United States preparing to mitigate those threats employing the elements on national power?
- Using case studies, discuss evidence of employment of irregular warfare (IW) using instruments short of large-scale military violence to achieve strategic objectives. Is there evidence that states as well as nonstate actors are conducting IW against the United States? Discuss evidence of cooperation among state or nonstate actors in such efforts.
- The United States and the Americas—assess the emerging actors, roles, and relationships in North, South, and Central America. Is Mexico our friend or foe, or disinterested neighbor? Is Mexico a staging ground for malevolent actors conducting IW against the United States?
- Do China, Russia, and North Korea have "Achilles' heels"? What are their centers of gravity? If each has one, how can it best be attacked/exploited?
- What do China, Russia, and North Korea view as the United States' "Achilles' heel" or center of gravity? How specifically are they attacking it?

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The Benning Revolution 2.0



Upgrading Army PME to Close the Adaptation Gap

Maj. Dalton T. Fuss, US Army

The Russo-Ukrainian War provides a daily reminder of how difficult battlefield adaptation is. The process is even more cumbersome if institutional education is built for compliance rather than speed. As the Combined Arms Command and Army University look to modernize professional military education (PME) for large-scale combat operations, they face an uphill battle against our own bureaucracy. Our schoolhouses are slow to modernize because the curriculum changes only after months or years of testing and validation, but the speed of battlefield innovation compels change. The Army must embrace bold action, not incremental curriculum updates, to address this. By requiring PME institutions to coach leaders to think on their feet and understand emerging technologies, Army University can build adaptation velocity before the first shot is fired.

We have suffered from this problem before. In the early twentieth century, the Russo-Japanese War gave observers a clear look at the future of industrialized warfare.[1] Reports detailed the devastating effects of long-range artillery, wireless communications, trenches, and machine guns.[2] The data was there, and war colleges studied the campaigns extensively. Yet less than a decade later, those same armies entered World War I with organizations and leadership expectations frozen in the past. Blinded by the "cult of the offensive," senior leaders gambled on swift maneuvers and inherited a meat grinder of industrial attrition instead.[3]

Today, the proliferation of first-person-view (FPV) drones and dynamic electronic warfare (EW) in Ukraine is challenging our combined arms doctrine. Similarly to how the great powers failed to internalize the lessons of the Russo-Japanese War in 1904–1905, modern militaries are falling behind the current pace of battlefield innovation. This growing gap is not simply the result of a technological shortfall. Instead, it reflects a systemic underinvestment in developing the cognitive agility of our future leaders. This may leave them poorly equipped to manage rapid tactical change. To build a force capable of surviving the next first contact, the Army University must shift the educational focus of PME away from static doctrine mastery and toward dynamic, fast-paced adaptation.

The Adaptation Gap: Alignment vs. Awareness

This historical disjuncture captures what retired Lt. Gen. David Barno and Dr. Nora Bensahel term the "adaptation gap": the distance between the war militaries believe they are preparing to fight and the war they actually encounter.[4] It is a gap of alignment rather than awareness. The European powers were not ignorant of what had occurred during the Russo-Japanese War. The problem was that their institutional concepts, education systems, and leader development models did not meaningfully adapt to their observations. Because these militaries failed to close the adaptation gap before and, perhaps more importantly, *during* the opening

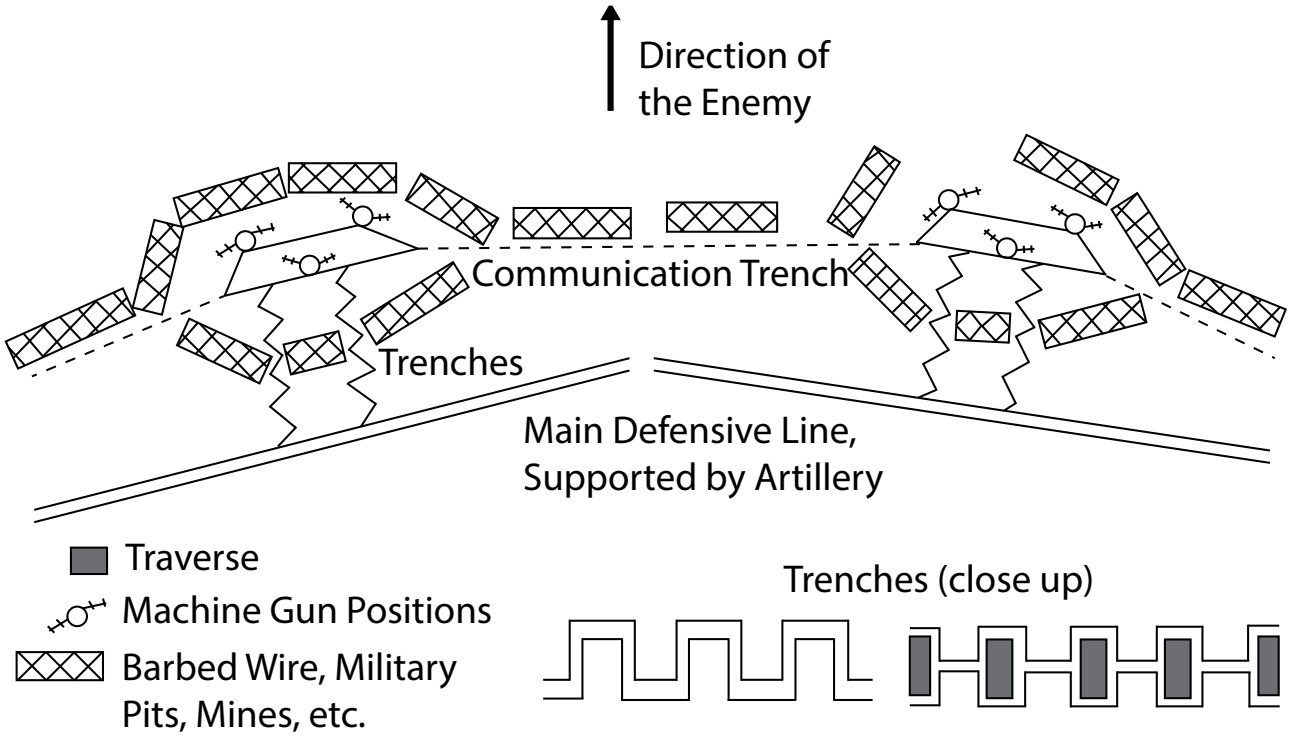


Ukrainian military specialists assemble drones and other high-tech defense-related equipment. In January 2025, the Ukrainian Ministry of Defense approved approximately 130 new weapons and military equipment models for use with nearly one hundred being made domestically. Ukrainian Deputy Minister of Defense Brig. Gen. Anatolii Klochko stated, "Considering the high intensity of combat operations on the front lines, our warriors require a broader array of weapons. There is a need for more innovative strike capabilities to expand the operational capacity of our warriors on the battlefield. Domestic weapon manufacturers quickly address contemporary demands, increase production volumes, and create state-of-the-art weapon models. Notable progress is particularly evident in developing unmanned systems of various types and purposes—air, ground, sea, and electronic warfare." (Photo from Adobe Stock)

campaigns of the First World War, they suffered incomprehensible casualties.

The adaptation gap is a persistent feature of military history. No force enters a major war with a perfect understanding of how that conflict will unfold. What has changed in the twenty-first century is the *rate* at which the character of war can evolve once fighting begins. Rapid technological advancements such as cheap uncrewed systems, commercial satellite imagery, and software-enabled EW enable combatants to modify tactics, techniques, and procedures in days and weeks, not years. The ongoing war in Ukraine illustrates this dynamic vividly. Both Russia and Ukraine are constantly adjusting how they fight. Both integrated new drone employment concepts, applied EW measures, dispersed their maneuver elements under pervasive surveillance, and leveraged AI to support decision-making and analysis of vast amounts of data. Adaptations like these occur regularly during each campaign season.

In this environment, it is no longer sufficient to recognize that an adaptation gap will exist at the outset of future large-scale combat operations. It certainly will. What will matter is how quickly that gap can be narrowed, and the timer starts upon commencement of hostilities. This concept is best understood as a force's "adaptation velocity." Adaptation velocity is the speed and effectiveness with which a force can reassess its assumptions, generate new practices, and spread successful adaptations to other units.[5] It includes all the ways a force adjusts to its battlefield environment. No matter how big the initial adaptation is, a military with a high adaptation velocity will



(Figure from Nicholas Murray, *The Rocky Road to the Great War: The Evolution of Trench Warfare to 1914*)

Figure. Russian Defensive System by 1905

be able to close the gap rapidly enough to preserve or regain the initiative.

To achieve a high adaptation velocity, the US Army must restructure how it develops leaders at all ranks. This article provides a framework for that reform by analyzing historical failures alongside contemporary realities. First, it evaluates the Russo-Japanese War to show how rigid doctrine and technological illiteracy can paralyze a force. Second, it examines the developments of the war in Ukraine to emphasize the speed of adaptation on the modern battlefield. Lastly, it describes how the US Army's PME developments between the two world wars provide a successful outline for change. Using this framework, the Army can cultivate creative leadership in uncertainty and technical fluency as a critical component of PME.

Poor Adaptation Velocity: British Observers in the Russo-Japanese War

The adaptation gap does not stem from a lack of data. Instead, it is usually caused by a combination of institutional stubbornness, a lack of mechanisms to incorporate new lessons, or the selection of the wrong lessons to apply. The British experiences following the Russo-Japanese War and during the first few years of the First World War offer a cautionary example. Great Britain was a global maritime power with vested interests in the region, so the country dispatched several capable officers to observe the conflict. Their reports on Gen. Nogi Maresuke's siege of Port Arthur provided chilling accounts of mass death. These observers described how the combatants employed quick-firing artillery, barbed wire, entrenched machine guns, and even wireless communications.[6] While they did not use this term, the British officers watched as Nogi's attacks were shattered by Russian firepower from fortified positions. His conviction and patriotic drive were admirable, but he failed to intuit the adaptation gap and lacked the cognitive flexibility to adapt to the new, industrialized way of war.

Even though the embedded observers provided detailed schematics of trench systems, as seen in the figure, and gruesome reports that the bayonet charge



Members of the balloon factory aero club watch a tethered observation balloon in 1908. The balloon was typical of pre-World War I observation balloons, and Russian Forces used them as aerial observation platforms during the Russo-Japanese War. (Photo from the Imperial War Museums, courtesy of Wikimedia Commons)

was becoming a suicidal relic, the British military establishment dismissed these findings.[7] Steeped in the cult of the offensive, British institutions criticized these cases as unique to the "dispirited" Russian defenders or the "patriotic fanaticism" of Japanese infantry.[8] British military institutions did not see the slaughter in Manchuria as a major shift in the character of war. They saw this as a tactical anomaly caused by two poorly disciplined militaries without a refined sense of tactics. Their rigid educational model prioritized *what* to think over *how* to think, fostering a deep reliance on the doctrinally approved answer. This mentality likely contributed to the failure to change. At the time, British military education focused on a single, approved tactic to solve battlefield problems. When the observers' reports contradicted this approved solution, the institution dismissed the findings. The War Office did not lack information. It lacked the cognitive flexibility to allow that information to challenge its core assumptions.

The Royal Navy held this institutional blind spot, too. Throughout the war, both the Russian and Japanese navies used long-range wireless telegraphy and centralized fire control measures. For the first time, fleets could coordinate maneuvers and synchronize massed fires across vast distances in real-time. Winning required technical literacy just as much as traditional seamanship. Technical fluency is a key driver of a force's adaptation velocity. During the siege of Port Arthur, the Imperial Japanese Navy routinely attempted to mass fires and sink the Russian fleet. However, technologically savvy Russian personnel at a coastal wireless station detected these Japanese communications.[9] Then, in one of the first instances of EW, the Russians flooded the transmitting frequencies with noise from their own radios. The Russian operators successfully disrupted the Japanese command-and-control network.[10] This EW countermeasure was primitive but effective. This technique temporarily prevented Japanese spotters from transmitting ballistic corrections, effectively neutralizing Japanese naval gunfire.[11]

Even with direct access to these changes, the British War Office and the Admiralty could not square what they observed with how they viewed themselves. The Royal Navy adopted cutting-edge ship designs like the HMS *Dreadnought*, but it completely ignored technological literacy within its own officer corps and failed to address their deeply ingrained institutional culture. The Royal Navy remained constrained to its centralized control philosophy instead of maximizing tactical flexibility.[12] Rather than leveraging wireless telegraphy to push authority to the edge, the Admiralty used the technology and flag signaling to micromanage fleet movements.[13] British officers did not see the potential that wireless communications offered to enhance disciplined initiative. At the Battle of Jutland in 1916, this over-centralized command structure broke down. Subordinate commanders hesitated to act without explicit orders, while Admiralty-directed radio traffic created severe information bottlenecks and delayed intelligence transmission.[14] This reliance on mission directives, exacerbated by the centralized misuse of communications technology, ultimately prevented the Grand Fleet from decisively destroying the German High Seas Fleet.

Over a century later, the contemporary battlefield of the Russo-Ukrainian War provides many of the

same lessons. Just as the advent of quick-firing artillery, wireless telegraphy, and aerial observation from balloons demanded a radical shift in cognitive flexibility during the first two decades of the twentieth century, the proliferation of uncrewed systems, algorithmic targeting, and ubiquitous EW in Ukraine requires an unprecedented adaptation velocity from modern military leadership. Ukrainian and Russian forces are continually adapting their tactics and integrating novel technologies, underscoring why rapid institutional learning remains a decisive factor in contemporary warfare.

The High Adaptation Velocity of Russia and Ukraine

Both the Russian and Ukrainian militaries provide examples of what a high adaptation velocity looks like in execution, since the life cycle of tactical innovation is often measured in days and weeks.[15] The Bayraktar TB2 drone emerged as a symbol of Ukrainian defiance in the opening months of Russia's full-scale invasion.[16] This one-way attack drone devastated Russian armored columns.[17] Early in the war, Russian formations were tethered to rigid war plans, and tactical-level Russian leaders were not empowered to execute disciplined initiative. This early Ukrainian success reflected a massive Russian adaptation gap, not an inherent superiority of the Bayraktar platform.[18] Although Russian air defense and EW units were theoretically superior, they initially failed to coordinate their mobile protection umbrellas. These openings created blind spots, allowing medium-altitude drones to operate freely.[19] However, this gap proved exceptionally short-lived. Only months later, by the summer of 2022, Russian forces consolidated their integrated air defense systems and refined their EW protocols. Perhaps the most important change the Russians made was the relocation of their high-tier jamming assets such as the Krasukha-4 to protect frontline sectors.[20] These adjustments effectively neutralized the Bayraktar as an effective strike platform, forcing this once-celebrated one-way attack drone into reconnaissance missions far from the forward line of troops.[21] This Russian adaptation caught many military pundits by surprise. Experts wrote that large tactical drones would dominate the sky and could achieve decisive effects, but this rule was broken in just a few months by a disciplined adversary's adaptation velocity.

By early 2023, the focus of the conflict shifted from medium-altitude platforms to the mass proliferation of FPV loitering munitions. While these tools were initially employed as niche tools by specialized volunteer units, soldiers on both sides readily adopted these tools, and FPV drones rapidly became the primary killers on the battlefield.[22] The omnipresence of FPV drones forced maneuver units to disperse for their protection. This shift created a crisis for prewar combined arms doctrine. During that summer counteroffensive, Ukraine attempted to execute high-intensity breaching operations, modeled on NATO standard doctrine, but the pervasive presence of loitering munitions made these ground assaults costly.[23] This doctrinal shortfall forced small-unit leaders to innovate to survive. They reduced the size of assault forces, developed ad hoc mesh networks for controlling drones to extend their range, and improvised "turtle tanks" with do-it-yourself cage armor to mitigate the effectiveness of armed FPVs.[24] These were not top-down solutions. They were creative, bottom-up responses to a transparent battlefield that had challenged and, in some cases, rendered standard concepts of combined armed maneuver obsolete.[25] Even though institutions were slow to modernize their doctrine in response to technical progressions on the forward line of troops, frontline leaders recognized the adaptation gap and pivoted to protect their formations.

Since then, the war has been characterized by a rapid EW-FPV arms race, driving the cycle of measures and countermeasures at an unmatched pace.[26] Throughout late 2023 and into 2024, the electromagnetic spectrum (EMS) became as contested as the physical trenches.[27] This shift requires EW effectors at every echelon, with the Ukrainians embracing these assets at the platoon level.[28] The Ukrainians may only be able to use a specific control frequency for a loitering munition for only a few days before Russian EW specialists identify its unique signature and jam it across that sector. This suppression relies on a layered architecture of specialized systems designed to sever the link between the operator and the munition. The R-330Zh Zhitel provides a "hard" kill of the command link by automatically detecting, direction-finding, and jamming bands used by satellite communications and drone control signals.[29] Simultaneously, the Pole-21 creates a pervasive Global Navigation Satellite System-denial environment.[30] This removes the precision navigation required for terminal



Russian six-inch howitzer artillery helped destroy the Japanese siege of Port Arthur during the Russo-Japanese War, 1904–5. The Japanese failed to adapt to the new, industrialized way of war. (Photo courtesy of Wikimedia Commons)

guidance on loitering munitions.[31] The Ukrainians were forced to develop a more frequency-agile system that can survive in a denied environment.[32] Recently, Russian EW units have moved from broad-spectrum jamming to more sophisticated "AI-based spectrum analysis tools" to spot-jam specific control links and communication hubs.[33] In response, Ukrainian drone teams have had to innovate beyond manual procedures, and they are now using firmware updates and multifrequency antennas to maintain link integrity.[34]

This progression highlights a clear shift in the complexity of battlefield adaptation, from changing tactics, techniques, and procedures to altering physical hardware, and now into EW and software modifications. The modern battlefield is no longer a contest between two forces following a better manual. This is a contest between two militaries that are not rigidly restrained by an inflexible, centralized bureaucracy that waits for an approved solution from the rear. Instead, battlefield dynamics are forcing both militaries to empower frontline leaders to experiment and iterate in contact. Such continuous adaptation requires that leaders at every echelon possess cognitive flexibility, openness to change, creativity, and technical fluency.

The Russo-Ukrainian War today may be a good analogy for the lessons that should have been learned from the Russo-Japanese War for early twentieth-century Europe. The Russo-Japanese War offered a violent demonstration of how new combinations of technology, organization, and doctrine could alter the conduct of high intensity war. The war in Ukraine may be doing the same today. The US Army is already drawing lessons from this conflict by revising doctrine, investing in counter unmanned aircraft system (UAS) capabilities, and updating training scenarios. While this is certainly necessary, it is not sufficient to increase the Army's adaptation velocity. It must also transform how it educates and develops its leaders.

Traditionally, US Army PME has been optimized to codify and transmit the institution's current understanding of war. It passes down its doctrine, planning processes, and combined arms concepts. However,

if future conflicts require a high level of adaptation velocity, PME must do more than transmit existing knowledge. It must also prepare leaders for the near certainty that their first conflict will not conform to their mental models. Their success will depend on how rapidly they can recognize, interpret, and close the emerging adaptation gap while in contact. This will require two distinct lines of effort: first, developing creative leaders who are cognitively prepared to challenge assumptions, recognize when reality is diverging from prewar models, and adapt and innovate under fire; and second, enhancing technical fluency in key battlefield technologies.

Line of Effort 1: Developing creative leaders—a Benning Revolution 2.0. PME institutions must forge creative leaders who are intellectually prepared to challenge fundamental prewar assumptions. Unlike Nogi in 1904 or the British Army in 1914, the United States cannot afford to learn lessons after tens of thousands of soldiers die. US Army leaders must identify and apply lessons faster than their early twentieth-century counterparts, outpacing even the adaptation velocities demanded during the Global War on Terrorism.[35] When reality begins to diverge from standard models, leaders must be able to rapidly identify an adaptation gap and close it. A leader's cognitive flexibility, or willingness to discard what was practiced when it proves ineffective, often determines an operation's outcome. PME institutions should build environments in which leaders are continually pressured to assess novel situations, generate and test new solutions, accept failure as a necessary byproduct of learning, and foster a culture that challenges existing procedures.

The Benning Revolution under George C. Marshall provides a historical blueprint for this type of institutional pivot.[36] When Marshall arrived at the Infantry School in 1927, he found a system obsessed with perfectionism and the meticulous drafting of multipage operations orders.[37] Recognizing that the next war would be defined by chaos and the fog of war, Marshall famously threw out the scripted solutions and static scenarios.[38] He forced officers to solve tactical problems with incomplete maps or under sudden, mid-exercise changes in the situation.[39] Marshall's intent was not to teach a specific maneuver but to develop the habit of mind required to act decisively when the manual was no longer applicable.[40] He understood that a leader's most dangerous foe is not the enemy's fire but their own psychological attachment to a plan that is no longer working.

Marshall demanded that leaders cultivate creativity and mental agility by forcing them into ambiguous scenarios. He knew that intellectual readiness in future large-scale combat operations required more than knowing the manual. He pushed his students beyond rote memorization. He wanted leaders to understand the fundamentals, but even more importantly, he wanted them to be able to apply these concepts in novel circumstances. Because Marshall introduced uncertainty and ambiguity into training events, his students were frequently compelled to devise solutions not codified in existing doctrines. There was no prescribed "right answer." Today's PME institutions need to do more to promote creative problem-solving. A Benning Revolution 2.0 must merge Marshall's foundational tactics with deliberate training in creativity and innovation. Only then will future leaders possess the intellectual framework required to maximize our adaptation velocity in contact.

Fortunately, the US Army has already taken steps to enhance creativity in its leaders. The author's firsthand participation in this imaginative research at Fort Belvoir underscores the institutional momentum on building creativity in the force.[41] Dr. Angus Fletcher's research, especially his field guide, *Creative Thinking*, provides actionable recommendations and exercises for building creativity by shifting the focus from rote deduction to narrative-based iteration.[42]

Furthermore, the US Army has integrated a creativity assessment known as the Individual Adaptability Measure (I-ADAPT) into Project Athena.[43] Project Athena itself was a program designed to promote self-awareness and encourage self-development to produce more effective leaders.[44] I-ADAPT is a diagnostic framework designed to quantify fluid adaptability, which the US Army is using as an initial leader assessment. Though this is an important first step, it is not enough. The Army should also conduct these evaluations longitudinally at every major PME milestone, tracking a leader's cognitive evolution from initial entry to senior leadership. Data from these longitudinal assessments should inform individualized coaching, not sit unused in personnel records. PME instructors should use this information to tailor their teaching

approach to fit their individual students. For students demonstrating high natural creativity, instructors can leverage complex historical vignettes to push the boundaries of their conceptual risk-taking. Conversely, for leaders who have a more rigid cognitive baseline, the instructors could emphasize how intellectual humility and psychological safety can create a climate where subordinates feel empowered to share their creative ideas. A commander does not personally need to be a creative visionary. They must, however, possess self-awareness to recognize their own cognitive limitations and to cultivate an environment in which subordinate innovation is protected and even encouraged.

This individualized approach to building adaptive command teams should reflect rank and personnel category. For the officer and warrant officer corps, spanning from their respective basic leader courses through Intermediate Level Education, PME can be used to shatter the chronic institutional obsession with engineering a flawless military decision-making process. Similar to Warfighter exercises, PME should employ high-fidelity virtual simulations where the adversary's capabilities and tactics dynamically evolve mid-exercise. Rather than evaluating students on the flawless execution of a rigid planning process under optimal conditions, evaluation metrics should measure how quickly a leader can pivot in response to a diverging battlefield reality. These compressed, high-pressure scenarios must integrate degraded communications, partial or conflicting intelligence, and an actively adapting threat. Like Marshall did during the Benning Revolution, PME should strip away the safety net of a "schoolhouse correct" answer and force leaders across all ranks to lead through acute uncertainty.

For too long, officer PME has been constrained by the tyranny of optimization during the military decision-making process. Instead of building a single, flawless plan, officers should be forced to build a plan, test it against a thoughtful, agile enemy, and then reassess. Instead of fighting multiple single, pitched battles, PME courses should fight a long, continuous campaign. Battles should progress from one engagement to the next without a tactical reset. In a continuous campaign, the consequences of an officer's initial blind spots or poor risk management are carried forward into the next fight. If a leader overextends their force or miscalculates an enemy capability in the opening battle, they cannot simply wipe the whiteboard clean for the next block of instruction. They must live with their attrition, degraded logistics, and broken communications, forcing them to modify their plans in real time.

This shift may be even more important in PME for the Noncommissioned Officer (NCO) Corps. By integrating the same persistent, multiday exercise framework into the NCO curriculum, the Army can encourage disciplined initiative. From the Basic Leader Course through the Master Leader Course, NCOs must be conditioned to dynamically alter troop leading procedures and battle drills in response to real-time battlefield disruptions. This could be accomplished through unexpected electronic jamming or localized drone saturation. When communication with higher headquarters is severed during a simulated campaign, the exercise must continue, forcing squad- and platoon-level leaders to rely on their own initiative.

NCOs will likely be the ones who innovate and identify opportunities to exploit enemy vulnerabilities on the transparent battlefield. Because small-unit leaders operate at the point of friction, they are often the first to recognize when a standard operating procedure has become a tactical liability. In a perpetual campaign spanning the PME curriculum, NCOs must be empowered not only to react to disruptions but also to invent field-expedient solutions. This might be modifying patrol formations to evade drone sensors, employing drones in new ways, or even reconfiguring local networks to ensure security. They must be encouraged to act and not wait for orders. Our NCOs will largely be responsible for recapturing the initiative and seizing opportunities for frontline innovation. When communications go down, victory goes to the platoon that can innovate immediately and effectively.

Additionally, the Army should integrate adversarial peer review and opposing force as mandatory, graded components across all PME events. Instead of merely presenting a tactical plan for passive evaluation, students must defend their operational logic against a peer group explicitly tasked with challenging assertions and questioning unfounded assumptions. This dialectical educational model could replicate the hyper-rapid feedback loops observed in the drone development labs of contemporary Ukraine. Regular exposure to rigorous peer critique habituates leaders to maintain a healthy skepticism toward institutional dogma. This

> Today, a tactical leader without a working knowledge of the electromagnetic spectrum is functionally blind.

debate-style format also provides instructors with opportunities to call out cognitive biases in real time. When peers ruthlessly dismantle a standard doctrinal approach, students will have to think creatively to salvage their mission. Even if this technique is uncomfortable, having peers professionally critique each other's ideas enhances the force's collective creativity. Building this habit of rapid iteration is the best way to outpace a modern adversary and decisively close the adaptation gap on the battlefield.

By structuring PME around a fluid, multiday campaign rather than isolated tactical vignettes, the Army can replicate the true velocity of modern war while promoting creative cognition under stress. This continuous campaign would provide instructors with an unbroken arc of successes and failures to use as narrative-based training tools. Like Marshall did at Fort Benning, PME instructors should challenge students to analyze *why* a creative risk succeeded or *how* a cognitive bias led to failure. Ultimately, this method trains leaders to outthink the adversary and increase adaptation velocity. Victory will belong to the command teams that can rapidly spot a failing situation, scrap obsolete assumptions, and invent novel alternatives under pressure.

Line of Effort 2: Enhancing leaders' technical fluency. The modern battlefield requires an unprecedented baseline of technical fluency. Leaders must understand, with some degree of precision, how technology influences a battle. Today, a tactical leader without a working knowledge of the electromagnetic spectrum is functionally blind. PME must therefore cultivate widespread, conceptual literacy that enables leaders to understand the capabilities, vulnerabilities, and integration mechanics of modern technology.

To systematically scale this fluency across the total force, PME should organize its technical curricula into technology focus areas (TFA). These could be UAS and counter-UAS, EW and EMS management, cyber and data security, distributed mesh communications, or artificial intelligence integration.[45] Or PME could align these focus areas with the Office of the Under Secretary for Research and Engineering's six critical technology areas.[46] Each TFA or critical technology area would provide an introductory familiarization with relevant technology. As junior soldiers, leaders should strive to develop a deep understanding of one of these TFAs. However, as they advance in rank and responsibility, this relationship must deliberately broaden to encompass an understanding across every focus area. PME institutions must cultivate this progressive technical acumen by deliberately shifting a leader's focus from a single specialty early in their career toward general proficiency in how these different systems integrate.

For junior officers and NCOs, technical training must be hands-on, tactile, and highly practical. At the squad and platoon levels, leaders should know how to assemble an FPV drone, troubleshoot a localized mesh radio node, and employ tactical counter-UAS jamming equipment. They must understand the immediate physical parameters of their gear such as battery life constraints, sensor ranges, payload limitations, and simple signature management.

As these leaders transition into midgrade staff and command positions, they must shift their focus away from operating a single tool. These leaders must explore how each core technology fits into the entire system. A major should no longer spend instructional hours learning how to fly an individual drone. Instead, they should focus on auditing data streams from disparate sensors to shorten the kill chain or mitigate their tactical operations center's electromagnetic signature. Their training must emphasize how disparate technical systems interact, where the single points of failure exist in a distributed network, and how to manage risk on a fluid battlefield.

At the senior leadership level, technical acumen must focus on systems architecture and capability management. A brigade commander does not need to know how to program a software patch, nor do they

need to know how to solder a wire. Instead, they must understand how the waveforms that control autonomous drone swarms can be manipulated, the structural defense of data pipelines against algorithmic manipulation, and the EMS signatures that their entire formation projects across multidomain sensors. By offering a ladder of technical literacy, PME ensures that leaders can confidently build technical fluency appropriate to their increasing scope of responsibilities.

Institutional training cannot realistically scale or maintain bleeding-edge technical expertise within a rigid classroom pipeline, but it may create an intellectual spark.[47] By exposing young soldiers to these technologies early, PME can help demystify seemingly complex technologies and drive long-term curiosity. Rather than trying to forge finished technical experts through a standardized curriculum, military education should seek to ignite a soldier's personal initiative by providing the foundational knowledge that they will need. By treating education as a gateway rather than a final destination, the Army can shift the burden of continuous technical mastery from static schoolhouses to dynamic, self-motivated individuals who are driven to explore these domains on their own.

To sustain this spark, the Army must lower the barriers to individual experimentation by reducing bureaucratic barriers and offering accessible avenues for self-development. For example, training institutions could integrate hands-on fabrication labs where soldiers physically build customizable, low-cost drones. This initiative would be even more effective if the Army allowed soldiers to retain the hardware upon graduation. This would allow soldiers to continue testing and modifying systems at their own pace. Or, to promote computer skills, PME institutions could help enroll students in advanced certifications and foundational programming courses. This could help soldiers maintain momentum in their cyber studies well after leaving the schoolhouse.

PME could also be the most important conduit, linking soldiers to exceptional but little-known resources such as the Department of Defense Information Analysis Centers and, specifically, the Defense Technical Information Center.[48] By reducing barriers between scientists and practitioners, PME institutions can increase the speed and precision of peacetime experimentation. Just by exposing soldiers to resources like the Defense Technical Information Center, PME can elevate the quality of military discourse by injecting detailed, unclassified technical reports into the conversation about lethality.[49] Finally, institutions must equip leaders with practical know-how to navigate and exploit tuition assistance programs for key technologies. This could include training on enhancing operational workflows such as prompt engineering, custom script writing, and large language model integration. With this exposure and these concrete, self-paced resources, the Army can transform casual interest into a competitive advantage.

Implications of These Changes

Implementing these reforms across PME demands structural changes to how the Army recruits talent and evaluates performance. First, the institution must incentivize duty as a PME instructor to ensure the highest-caliber leaders are in front of the classroom. The Army can no longer afford to treat schoolhouse assignments as a career plateau. Instead, successful instructor tenure must be explicitly linked to fast-track promotions and key developmental selections. Furthermore, the Army must update institutional reading lists across all levels of PME to anchor this cultural shift in modern reality. Standard historical texts should be paired with modern research by David Barno, Nora Bensahel, and Liam Collins. These scholars specialize in battlefield adaptation and wartime innovation, and they offer specific examples of high and low adaptation velocities. Integrating their work alongside Fletcher's foundational cognitive research encourages leaders to study how formations successfully pivot under fire. This ensures the force is analyzing examples of real-time adaptation rather than memorizing outdated operational paradigms.

Second, the Army must modernize its talent management systems to capture creativity and innovation on evaluation reports more explicitly. There are limited incentives for leaders to challenge the status quo. Current evaluation metrics heavily favor predictable execution and risk aversion. But this institutional conservatism has noble roots. It stems from the need to ensure discipline, standardization, and repeatable performance. However, when left unchecked, these rigid traditions inadvertently penalize experimentation and agility. By creating distinct, measurable performance

indicators for creative problem-solving and adaptive leadership on officer and NCO evaluations, the institution will naturally incentivize prudent risk-taking. What gets measured gets managed. Assessing and rewarding adaptability ensures that creative thinkers are promoted into positions of operational influence.

Finally, we must improve our foundational military doctrine. The Army must formalize "adaptation" as an official principle of war.[50] The current principles are enduring and timeless, but they also largely assume a stable baseline of battlefield conditions. Adding adaptation acknowledges that the ability to rapidly discard obsolete assumptions and iterate on solutions is itself an important skill that must be deliberately cultivated. Codifying adaptation as a principle ensures that PME institutions emphasize its importance throughout the curriculum. This update would also encourage military planners to build additional flexibility into regional war plans.[51] Ultimately, this addition would celebrate creativity as a doctrinal requirement for victory in the next conflict, no longer an optional leadership trait.

Conclusion

While the US Army has a strong lineage of successful adaptation under fire, the compounding speed of change in modern conflict demands a radical transformation in education. Marshall's interwar tenure at Fort Benning provides the ideal historical template for this shift. Launching a Benning Revolution 2.0 requires a holistic approach that seamlessly merges Marshall's signature emphasis on tactical ambiguity with a requirement for technical fluency. PME must transcend its traditional role as a conveyor belt for established doctrine and must instead habituate leaders to continuous reassessment and rapid conceptual iteration. Ultimately, the survival of the future force rests on an educational ecosystem that trains leaders not *what* to think but teaches them *how* to outthink a volatile, shifting battlefield.

The historical cost of failing to upgrade our institutions is steep. A century ago, British observers meticulously documented how industrialized technology shattered legacy tactics during the Russo-Japanese War. Yet, the British military establishment failed to translate these accurate field reports into an agile institutional learning loop. As a result, its leaders were unable to adapt rapidly during the opening months of the First World War. The British experience serves as a stark reminder that possessing battlefield data is fundamentally meaningless if an educational system lacks the mechanisms to operationalize it.

Today, the United States must use the conflict in Ukraine as a critical window into the future of peer warfare. However, this modern window offers no luxury of complacency, as the Russo-Ukrainian War demonstrates that the life cycle of tactical innovation is now compressed. In future large-scale combat operations, technical and organizational sluggishness will be punished instantly and severely. Future battlefields will reward militaries that have high adaptation velocity and punish those that adhere to legacy doctrine for too long. Implementing a Benning Revolution 2.0 is now an operational necessity to ensure that future leaders, especially commanders, possess both cognitive agility and technical literacy. By deliberately embedding uncertainty and technological fluency into leader development today, the Army will ensure that our next first contact results in rapid adaptation rather than institutional tragedy. ■

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Normandy
Virtual Staff Ride

Normandy virtual staff ride products are available for download from the Army University Press website. This study focuses on the American side of the invasion to include the airborne assault, Omaha and Utah Beaches, Pointe du Hoc, and a study on sustainment and the artificial harbors. Materials include instructor notes, participant read aheads, and the virtual terrain. These products will enable organizations with access to Virtual Battlespace 3 to conduct their own virtual staff ride or to conduct their own professional development sessions without the terrain.



Omaha Beach



German Defenses



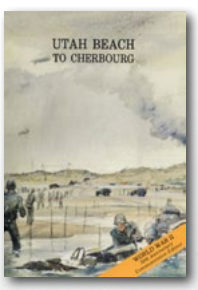
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Omaha Beach, 6 June 1944 (Photo courtesy of the National Archives)



Monitors display drone footage on 8 December 2025 during a dry run for the Eastern Flank Deterrence Line (EFDL) Task Force X Pilot Series demonstration at Riihimäki, Finland. The EFDL concept is a distributed persistent system of capabilities across domains and national forces that leverages unmanned systems, backed with an AI-enabled mission command network using live data to accelerate decision-making to detect and disrupt adversary attacks. (Photo by Staff Sgt. Karla Parra, US Air Force)

Command Without Control



Mission Command in the Age of Artificial Intelligence

Maj. Prabhat Mishra, Indian Army

War is a human endeavor—a clash of wills characterized by the threat or application of force and violence … It is not a mechanical process that can be precisely controlled by machines, calculations, or processes.

—ADP 6-0, *Command and Control*

As artificial intelligence (AI) increasingly shapes how militaries perceive, decide, and act, the assumption that war resists mechanization is being tested in ways doctrine did not anticipate. Across recent conflicts, from Ukraine's AI-enabled drone networks to the US-Israel campaign against Iran, machine-assisted systems are compressing decision cycles to speeds that strain human cognition, enabling sensor-to-shooter linkages and generating targeting options at a scale and tempo previously unattainable. In such environments, commanders are shifting from making decisions to choosing among machine-generated recommendations under significant time pressure. The question is no longer whether AI will shape military operations; it already does. The question is, What does it do to the humans who command them?

This shift strikes at the foundation of military leadership doctrine. Mission command, as articulated in Army Doctrine Publication 6-0, *Command and Control*, is built upon mutual trust, shared understanding, and the exercise of disciplined initiative guided by the commander's intent.[1] The operations process similarly places the commander at the center of an iterative, human-driven cycle of understanding, visualizing, and directing.[2] Yet, as AI systems mediate situational understanding and influence decision outputs, the locus of trust risks shifting from human leaders to algorithmic systems, while authority becomes increasingly diffused across human-machine networks. Responsibility, however, still rests with the commander.

Beyond the battlefield, these dynamics extend into routine organizational leadership. At the organizational level, leadership is fundamentally an act of influence grounded in trust, climate, and professional judgment.[3] As AI-enabled tools become embedded in planning, analysis, and decision support, subordinates may begin to give greater weight to machine-derived outputs than to the commander's intent, as algorithmic confidence gradually displaces human judgment. The concern is less about disobedience and more about the gradual sidelining of human judgment.

This is the condition I describe as "command without control," authority that remains formally intact while the human judgment animating it is gradually hollowed out by systems optimized for speed rather than wisdom. AI accelerates military decision-making and, more importantly, restructures who, or what, is trusted to make decisions, while accountability remains unchanged. The resulting tension with the doctrinal foundations of mission command is growing, underexamined, and consequential, and addressing it requires leaders not to reject AI but to reassert, deliberately and continuously, the primacy of human judgment within systems designed to make that reassertion feel unnecessary.

Mission Command Under Algorithmic Pressure

I have sat in planning rooms on two continents. In one, the map was hand-marked, the radio was the only link to higher, and decisions came from the commander's personal reading of ground he had walked himself. In the other, AI-generated overlays filled the screen faster than the staff could examine them, and the instinct was to build on the output rather than question it. The contrast had little to do with competence. It came from how quickly the machine made uncertainty seem settled.

Mission command rests on a simple but demanding premise: Commanders build cohesive teams through mutual trust, create shared understanding, provide clear intent, and empower disciplined initiative.[4] This philosophy assumes that humans interpret uncertainty, exercise judgment, and act within a framework of trust. AI challenges these assumptions indirectly by mediating how understanding is constructed, how trust is formed, and how decisions are executed within the command system, rather than by simply replacing leaders.

The first point of friction lies in shared understanding, a cornerstone of mission command. AI-enabled systems increasingly shape how units perceive the battlefield by filtering, prioritizing, and presenting information. Contemporary research on AI-enabled military decision-making highlights that such systems aggregate vast datasets and produce synthesized operational pictures at speeds beyond human processing capacity.[5] In Ukraine, AI-enabled drone networks



An M2A3 Bradley Infantry Fighting Vehicle assigned to 6th Squadron, 1st Cavalry Regiment, engages an unmanned aircraft system with its M242 25mm Bushmaster chain gun during Project Flytrap at Pabradė Training Area, Lithuania, 10 May 2026. During Project Flytrap, soldiers integrate counter-unmanned systems, AI-enabled command and control, and live data networks to move faster, decide faster, and fight more effectively across all domains. (Photo by Sgt. Max Elliott, US Army)

compressed sensor-to-shooter cycles from hours to seconds, demonstrating how machine-mediated ISR fundamentally alters the pace at which situational awareness is developed and shared at the tactical level.[6] The ongoing US-Israel campaign against Iran extends this pattern to operational scale, with AI targeting systems processing over one thousand strike targets within the first twenty-four hours at speeds that structurally preclude deliberate human evaluation of each decision.[7] As a result, the traditional staff-driven process of developing situational understanding is being supplemented, and in some cases supplanted, by machine-curated outputs. While this enhances speed and data integration, it also introduces a dependency on algorithmic filtering, raising critical questions about visibility, bias, and interpretability within the command process.

Second, AI reshapes the execution of the military decision-making process and related decision frameworks. Analysis of AI integration into command and control suggests that these systems compress planning and execution timelines, enabling faster course-of-action generation and selection.[8] Instead of working through a deliberate staff process, commanders are more often presented with options that have already been filtered and shaped by AI systems. This compression challenges the doctrinal expectation that commanders develop understanding through continuous dialogue, analysis, and refinement.

Third, AI alters the foundation of mutual trust within military organizations. Mission command assumes trust flows among commanders, subordinates, and staffs, enabling decentralized execution. However, studies on AI-supported decision-making in military contexts indicate that heavy reliance on AI tools can erode leaders' intrinsic decision-making skills and complicate their ability to retain control over complex systems.[9] Over time, this shifts trust away from people and toward the perceived reliability of the system. Ferenc Fazekas notes that AI-enabled command systems may unintentionally centralize informational authority, even as they are intended to support decentralized execution.[10]

Fourth, AI intensifies the tension between decentralization and centralization, a defining feature of mission command. Some scholars argue that AI can reinforce decentralized execution. Sorin Matei and Kyle Reed contend that AI-enabled command systems, through a process of data condensation and

distillation, can deliver decision-quality information to both senior commanders and the tactical edge simultaneously, thereby enhancing, rather than eroding, the intent-driven logic of mission command.[11] In this view, AI's capacity to manage information asymmetry is not a vulnerability but an enabler of distributed initiative.

Yet, war-gaming evidence from the US Army War College challenges this optimistic assessment. Aaron Wilcox and Chase Metcalf document how AI-enabled planning systems produced cognitive anchoring among staff officers, who deferred to machine-generated courses of action even after those outputs were demonstrated to be operationally flawed.[12] Rather than empowering decentralized initiative, the AI system functioned as a centralizing cognitive authority, not through command hierarchy but through algorithmic influence over how options were framed, sequenced, and accepted. This means commanders have to work harder to maintain the balance between control and initiative rather than assume it will emerge naturally.

Finally, AI complicates the relationship between authority and accountability within mission command. While decision-support systems shape recommendations and influence outcomes, responsibility remains with the human commander. Research on AI-enabled military decision-making highlights that such systems can be difficult for users to fully interpret, particularly when outputs are generated through complex models and probabilistic reasoning, complicating validation and trust.[13] This creates a gap. Decision inputs are spread across systems and actors, but accountability remains with the commander.

These structural pressures leave mission command intact on paper while exposing the conditions under which its human foundations begin to erode.

AI, Trust, and the Erosion of Human Judgment

A staff officer reviewing fifty AI-generated strike options in a thirty-minute window is not making decisions in any traditional sense; he is administering them. This distinction, subtle in peacetime planning and catastrophic in combat, lies at the heart of what AI does to military judgment. Doctrine consistently emphasizes that command is an exercise in judgment under uncertainty, requiring leaders to interpret incomplete information, accept risk, and act despite ambiguity.[14] AI leaves uncertainty in place yet creates an illusion of certainty that can distort how leaders and subordinates think, decide, and act.

A central risk is the growing prevalence of automation bias, where individuals tend to overtrust machine-generated outputs, particularly in high-tempo environments. Empirical research demonstrates that users frequently defer to algorithmic recommendations even when those recommendations are incorrect, especially under conditions of time pressure and cognitive load.[15] In military contexts, where AI systems are increasingly embedded in targeting, intelligence analysis, and planning tools, this bias can subtly shift decision-making authority away from human judgment. Critically, this tendency has been documented specifically in targeting contexts, where overreliance on automated recommendations raises not only operational but also legal concerns about compliance with the obligations of distinction and proportionality under international humanitarian law.[16] Leaders may begin to accept AI outputs not as inputs to be questioned but as conclusions to be validated.

This shift has important implications for organizational leadership and trust. Leadership within military organizations depends on influence grounded in professional judgment and experience. However, as AI systems become embedded in planning and analysis processes, the basis of that influence may evolve. Joseph Huitt argues that AI can shape how decisions are made and interpreted, suggesting that leaders must guard against overreliance on system-generated outputs, particularly when such outputs appear analytically robust.[17] In such contexts, subordinates may increasingly rely on system outputs when forming their own assessments. This is not disobedience; it reflects a reconfiguration of trust within the organization, with technology becoming a more prominent intermediary in decision-making processes.

This vulnerability is not confined to cognitively fatigued soldiers under fire. At the US Army Command and General Staff College, AI integration across planning exercises, war games, and academic assessments has proceeded rapidly enough to prompt formal institutional intervention. CGSC Bulletin 921, issued in August 2025, explicitly cautions that generative AI tools "lack the context and judgment to reliably turn summary or comparison into sound analysis or advice,"

and prohibits their unsanctioned use in graded assessments, a restriction whose necessity among field-grade officers with operational experience is itself instructive.[18] The fact that this shows up even in a professional military education environment suggests that exposure to AI is sufficient to displace independent judgment regardless of rank or experience.

AI also affects how leaders engage with uncertainty and risk, both of which are central to effective command. Military leadership traditionally requires comfort with ambiguity and the ability to make decisions without complete information. However, AI systems, by presenting structured outputs, probabilities, and prioritized recommendations, can create a perception of clarity that may not reflect the underlying complexity of the operational environment. Research notes that such systems can obscure their own limitations, particularly when users lack visibility into how outputs are generated.[19] The greater danger lies in AI's tendency to mask uncertainty, encouraging overconfidence in decisions that remain inherently uncertain.

Authority, Accountability, and Ethical Risk in AI-Enabled Warfare

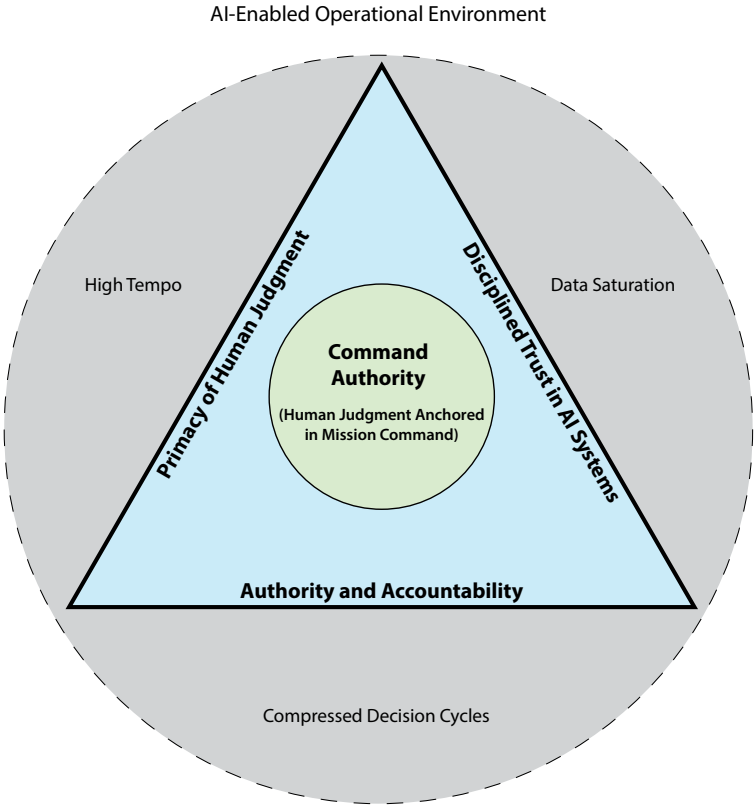
AI does not reduce a commander's responsibility. If anything, it makes it heavier. While AI systems increasingly inform, prioritize, and even shape operational decisions, the ethical and legal burden of those decisions remains firmly with human leaders. The result is a command environment in which authority expands while accountability stays fixed.

This tension emerges from the relationship between decision support and decision ownership. AI systems can process vast datasets and generate targeting options at speeds that exceed traditional human analysis. This increases operational efficiency, but it also places pressure on decision-makers, who must evaluate system-generated outputs rapidly. This pressure becomes more pronounced in high-tempo targeting environments. In the ongoing Iran conflict, the US military confirmed the use of the Maven Smart System (fusing over 170 intelligence sources) alongside commercial AI tools to generate and rank strike targets at a scale and tempo that defense analysts described as structurally incompatible with deliberate legal review.[20] Researchers observing the campaign also noted that while AI accelerated targeting significantly, it did not reduce, and may have intensified, the requirement for human judgment, precisely because the volume and speed of machine-generated options compressed the space available for evaluation.[21] In such conditions, decision-making risks shifting from deliberate assessment to procedural validation, where the role of the human commander is to confirm rather than critically interrogate system outputs.

At the same time, AI introduces a fundamental challenge to accountability structures. Traditional military accountability relies on a clear chain linking decisions to identifiable actors. However, AI-enabled systems distribute decision inputs across a network of designers, data sources, operators, and commanders. The Iran campaign has made this diffusion visible in an unprecedented way: the US military's integration of Anthropic's Claude (a commercial large language model) with Palantir's Maven targeting system for strike prioritization generated a public dispute over the ethical limits of that use.[22] This fracture illustrates precisely how responsibility fragments across commercial developers, military operators, and commanders when decisions emerge from human-machine networks rather than identifiable human actors. The result is a growing gap between who influences decisions and who is held accountable for them.

Military leadership rests on a principle that commanders are responsible for everything their units do or fail to do, including the ethical implications of their decisions.[23] AI complicates the application of this principle in practice. Concepts such as distinction and proportionality require contextual judgment and careful verification that AI systems cannot reliably provide. An analysis of algorithmic targeting in the Iran-Israel confrontation emphasizes that without sufficient human oversight, AI targeting systems can contribute to civilian harm and raise concerns regarding compliance with international humanitarian law.[24] AI targeting systems relying on statistical inference rather than verified legal certainty undermine the principles of international humanitarian law—obligations of distinction and proportionality—creating accountability gaps that existing command structures are not designed to address.[25]

When accountability becomes procedural in the form of a click on a targeting list rather than a substantive considered judgment, command without control is no longer a theoretical risk, it is the operational condition.



(Figure by author)

Figure. Reframing Military Leadership in the AI Era

Reframing Military Leadership for the AI Era

AI does not require a new theory of military leadership; it demands a more deliberate application of existing principles under altered conditions. The preceding sections establish a pattern: AI is less likely to break command outright and more likely to erode it from within by making the displacement of judgment feel like efficiency. Restoring meaningful command authority requires deliberate action along three reinforcing lines (see the figure).

First, leaders must deliberately reassert the primacy of human judgment. AI systems can generate and prioritize options, but they cannot replace contextual understanding, experience, or ethical reasoning. Leaders must resist the tendency to equate speed with sound decision-making and ensure that critical decisions retain a deliberate human checkpoint. This is particularly important in high-tempo environments, where the pressure to act quickly can reduce decision-making to selection among preprocessed options. Reasserting judgment does not require rejecting AI; it requires actively interrogating its outputs, questioning assumptions, and ensuring that decisions remain grounded in operational context.

Second, leaders must develop disciplined trust in AI systems. Trust remains central to mission command, but in an AI-enabled environment, it must extend beyond interpersonal relationships to include human-machine interaction. This trust cannot be uncritical. Leaders must understand the capabilities and limitations of the systems they employ and ensure that subordinates treat AI outputs as inputs for consideration rather than conclusions for acceptance. Building this discipline requires training, repetition, and command emphasis. Units must learn not only how to use AI but also how to challenge it appropriately, particularly when outputs conflict with contextual understanding or commander's intent.

Third, leaders must actively preserve command authority and accountability. AI-enabled systems can distribute decision inputs across networks and

processes, but they must not dilute the commander's role as the final decision authority. Leaders must reinforce the primacy of commander's intent, ensure clarity of purpose, and maintain ownership of outcomes. At the same time, they must remain accountable for decisions informed by AI, recognizing that responsibility cannot be delegated to systems. This requires conscious effort to retain moral and professional ownership of decisions, particularly in environments where technology accelerates action.

These adaptations are not theoretical; they are leadership practices. In doing so, they reinforce the human foundations of command while integrating new capabilities.

Ultimately, the effectiveness of AI-enabled forces will not depend solely on technological sophistication, but on the ability of leaders to remain in control of how decisions are made. The measure of command in the AI era will not be how quickly a leader can process machine outputs; it will be whether, under every pressure to defer, he chose not to.

Conclusion

The ongoing conflicts have crystallized what this article argued across doctrine, wargaming, and institutional observation: AI does not threaten command from the outside but hollows it from within. When a single operation generates thousands of AI-prioritized targets within hours, integrates a commercial large language model into strike decisions, and produces a public dispute over who bears ethical responsibility, the gap between doctrinal accountability and operational reality has ceased to be theoretical. It has become structural.

This is the condition of command without control. Authority remains formally intact. Commanders retain their titles, their signatures, and their legal responsibility. What erodes is the human judgment that gives command its meaning, displaced not by malice or negligence, but by systems optimized for speed, scale, and analytical completeness that make the abdication of judgment feel indistinguishable from efficiency.

Yet the enduring principles of military leadership remain valid. War continues to be a human endeavor, and effective command still depends on judgment, trust, and responsibility. An officer who has commanded soldiers in terrain where no machine reaches and no signal confirms understands this not as doctrine but as experience. That understanding is what the AI era demands and precisely what it is most at risk of eroding. The task for contemporary leaders is to ensure that these principles are not displaced by technology but reinforced through it.

In an era defined by machine speed, the decisive advantage will not belong to the force that processes information fastest, but to the one whose leaders retain control over how that information is understood, interpreted, and acted upon. The future of military effectiveness will depend not only on technological superiority but on the ability of leaders to resist command without control, and to remain, in every sense, in command. ■

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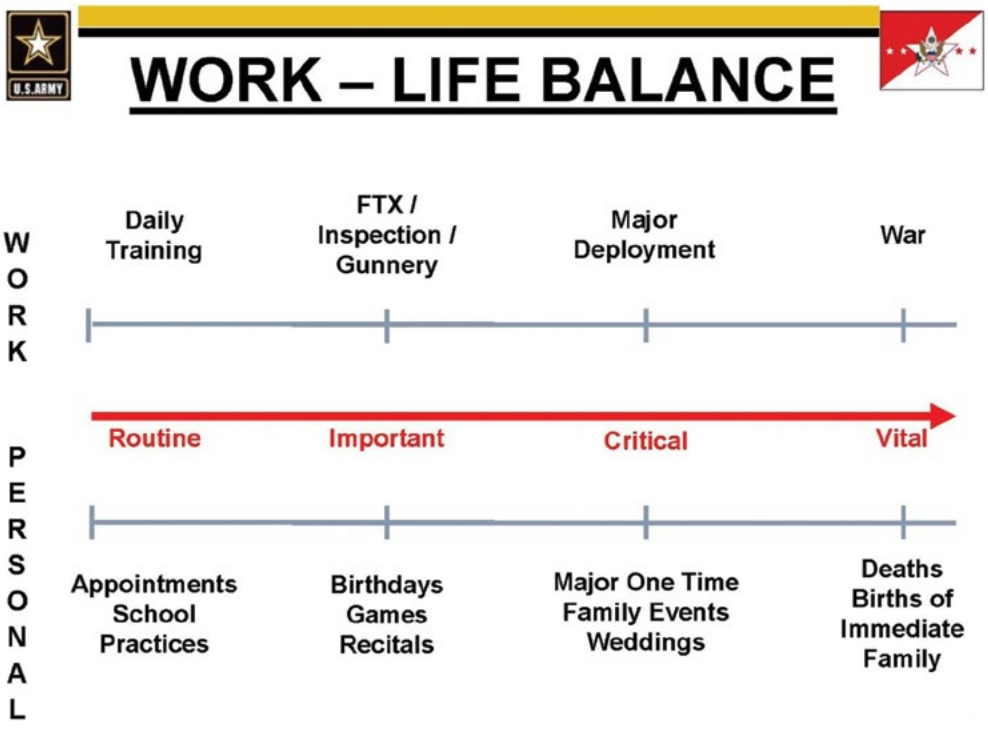


(Photo by Tommy Gilligan, US Army)

The Army's Work-Life Balance Problem

Research with Senior Leaders

Col. Andrew C. Steadman, US Army
Lt. Col. Thomas W. Britt, PhD, US Army Reserve
Leslie B. Hammer, PhD



(Figure by James McConville)

Figure 1. McConville's Work-Life Framework Comparing Routine, Important, Critical, and Vital Work and Personal Events

Editor's note: *This article is derived from Andrew Steadman et al., "Reframing Work-Life Balance for Senior Leaders," chap. 8 in* Maximizing Senior Leaders Health and Wellbeing, *ed. Michael Hosie et al. (Strategic Studies Institute, US Army War College, 2023).*

As a major, James McConville watched a fellow officer miss the birth of his child because the officer was participating in a training exercise in Thailand. The unit had an opportunity to return the captain to Hawaii, but the brigade commander would not let him go. McConville later described that decision as a travesty. The incident shaped his views about the relationship between military service and family life, and he vowed never to allow the same thing to happen on his watch.[1]

Years later, as the fortieth Army chief of staff, Gen. McConville developed a framework that associated important personal events with equivalent work events (see figure 1). Notably, he placed the births and deaths of immediate family members at the same level of importance as war. The framework did more than encourage soldiers to spend time with their families. It established an organizational boundary, elevated the importance of the home domain, and created conditions for soldiers to be present for critical events and fulfill their roles outside work.[2]

The idea of work-life balance is as popular as it is elusive. As mobile technology and globalized information have blurred the lines between home and workspaces, many organizations have emphasized the need to attain balance. A familiar refrain heard among professionals is to find balance, avoid bringing work home, and spend more time doing the things they love with the people they love. These sentiments hint at a point of harmony where individuals can place work into equilibrium with all else, resulting in a long-awaited and fulfilling quality of life.

Work-life balance, however, is neither a balancing act nor a function of delicately equalizing time throughout a day, month, year, or lifetime. A more useful approach is to examine the work-home interface: how work and all other areas of life interact, interfere with one another, and sometimes facilitate one another.

This perspective shows that the work-home interface is a function of apportioning personal energy resources, not simply managing time. When individuals invest in cognitive, emotional, behavioral, and physical resources at work, they retain fewer of those resources to invest elsewhere. This deficiency can lead to conflict between the work and home domains.[3]

To explore this dynamic among senior military leaders, we conducted research in 2021 with resident students at the US Army War College. This article presents the study's findings and shows that for the vast majority of senior military leaders, work holds clear primacy over other areas of life. Work also routinely interferes with home life; yet home responsibilities rarely interfere with work. The research also showed that although military leaders broadly communicate the importance of work-home well-being, many senior leaders do not believe they have the freedom to establish even basic boundaries protecting their own lives outside work. This impoverished quality of the work-home interface among senior military leaders represents a readiness issue that needs to be addressed.

Improving this condition requires senior military leaders to understand how their life roles interact, determine how they want those roles to relate to one another, establish personal boundaries, and create organizational conditions that permit their subordinates to do the same.

The Nature of the Military Work Environment

According to anecdotal and quantitative research, most senior military leaders find that work creates rather than reduces tension and conflict at home.[4] It is rare to hear a senior military leader who is content with the way work interfaces with home life. This narrative is so consistent that it has become a cultural norm and an institutional expectation of military service. Spouses grow frustrated with unpredictable work schedules that cause senior military leaders to miss family events such as dinner, date nights, and children's activities. Dissatisfaction compounds when service members routinely arrive home cognitively and emotionally drained; therefore, unable to engage fully. Six factors help explain the demanding nature of the military work environment.

First, the lethal nature of military work elevates its importance and often gives primacy over other areas of life. Military professionals are responsible for applying discriminate lethal force in defense of the Nation. This mission requires realistic, complex, and time-intensive training that often carries the risk of catastrophic loss. Service members understand the weight of the task, but leaders who plan, resource, and supervise training and combat operations feel especially responsible. Commanders routinely ask themselves, "What might get a soldier killed, and how must I prevent that?" This mindset causes leaders to remain deeply committed and mentally engaged during both duty and off-duty hours.

Military culture places the mission first, but within that mindset there can exist the notion that everything supporting the mission also comes first. This includes planning efforts, coordination meetings, resourcing actions, and the long list of pre-event tasks that prepare service members to achieve the standard. When friction arises or critical tasks stall, leaders often feel they must engage to provide oversight and guidance, regardless of the hour.

Second, the military is committed to providing care for its people. Commanders and leaders are responsible for the safety and welfare of those they lead. This responsibility preserves a ready force and reinforces the trust required for units to fulfill their missions. As a result, leaders quickly adopt a mindset of twenty-four-hour responsiveness and must be ready to dedicate cognitive and emotional resources at a moment's notice. Training accidents, service member indiscipline, and other unfortunate incidents regularly fixate the chain of command, disrupt engagement in other efforts, and pull senior military leaders away from planned commitments at home.

Third, there is simply too much work. The Department of War maintains a massive bureaucracy to manage the people, processes, materiel, and activities required to fulfill its mission. Leaders at every echelon compound that bureaucracy by imposing formal and informal requirements that tend to increase rather than diminish organizational workloads. Supporting service members and families also comes at a bureaucratic cost, as the military sustains programs and resources to house, feed, support, and protect its 2.91 million members.[5] The result is that

leaders perennially have more tasks than the workday allows.

Fourth, military leaders drive their organizations toward readiness without an objective measure of what it means to be ready. Standards for specific tasks may be clear, but an understanding of unit readiness often rests on subjective commander assessments. Readiness has no finish line. This feeds the notion that the unit is consistently behind and requires more training, more events, and more work to be prepared for war. High personnel turnover also creates perpetual cohesion and experience gaps that leaders continually seek to overcome.

Fifth, the up-or-out promotion system weighs a leader's success heavily on predetermined performance periods. Promotion potential rests on succeeding in command positions and key staff assignments. Leaders know their time is coming and consequently adopt a mindset that these jobs will demand peak performance and intense personal investment. They prepare for years and arrive in these positions with a flurry of ideas, initiatives, and expectations. This influx of new leaders occurs at every echelon, creating a constant cycle of organizational change that adds uncertainty, bureaucracy, unpredictability, and new tasks into the system.

The reward for good work becomes more work. Leaders who perform well while surging through demanding jobs signal that they are ready for even more difficult positions. Their seeming resilience in the face of challenging duty is precisely what reveals their potential for senior leadership.

Sixth, the military has a culture that espouses hard work and busyness as ends in themselves, sometimes regarded as ritual badges of honor in the process of leader development. When experienced leaders use phrases such as "Iron Major" and "rowing on staff," junior leaders learn to anticipate that certain career periods will be characterized by long hours and rote administrative work. The experience easily becomes a self-fulfilling prophecy when this becomes the expectation.

Successful leaders who have survived these experiences often look for junior leaders who display the same habits and tendencies. They may also demonstrate similarity bias, favoring subordinates whose habits, values, and beliefs resemble their own.[6] Leaders who remain at work late, respond continuously, and make similar sacrifices may consequently appear more committed than those who establish boundaries around work.

Some of these factors stem from organizational culture and some arise from leader influences, yet irrespective of the source, they contribute to a military experience that routinely spills over into home life. The question is not merely how service members can better manage their schedules. It is how work and home roles affect one another and how leaders can manage the resources and boundaries between them.

From Balance to the Work-Home Interface

The common notion of balance in work and life appears increasingly outdated and irrelevant. One early expression of balancing daily activities came from nineteenth-century Welsh manufacturer, philanthropist, and social reformer Robert Owen. Owen championed labor rights and coined the phrase, "Eight hours labour, eight hours recreation, eight hours rest."[7] The concept later fueled the labor rights movement in the United States and carried forward the implication that work, rest, and recreation should exist in equilibrium.

The Industrial Revolution changed the nature of work and nonwork roles by separating the workplace from the home and introducing the concept of regulating work time. This transition created the cultural notion of public and private life, along with temporal and physical boundaries between work and family.[8] During the twentieth century, tension between work and nonwork roles increased as women made up more of the workforce and dual-earner households became more prevalent.[9] Continued globalization and technological access to information have since eroded the line between work and nonwork activity, contributing to the on-demand economy and continuous on-call culture many workers face.[10]

This intersection of work and home has been a longtime focus of occupational health psychologists, who have examined the relationships among the various domains of life, such as work, self, family, and community, and how those interactions affect health, well-being, and productivity. The intersection of work and family roles is generally known as the work-family interface. Given the variety of identities leaders adopt, however, the term work-home interface is more inclusive and appropriate for describing the relationship between



A leader who consistently works late can set a poor example for subordinates and should embrace work habits that facilitate a more reasonable work-life balance. (Photo by Spc. Jacob Nunnenkamp, US Army)

work and everything else.[11] Work-home well-being is a perception of a positive work-home interface that contributes to the well-being of the individual.

The phrase work-life balance appears in scientific literature but often in generic terms that lack measurable qualities.[12] One study defined it as the degree to which an individual can "balance the temporal, emotional, and behavioral demands of work and family responsibilities."[13] This reasoning creates an unrealistic condition for achieving balance. It suggests an equilibrium be pursued while providing no common scale for measuring progress toward it. Herein lies the myth of balance.

Interference, Facilitation, and Personal Resources

Rather than pursuing equilibrium, it is more useful to understand how the domains of life interact and affect a person's ability to fulfill obligations, build healthy relationships, and sustain well-being. Demands in one area can easily spill into others.[14] Work commitments may cause physical, mental, and emotional exhaustion that diminishes a person's ability or desire to engage at home. This is known as work interference with family, or work-to-home interference.[15]

The reverse can also occur. When stress in other areas of life affects performance at work, the interaction is called family interference with work, or home-to-work interference.[16] This negative spillover is evident when thoughts about a sick child occupy an individual's mind during a meeting or when poor sleep affects productivity. Home-to-work interference may appear less common because many workplace cultures discourage personal matters from visibly interfering with work. Even when they are distracted and less productive, workers may be reluctant to disclose stress at home or request assistance.[17]

Research has repeatedly shown that workers are more engaged and experience less stress when supervisors provide flexibility and support in the face of home-to-work interference.[18] These actions, known as family-supportive supervisor behaviors, contribute to positive climates,

greater job satisfaction, lower turnover intentions, and an improved work-home interface.[19]

Interference is not the only possible outcome when life domains intersect. One area of life can also enhance another through facilitation, or positive spillover.[20] Home-to-work facilitation may occur when a supportive spouse inspires a service member to become more engaged at work or when a relaxing vacation helps an executive think more strategically. Work-to-home facilitation occurs when knowledge, experiences, or benefits gained through work enhance life at home. Facilitation and interference can be interactive, concurrent, and multidirectional.[21]

Conservation of resources theory offers a framework for understanding these effects. It states that an individual's resilience depends partly on the possession or absence of psychological and behavioral resources.[22] Psychological resources help people maintain mental health and well-being in adversity, while behavioral resources help them remain effective, focus on relevant goals, and carry out important tasks.[23]

The theory provides two important insights. First, personal resources are finite. Deep investment at work can deplete resources available for investment at home. This principle applies to cognitive, behavioral, emotional, and physical resources and explains why leaders sometimes feel there is nothing left in the tank after a difficult workday. Second, the possession or absence of resources can influence a person's response to stress. Individuals with resilience-building resources such as hardiness and self-efficacy are more likely to navigate stressful conditions successfully when life roles conflict.[24]

Emotion regulation, the process through which individuals influence the emotions they experience and how they express them, can also deplete resources.[25] Military service has deeply ingrained rules, norms, and expectations that require members to modify feelings, suppress reactions, or display emotions appropriate to their roles. This regulation is energetically costly. Research has associated emotion regulation at work with insomnia, fewer positive feelings at home, and dissatisfaction among partners.[26]

Recovery is the antidote to strain.[27] Positive experiences in one domain can offset negative experiences in another. Recovery during the workday, including breaks and rejuvenating distractions, can improve immediate work experience. Vacations, exercise, sport, and relaxation can reduce the cumulative effects of resource depletion.[28] The best recovery activities allow a person to detach psychologically from work, avoid producing new stress, and inspire more positive thoughts about work.[29]

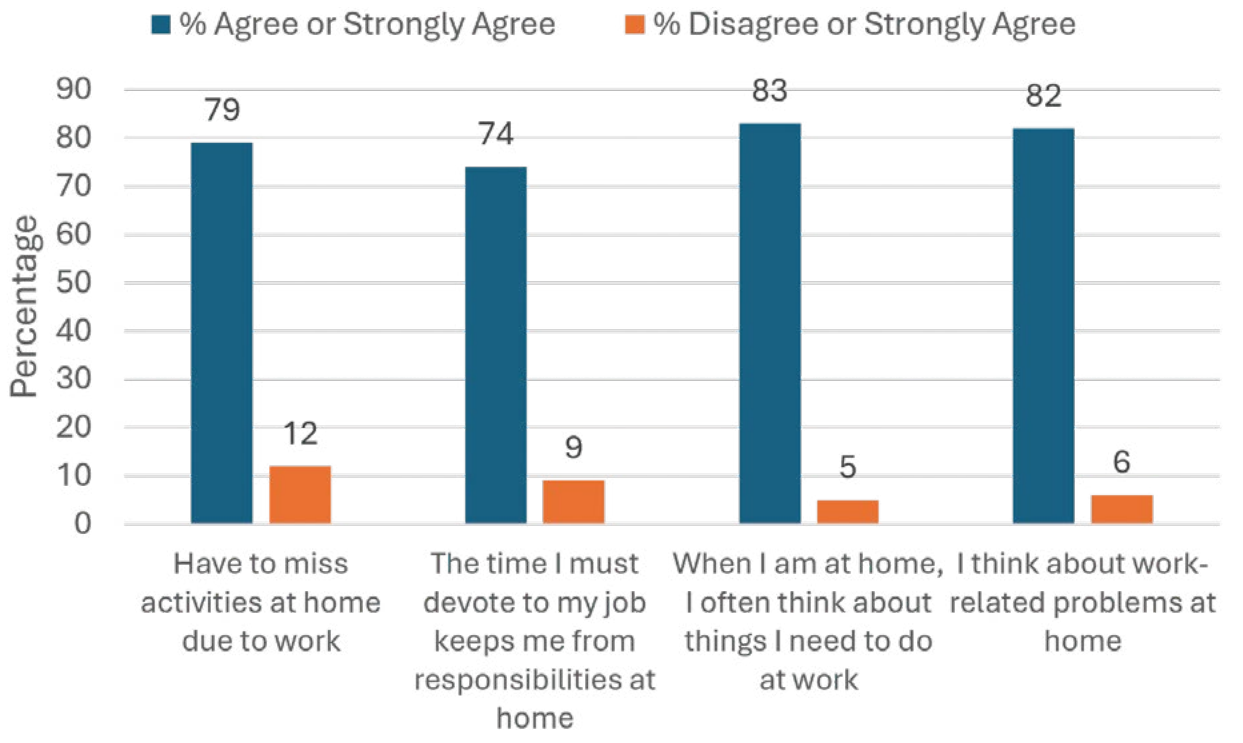
The key to psychological detachment is boundary management, a broad construct reflecting mental models about the permeability among life roles, personal preferences for managing those relationships, and the choices and constraints involved in acting on those preferences.[30] Boundaries can be cognitive, physical, behavioral, emotional, spatial, temporal, or social and may form deliberately or unintentionally through habit. In either case, they indicate a person's preference for segmenting or integrating life roles.[31]

A person who prefers low work-to-home permeability might change clothes before leaving work, ignore work tasks while at home, and avoid developing social relationships with colleagues. This type of distancing can promote the psychological detachment required for recovery.[32] That same person may prefer high home-to-work permeability by displaying family photographs, bringing family members into the workplace, or completing home-related tasks during the day. There is no single ideal arrangement; boundary preferences are individualized.

Senior military leaders also have a responsibility to enable the work-home well-being of subordinates through organizational boundary management. This approach seeks to improve the way workers address the interface between work and home through policies, hierarchies, rewards, and organizational narratives.[33] When workers have more control over the work-home interface, including flexible work hours and freedom to leave for important personal events, they are more likely to remain engaged and demonstrate fewer stress indicators.[34] Organizational boundary management can also challenge the assumption that work must be the central identity in a person's life, an assumption described as the hegemony of work.[35]

Research at the US Army War College

To understand how the quality of the work-home interface affects senior leaders, the research team surveyed resident students from the Academic Year



(Figure by authors)

Figure 2. Senior Leader Reports of Work Interfering with Home

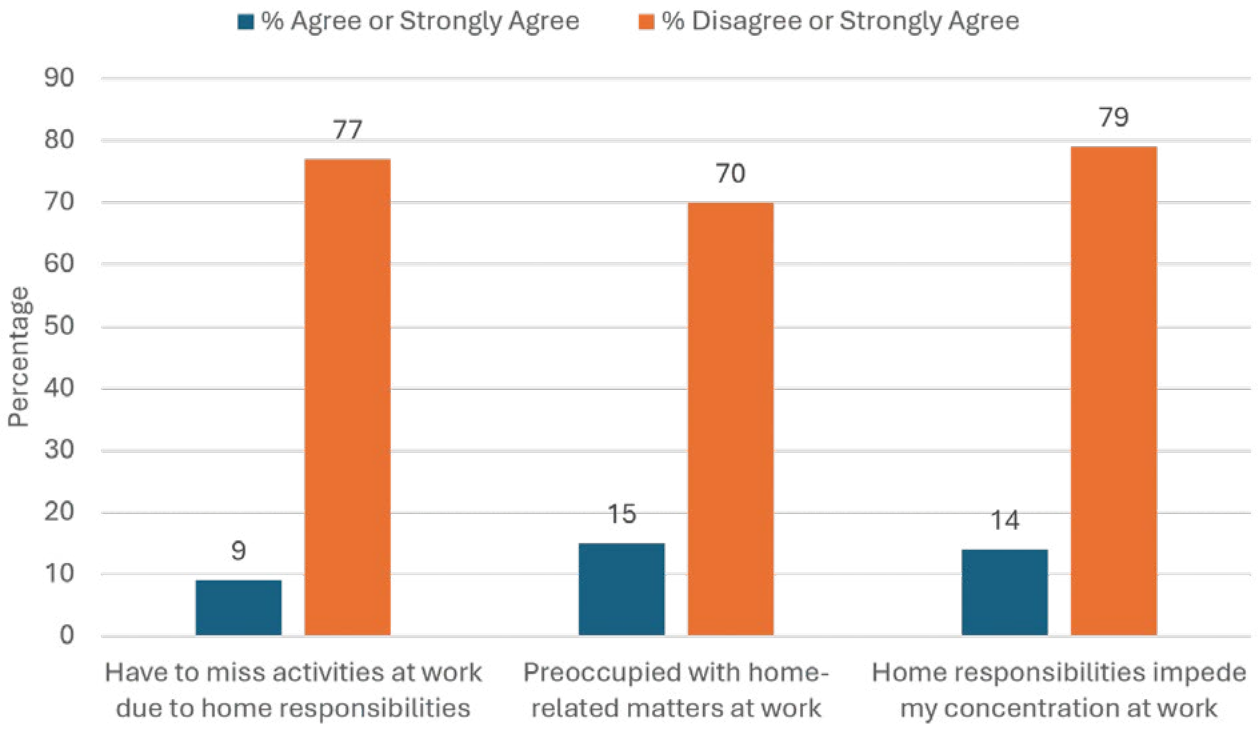
2021 US Army War College cohort. The voluntary survey was distributed to 360 resident students, and 108 officers and civilian leaders participated, producing a response rate of 30 percent.[36] Of the respondents, ninety-four were male and fourteen were female, ninety-eight were married and the remaining were divorced or single, and ninety-seven had children.[37]

The survey contained demographic questions, three sets of questions adapted from previously published work-home research, and original questions tailored for senior military leaders. Participants assessed how work and family roles interfered with or benefited one another. They answered questions concerning personal boundaries, organizational climates, leader messaging, and policies designed to support the work-home interface. Participants also provided 397 written comments, adding personal context to the quantitative findings.[38]

The results revealed five central findings: work-to-home interference is a normative reality for senior military leaders; home life rarely interferes with senior military leader work; work-related fatigue is correlated with conflict at home; home life is an important source of recovery; and most senior military leaders observe and use institutional messaging that supports work-home well-being, but few are able to improve the interface in their own lives.[39]

Work routinely interferes with home life. The survey showed that work had established clear primacy over home life.[40] As seen in figure 2, 79 percent of respondents agreed that work caused them to miss activities at home, while 74 percent agreed that the time devoted to their jobs kept them from responsibilities and activities at home.[41] As one respondent explained,

> My entire career I have avoided scheduling anything before 1900 during the week because I know that I will probably be late to even those activities. I have not been able to consistently attend a weekly activity that starts earlier than 1900 due to emerging work requirements. I have found that it is not worth the headache of explaining to my wife why I won't make it and instead just don't schedule such activities. I always try and schedule events on Friday night since I



(Figure by authors)

Figure 3. Senior Leader Reports of Home Interfering with Work

> have the most success. My entire family's life revolves around my job and the fact that I will be working nights and at least one day on the weekend. When I am home, I will be responding to calls and e-mails.[42]

Another senior military leader commented, "In the last two years I have spent a total of 9 weeks with my family. I've lost count of the things I've missed."[43]

Interference was not just limited to physical absence. Of senior military leaders, 83 percent agreed that they thought about work tasks while at home, while 82 percent reported thinking about work-related problems at home.[44] Only 13 percent were satisfied with how frequently work detracted from activities at home.[45] Seventy percent reported previously implementing techniques or placing boundaries on work to protect home life.[46] Yet these same respondents continued to report high levels of interference. Of the sixty-six leaders who reported establishing boundaries, only seven felt that work allowed them to enjoy home activities adequately.[47]

One leader summarized the challenge: "There is no escaping the iPhone once one is issued. Anyone who tells his/her boss that they are turning off the iPhone at 1800 is committing professional suicide."[48]

Home rarely interferes with work. While work regularly entered the home domain, the reverse rarely occurred. As seen in figure 3, only 9 percent reported missing work activities because of responsibilities at home, 15 percent reported being preoccupied with home matters while working, and 14 percent reported that home responsibilities impeded concentration at work.[49]

The boundary from home to work appeared far less permeable than the boundary from work to home. Senior military leaders remained at work, fulfilled professional responsibilities, and protected their concentration despite competing demands elsewhere.[50] The normative state was to compartmentalize home-related stress to protect work performance.

Work-related fatigue and conflict. The survey also included measures of general work-family balance, perceived work fatigue, and facilitation and conflict between work and family domains. Analysis showed strong correlations between reports of work-to-home conflict and physical, mental, and emotional fatigue.[51]

Senior military leaders who reported higher levels of work-to-home conflict also reported higher levels of all three types of fatigue.[52]

Although family-to-work conflict was also positively correlated with fatigue, the relationships were not as strong.[53] The evidence suggests that work-related fatigue in senior military leaders is more consequential for work-to-family conflict than family-to-work conflict.[54]

Home life supports work performance. Although home matters rarely interfered with work, respondents identified home life as an important source of recovery and facilitation.[55] Of the respondents, 72 percent reported that activities at home helped them arrive at work in a good mood, and 64 percent believed that the energy regained at home improved their concentration and focus at work.[56]

Home life can restore depleted resources, motivate leaders to complete work efficiently, improve mood, and help leaders return to work prepared to engage.[57] Family and home are not merely competing demands; they can make leaders more effective at work.

Messaging does not match organizational reality. The survey also revealed a gap between what leaders say about the work-home interface and what organizational climates permit: 70 percent reported that their leaders emphasized the importance of harmony between work and home; 93 percent reported personally encouraging subordinates to pursue a healthy work-home interface; and 72 percent reported instituting policies, rewards, or guidance intended to promote a healthy interface for subordinates.[58] Yet, participants continued to describe a culture that rewards long hours, discourages boundaries, and equates visible sacrifice with commitment. One participant wrote,

> I don't regret the career, but I also think we need to be honest that a career as an Army officer is inherently unbalanced if you want to do well. I realize that is not the party line, but all the successful ones I have observed do not have that balance. In fact, it is the opposite, we negatively view the officer leaving at 1700 while everyone else is still working. This is not about techniques or efficiency, but this is a culture issue that we have where no one dares to say no.[59]

Respondents also expressed frustration with senior leaders who encouraged others not to repeat the sacrifices they had made. One wrote, "Senior leaders seem to speak from a position of regret regarding work and life. 'I should have spent more time. Don't make the same mistakes I did.' However, they forget that if they did not prioritize work in key positions of their junior years, they would not have been in position to contribute or solve the problem the organization needed."[60] Another wrote, "I would stop the same speech by every General, 'Don't do what I did.' Thanks. We all know that you put your career first and it worked out. Let's find examples of the O5s and O6s who put their family first and retired and were still successful in life."[61]

The institution therefore faces a dilemma. If the military cannot reasonably support work-home well-being because of operational requirements, leaders should align their messaging with that reality. If work-home well-being is essential to health and personal satisfaction, leaders must establish and enforce organizational boundaries that make it possible.

From Research to Command Practice

The research did not remain an academic exercise. After completing the Army War College study, the first author, Col. Andrew Steadman, used its findings to inform his leadership of over four thousand soldiers while in brigade command. He found in practice what the survey results reinforced: Leaders cannot improve the work-home interface through encouragement alone. They must establish organizational boundaries that give subordinates greater predictability and permit them to detach from work.

As commander of the 1st Stryker Brigade Combat Team, 4th Infantry Division, Steadman published "Raider Brigade Guidance for Improving Work-Home Health." The purpose was to create conditions for soldiers, families, and civilians to improve the interaction between work and home. The guidance began with a direct acknowledgement: Military work routinely takes precedence over home life because of the responsibility to defend the Nation, care for people, and improve readiness. It also stated that the prevailing culture gave work default primacy and left many people feeling that home was not protected time.[62]

The guidance addressed a question that McConville's framework did not fully resolve. Most soldiers understand that they may miss work for the birth of a child. The more routine question is whether they have the freedom to leave work at a reasonable hour to spend time with that child and whether their supervisors will avoid intruding on that time for routine matters. Steadman believed that too many soldiers, families, and civilians felt obliged to remain available for a supervisor's call and that supervisors felt an inherent right to email, text, chat, or call subordinates outside normal work hours.[63] To improve the situation, the brigade implemented four changes.

First, leaders were directed to "lead in person" whenever possible and avoid defaulting to digital communication. The guidance emphasized that synchronous exchange, nonverbal cues, and emotional interaction become diluted when leaders attempt to connect primarily through text, chat, or email.[64] Leadership is most effective when it is a personal experience.

Second, the brigade established a routine communications window from 0500 to 1900 on workdays. Routine communication included normal duty matters and non-time-sensitive coordination. Outside that window, leaders could contact subordinates for emergencies, time-sensitive requirements, serious incidents, welfare checks, personal support, and short-notice changes to training or battle rhythm.[65]

The communications window was not a directive for a fourteen-hour workday and did not prevent leaders from working on their own time. Leaders who chose to check email or complete routine work during off-duty hours were expected to delay communication with their teams until the next routine communications window. The intended end state was for subordinates to have the freedom to physically and psychologically disengage from work so they could invest in home-related activities.[66]

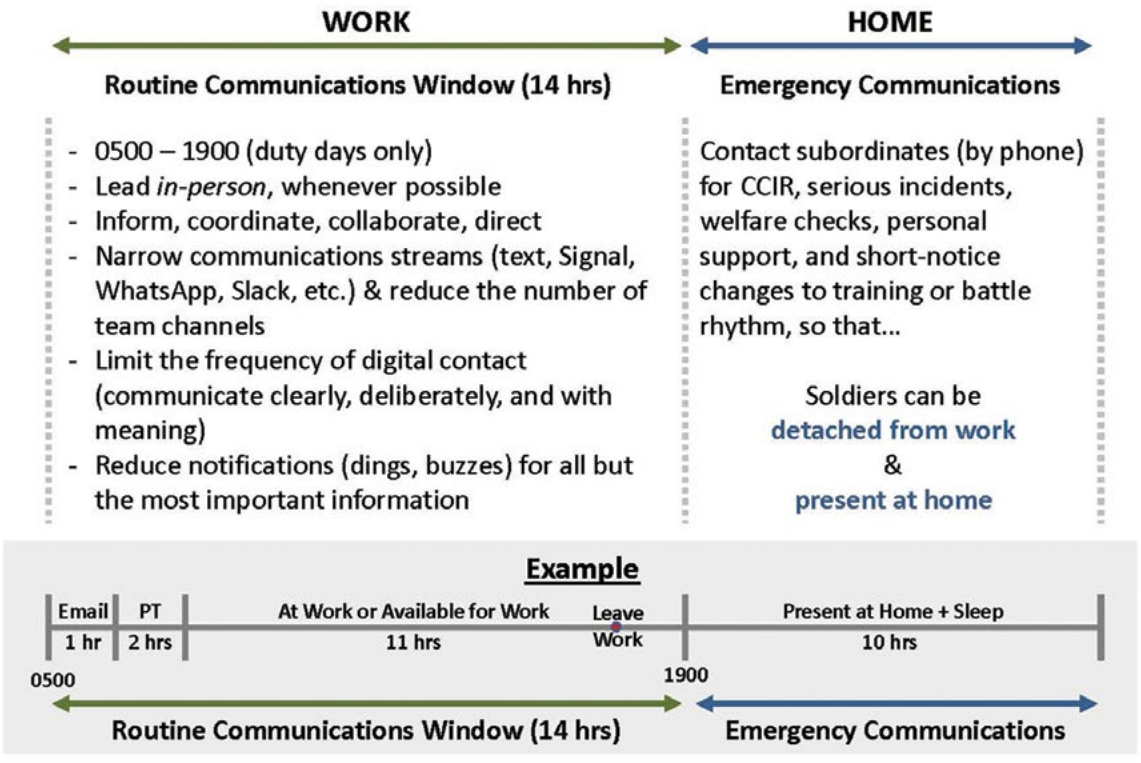
Third, leaders were directed to reduce the number of communication streams. The guidance noted that soldiers and leaders were often required to monitor numerous group threads across email, text, Teams, WhatsApp, Signal, Slack, GroupMe, and other applications. Leaders were expected to narrow those streams, reduce unnecessary group channels, separate work communication from unofficial conversation, and use a clear communication plan for urgent matters.[67]

Fourth, soldiers and leaders were directed to reduce nonessential notifications. The guidance emphasized that much military work, particularly leading, requires focus and presence. Audible alerts and app-generated notifications fragment attention and encourage reactive communication. Leaders were expected to prioritize notifications according to their communication plans and engage in email and chat during planned periods rather than responding continuously throughout the day.[68] The one-page "Improving Work-Home Health" graphic summarized these expectations in figure 4. It depicted the routine communications window, the limited purposes for emergency communication outside that window, and the desired outcome that soldiers could be detached from work and present at home. It also reinforced the expectations to lead in person, narrow communication streams, limit the frequency of digital contact, and reduce notifications.[69]

The guidance had a positive impact because it gave the brigade a common framework for discussing communication, subordinate time, and leader restraint. It made expectations more explicit and shifted the burden from subordinates asking for protection to leaders creating the conditions for it. It also preserved the brigade's ability to respond to emergencies and changing mission requirements.

The experience reinforced the central finding of the research: change begins with leaders. Subordinates are unlikely to ask supervisors to release them at a reasonable time or stop contacting them late at night. Leaders must therefore set an example and establish work conditions that facilitate a more reasonable work-home interaction. As the guidance stated, "If we are to truly care for our people, we must recognize and respect who our people are outside of work."[70]

The brigade approach was not presented as a perfect or universal model. It was an effort to translate research into command practice and to create greater discipline in how leaders communicated. Its ultimate intent was straightforward: to become more efficient and disciplined at work so soldiers, leaders, families, and civilians could be more present and engaged at home.[71]



(Figure by Andrew Steadman)

Figure 4. Improving Work-Home Health

Recommendations for Individual Change

Senior military leaders can improve the work-home interface through three actions. They must assess how their life roles interact, determine how they want those roles to relate to one another, and establish boundaries to reduce interference and fatigue.

Assess the work-home interface. The first step is awareness. Senior military leaders should examine how work and home roles interact and identify sources of interference, conflict, strain, reduced engagement, and facilitation. They should consider not only the time devoted to each role but also the cognitive, emotional, behavioral, and physical resources consumed by those roles.

This process should include conversations with spouses, children, close friends, and trusted colleagues. Leaders may not fully recognize the effects of their work habits on others. Those closest to them can provide important perspectives regarding distraction, emotional availability, fatigue, missed obligations, and the quality of engagement at home.

Leaders should also examine the positive interactions among their roles. Home activities, relationships, exercise, recreation, and periods of recovery may improve performance at work. Work experiences and benefits may likewise strengthen life outside work. The objective is not to calculate a perfect distribution of time. It is to understand how each role consumes, restores, or transfers personal resources.

Determine the desired relationship among life roles. Senior leaders should determine the extent to which they prefer to segment or integrate work and home. Those seeking greater segmentation might limit work communications at home, change out of uniform before leaving work, avoid bringing work equipment into personal spaces, and establish periods when work tasks will not be performed.

Those preferring greater integration might contact family members during the workday, display family photographs, bring family members into appropriate workplace activities, discuss personal experiences with colleagues, or participate in unit social events. There is no single arrangement that applies to every

leader. The important point is to choose an approach deliberately and, where appropriate, with input from those affected.

Establish boundaries that reduce negative spillover. Leaders should establish cognitive, behavioral, temporal, physical, and technological boundaries that reduce work-to-home interference. Information and communication technologies deserve particular attention. Mobile devices and email enable work to follow leaders continuously, creating a sense that every message requires immediate attention.

Leaders can turn off nonessential notifications, batch email into defined periods, establish expected response windows, and restrict activities that occur over email. Meeting coordination, brainstorming, and casual conversations do not always require email exchanges. Senior leaders can also state clearly that no response is required until the following business day or establish a daily period after which they will not review email.

One survey participant described an organizational approach:

> In one job, I outlawed downstream tasks/due outs after 1600 and nothing except emergencies, my decision point, were allowed via phone call after 1600 or requiring a response before 1000 the following morning. I also outlawed working past 1800. Any presence after that time required my approval with the opportunity to work at 0300 the next morning instead, when kids and spouses wouldn't care you had to work. Not a surprise that work efficiency went up and very few people truly needed to work late.[72]

Senior military leaders can also reduce fatigue through improved workday habits. Routine breaks and transition periods between events can ease an exhaustive pace. Short walks can be restorative while providing opportunities to reflect, interact with colleagues, and visit unit areas. Planned reading time can provide a sense of personal development and self-care. Short naps can restore energy and improve efficiency.

Sleep requires particular attention. Of surveyed senior military leaders, 81 percent reported compromising sleep quantity or quality during the preceding five years because of work demands.[73] Leaders should avoid treating sleep as a source of additional work time.

Recommendations for Institutional Change

Individual techniques will have limited effect if organizational climates do not permit leaders to use them. Senior military leaders influence policies, schedules, rewards, expectations, and organizational narratives. They can create conditions that allow subordinates to establish boundaries without fearing that doing so will harm their reputations or careers.

Leaders should establish limits on routine working hours, prevent unnecessary after-hours taskings, and define what constitutes an emergency. They should establish reasonable expectations for email responses and avoid sending downstream requirements late in the day. Units should permit personnel to leave for important family events and provide flexibility over where and when appropriate work is performed.

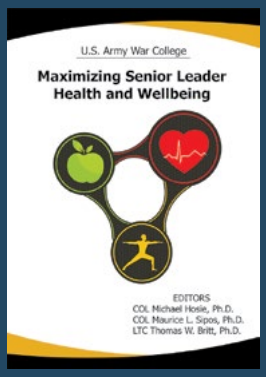
They should also distinguish between genuine operational demands and habitual busyness. War, deployments, emergencies, and critical readiness requirements will sometimes require substantial sacrifice. Routine administrative work should not automatically receive the same priority. McConville's framework is useful because it differentiates among levels of importance in both work and personal life.[74]

Organizational boundaries remove doubt before conflicts arise. When a policy directs personnel to leave at a specified time unless authorized to remain, individuals do not have to assess whether departure will be interpreted as a lack of commitment. When leaders state that newly arriving personnel will not begin normal duties until their households are established, the organization removes tension between settling a family and demonstrating professional dedication.

Senior leaders must also model the behavior they expect from others. They should not encourage subordinates to protect their families while personally celebrating fifteen-hour days, sending routine messages late at night, or remaining continuously accessible. Their behavior communicates organizational priorities more powerfully than speeches or written policies.

A Call to Action

For senior military leaders, work holds hegemonic primacy over other areas of life. Empirical data from the US Army War College resident population supports this reality. Occupational psychology research



Military Review Recommends *Maximizing Senior Leader Health and Wellbeing*

To read more on the topic of senior leader health and well-being, see Michael Hosie, Maurice L. Sipos, and Thomas W. Britt's book *Maximizing Senior Leader Health and Wellbeing* (Strategic Studies Institute, US Army War College, 2023), available online at https://ssl.armywarcollege.edu/DCLM/Publications/readings/Maximizing%20Senior%20Leader%20Health%20and%20Wellbeing%20(2023).pdf.

shows that this interference can produce strain, conflict, and lack of engagement, ultimately affecting readiness.

The popular notion of work-life balance fuels the belief that well-being arises from spending equal time among life domains. Faced with overwhelming workloads, senior leaders often lament that work demands too much time and forces them to miss activities at home. The critical variable, however, is not merely time. It is energy.

Life roles can both stimulate and consume cognitive, emotional, behavioral, and physical resources. They can facilitate one another by generating energy, but interference occurs when deep investment in one role leaves fewer personal resources to invest elsewhere. Work fatigue can make senior military leaders less capable of meeting obligations at home.

To improve the work-home interface, senior military leaders must first understand how their roles interact and how work affects themselves, their friends, and their loved ones. They must then determine the degree of segmentation or integration they desire. Finally, they must establish boundaries that reduce fatigue and mitigate negative spillover.

Senior military leaders also have a responsibility to enable those they lead to take the same actions. They must align institutional messaging with organizational behavior and give subordinates the authority and freedom to protect roles outside work.

This requires more than encouraging people to find balance. It requires leaders to establish policies, schedules, expectations, rewards, and personal examples that allow people to recover, fulfill obligations outside work, and remain engaged in the important roles of their lives.

It is time for senior military leaders to imagine a different culture, one in which leaders are attuned to demands across all life roles and establish appropriate boundaries to meet them. Their example can facilitate work-home well-being for subordinates and allow others to do the same.

The individual and organizational freedom to support life roles outside work does not currently exist for many senior military leaders. With intention, leaders at every echelon can create it. ■

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Rows of combat boots line Ford Island during the 11 September 2025 Boot Memorial Hero and Remembrance Run at Pearl Harbor, Hawaii. The boots were adorned with photos, flags, leis, and personal messages to honor the lives and sacrifices of fallen service members. Every boot in the display bore the name of a fallen hero, from combat casualties to those lost to training accidents, illness, or suicide, reminding visitors that behind each pair of boots is a face, a family, and a legacy worth remembering. (Photo by Staff Sgt. Andre D. Taylor, US Army)

Thinking Differently About Army Suicide

A Social Diagnosis

Maj. Robert G. Rose, US Army

In 2001, the annual suicide rate per 100,000 active-duty soldiers in the US Army was 8.7.[1] By 2023, the rate had quadrupled to 34.8.[2] Wars have started and ended. Behavioral health services have expanded. Annual suicide awareness training has occurred. Units have surveyed "high-risk" soldiers.

The Army's "People First" philosophy came and went.[3] Nevertheless, US Army suicide rates have continued to climb. It is time for the Army to question why its suicide rate has outpaced that of both American society and similar armies.

The work of Émile Durkheim may provide insight as to why the Army is an outlier. Often considered the father of sociology, he studied why suicide rates soared across Europe during the nineteenth century. Instead of focusing on individual factors, he took a statistical approach to compare populations. He provided a different way to think about suicide: not as an individual event but as a societal one. He concluded that the social alienation of the Industrial Revolution increased suicide rates, a finding with implications for today's Army.

Our Fraying Bonds

As a commander, I faced my most emotionally taxing moment when I gave a speech to memorialize a soldier who died by suicide. As I looked into the eyes of the soldier's family, sitting in the front rows of the post chapel, what could I say? How could I provide solace? How could I provide meaning?

Staring out at the family and everyone filling the chapel, I could see how many people were connected to this soldier, so I turned to John Donne's poem "No Man Is an Island" as a metaphor to illustrate how many lives he affected in his time in our world:

> No man is an island,
> Entire of itself;
> Every man is a piece of the continent,
> A part of the main …
>
> Any man's death diminishes me,
> Because I am involved in mankind.
> And therefore never send to know for whom
> the bell tolls;
> It tolls for thee.[4]

I tried to describe how the soldier had impacted our lives, but I did not truly know him. Although he had served in the Army for nearly two decades, he had been in our unit for only a few months. I knew he was an enthusiastic and talented member of our football team, but I had not deployed to combat with him. I had not joined in his triumphs and losses. Most of us were merely his work acquaintances. We were not comrades. We had not forged a band of brothers. I will never know if he felt obligated to live for us. He may have felt like he was on an island.

Throughout modern society, social bonds have been fraying. In 2000, Robert Putnam identified that the United States was experiencing a collapse in civic engagement as people transitioned from group to individual recreational activities. American participation in civic clubs, fraternal organizations, veterans' groups, and even bowling leagues (hence, his book's title *Bowling Alone*) all declined between the 1950s and 1990s.[5] Since 2000, the decline in social integration has accelerated as the internet spread, social media emerged, and the isolated watching of reels on TikTok or YouTube dominated entertainment habits.

In 2023, US Surgeon General Vivek H. Murthy published an advisory titled *Our Epidemic of Loneliness and Isolation*. He worried about the number of Americans who expressed, "If I disappear tomorrow, no one will even notice."[6] The advisory highlighted a study reporting that "social isolation is arguably the strongest and most reliable predictor of suicidal ideation."[7] From Putnam's book in 2000 to the surgeon general's advisory in 2023, annual suicide rates per 100,000 Americans increased from 10.4 to 14.1.[8]

During that same period, the Army's operational tempo and frequency of personnel moves exacerbated the wider epidemic of loneliness, and correspondingly, the Army's increase in suicide outpaced society's. While we cannot prove causation for suicides, increasing social isolation provides a useful correlation to understand the problem.

War Is Not the Cause

Often other factors such as trauma from recent wars are used to explain the rise in Army suicides. However, the suicide rate actually decreased during the Global War on Terrorism's (GWOT) initial years. After a few years, the rate began increasing, but the rise persisted long after combat operations ceased. Tellingly, soldiers who were in combat have been no more likely to kill themselves than those who were never under fire.[9] Furthermore, other armies that participated in Iraq and Afghanistan did not experience the same rise in suicide rates. During the first decade of the GWOT, the British Army witnessed a consistent decline in

suicide rates. It reached its lowest recorded rate of suicides in 2008–2010 of 5 per 100,000 soldiers.[10]

During previous wars, Army suicide rates generally declined because war provides a sense of purpose and strengthens bonds between soldiers. Until the Vietnam War, the US Army experienced much lower suicide rates when it went to war.[11] Across the entire Civil War, the Army recorded 278 suicides, averaging 69.5 per year of war.[12] In 2023, with fewer soldiers under arms, most of whom had never experienced combat, the Army's components totaled 279 suicides.[13]

An Outlier Among Armies

In understanding the US Army's suicide problem, we should recognize that it is an outlier compared to similar armies. Its suicide rate is at least twice that of most other NATO armies. Accounting for demographics, the US Army has a higher suicide rate than the American population, while other armies have lower suicide rates than their countrymen.[14] While in 2023, the US Army had a suicide rate of 34.8 per 100,000, in recent years, the British Army's suicide rate has been 12 per 100,000, the Canadian Armed Forces' rate was 17.6, the German military's rate was 9.1, and the French Army's rate was 20.3.[15]

The fact that the French Army has the highest rate of suicides of major European armies may shed light on the US Army's high suicide rate. Like the US Army, the French Army employs a centralized personnel system involving frequent moves to new duty stations. For a century, the French Army has prioritized centralized efficiency over the regimental systems employed by many European armies. This approach fractured soldiers' solidarity and led to worse battlefield outcomes. France's poor unit cohesion compared to Germany's was a significant contributor to its defeat in 1940.[16] Solidarity matters for fighting prowess, but it also provides resistance against suicide by turning individual islands of men into a cohesive whole.

Understanding Suicide as a Societal Problem

In the 1890s, Durkheim argued that societal factors best explained suicide rates. He found little correlation between suicide rates and individual factors like poverty, stress, alcoholism, or mental health issues.[17] Today, the Army tends to focus on such individual factors to explain suicide. Researchers have worried that the Army has not conducted a concerted "effort to understand suicide in the larger historical context of society and culture. Rather than understanding military suicides as being primarily driven by combat trauma or PTSD, to be effective, prevention efforts should strive to understand military suicides through the lens of larger societal factors."[18] By focusing on individual instead of societal or institutional factors, the Army misses the potential drivers of suicide rates.

Durkheim wrote that during the nineteenth century, suicide increased as people shed their traditional attachments. Religious structures no longer dictated lives. The Industrial Revolution alienated its laborers. Guilds dissolved. Villagers abandoned their ancestral homes and packed into industrial towns. Workers had few bonds in their new environments. Durkheim highlighted three factors that had traditionally provided those bonds: religion, family structure, and political movements.[19]

With religion, he found a higher rate of suicide among Protestants, a lower rate among Catholics, and the lowest among Jews.[20] He clarified that religious beliefs were not driving these differences: Jews had less of a prohibition against suicide than Christians. Rather, he attributed the reduced proclivity for suicide to the social aspects of religion.[21] Protestantism arose from an individualistic revolt against hierarchy, while other religions maintained more structured and socially integrating practices, such as Catholic feast days or the Jewish Seder. Today, researchers continue to find that religious service attendance reduces suicide. One study found that among women, those who attended church weekly suffered five times fewer suicides than those who never attended. Like Durkheim, the researchers found that practicing Catholics experienced stronger resilience against suicide than Protestants.[22]

Additionally, Durkheim noted that Jews, as well as Black Americans, had relatively low rates of suicide, even though they faced persecution. Their tighter-knit communities yielded resilience in the face of oppression. Currently, Black Americans suffer suicide at half the rate of White Americans.[23] The resilience of marginalized groups indicates that focusing on "stressors," as the Army tends to do, provides a poor indicator of suicide proclivity.

Furthermore, Durkheim found that family structure influenced suicide rates. Compared to single



Troopers assigned to 1st Cavalry Division Artillery attend an immersive, scenario-based training provided by the Overwatch Project, a peer-intervention education program focused on firearm-related suicide prevention at Fort Hood, Texas, on 7 January 2026. The training utilized live-action scenarios to equip troopers with the knowledge and skills needed to recognize risk factors, intervene effectively, and help prevent suicide. (Photo by Sgt. Lyca Williams, US Army)

men, married men had a third fewer suicides, and married men with children had two-thirds fewer.[24] He explained that denser family relations (extended relatives living together) produced lower suicide rates. When families were dispersed, they had less influence on an individual's conscience and so provided less resilience.[25]

Finally, he found that political movements reduced suicide. During revolutions, political crises, and wars, suicides declined. The more the population was mobilized toward a cause, the less they fell to suicide.[26] Such periods of crisis might increase stress, but they amplified social bonds and provided purpose, which resulted in stronger resilience to suicide.

Work, religion, and political movements integrate individuals into society, enabling them to build bonds of obligation with their fellow humans. Such bonds make individuals feel that their life is not theirs to willfully terminate. Being a part of a society arouses a sense of sympathy for others. We feel a part of a collective whole—a part of the language, art, morals, environment, and civilization passed down to us and continued beyond our short mortal time to future generations. When humans are socially integrated, we find meaning to live. As Viktor Frankl, a psychologist and concentration camp survivor, wrote, referencing Friedrich Nietzsche's words, "He who has a why to live for can bear almost any how."[27] We resolutely hold onto life as a duty to those we love.

Alternatively, when we lose that attachment and feel that we are individuals who are masters of our destinies, we may feel that it is our privilege to terminate our lives. Unlike our ancestors, who lived in close-knit communities with a deep well of social duties, modern man is liberated from those obligations. However, humans are social animals. We evolved language to allow for complex cooperation to survive and thrive. When we are not a part of a group, we are likely to feel lonely because we sense that an essential element of our humanity is lacking.

Psychologically, the detached individual can lose a sense of belonging to an eternal whole. They can fall into a sense of melancholic meaninglessness. When they face life's suffering, they can feel no obligation to cling to it. Rather, they simply let go.

So how can society reconnect individuals? Durkheim argued that relying on education and prohibitions against suicide could never resist wider social currents. Suicide could only be reformed if society were reformed.[28] Since humanity cannot turn back the clock of progress, he argued that the best means to reduce suicide was to leverage the new social organizations emerging among industrial workers. With work consuming so much of modern man's life, increasing cohesion among occupational groups could rebuild a sense of social solidarity.

While Durkheim was describing civilian unions and workers' clubs, he could have similarly discussed armies' regimental systems. Professional armies have always pulled soldiers out of their childhood homes, but traditionally, they have offered a new family in unit solidarity—in a band of brothers.

The Problem of Military Moves

When I arrived at my first unit, they had just returned from a bloody deployment to Ramadi, Iraq. The battalion had lost dozens of soldiers. Their comrades hung their pictures on a memorial wall. And then, the Army broke apart the surviving band of brothers. Soldiers who had grown close in combat were dispersed. The battalion was left deserted. Hardly anyone remained who had actual memories of the soldiers pictured on the wall. And then, one of the remaining veterans shot himself. I did not know who he was, but I attended his service. The chapel teemed with new arrivals who, like me, had no idea who he was or what he had gone through.

If he had still been close to the compatriots he had deployed with, he may have felt an obligation to continue living for them, but he had lost those connections in the Army's constant personnel moves. The relentless churn of soldiers strips off their protective layers. They become individual particles or atomized family units bouncing between installations. Moving might be exciting and may help a career, but dislocation shatters social bonds. As far back as 1865, researchers found that most of the suicides in Paris were recent migrants from the countryside. Writing in 1991, Joseph Rothberg, who researched military suicide over two decades, warned of higher suicide rates in the US Army due to the frequency of moves.[29] At the time, he described the average soldier moving once every four years. Today, few soldiers have the luxury of staying that long.

To understand the value of stability, look toward the Army Reserve and National Guard. Their soldiers consistently suffer around a third less rate of suicides than active-duty soldiers: in 2023, per 100,000 soldiers, active duty had 34.8 suicides compared to 24.9 for the Reserve and 23.7 for the National Guard.[30] Reservists likely have less access to the Army's suicide prevention programs, but they stay in the same location and with the same unit for longer. By not moving, they have more opportunities to integrate into their local community, their family, and their unit.[31]

The History of the Problem

Before the 1940s, soldiers often served entire careers with the same unit. It was only when the Army applied scientific management models to its personnel system that soldiers began to feel like interchangeable parts. During World War II, the Army shifted to an individual training and replacement system that eased centralized administration but came at a high cost to unit cohesion.[32] From the inception of this system, it degraded the psychological experience of soldiers.

In World War II, American replacements began deploying as individuals rather than as whole regiments. Soldiers complained of being "herded like sheep" or "handled like so many sticks of wood."[33] Conducting hundreds of interviews as part of an investigation into the European Theater's personnel policies, Col. Howard E. Kessinger found that soldiers "wanted most of all to be identified with a unit."[34] They had a psychological need for solidarity. Without this need being met, they suffered distress. Employing the slang for replacement depots, medical officers wrote of "repple depple syndrome" as a new kind of mental disease.[35] From June to November 1944, units fighting in France experienced a psychiatric casualty rate of 26 percent, which was ten times higher than the German units they faced.[36] Unlike the US Army, the German Army emphasized unit cohesion. In late 1944, more soldiers were being discharged from the Army for psychological reasons than were coming in as replacements, which led Gen.

George C. Marshall to set up an investigation committee into the Army's psychological issues.[37]

The committee's researchers focused on understanding how soldiers developed psychological resiliency. They found that "it appears that the most significant factor is the soldier's position in the constellation of his social group, the combat team."[38] Soldiers could bear extreme strains when members of a cohesive group. When soldiers no longer felt they belonged to a group, their resiliency collapsed. The researchers wrote, "The soldier [who] lost his group relationship … forfeited all the strengths and comforts with which it had sustained him. As a member of the team, he would have been able to take it; alone, he was overwhelmed and became disorganized … The disruption of group unity is, in the main, a primary causal factor, not a secondary effect, of personality disorganization."[39]

Forewarning the contemporary blind spot for societal factors, the researchers worried that Americans "have considerable difficulty in grasping the significance of the group as the core of personality organization for the soldier in his fighting function. Our modes of psychiatric thought and practice and our general American ideology have been almost exclusively developed on the consideration of the individual."[40] They concluded that soldiers should not be sent as individual replacements but rather as preformed units, allowing soldiers time to develop cohesive bonds.[41]

The Army did not heed the committee's recommendations. After World War II, it doubled down on the scientific management of its personnel. Congress passed the 1947 Officer Personnel Act, which established a predictable path for an Army officer's career with regular moves between positions.[42] Enlisted careers soon followed a similar trail. Before the 1940s, a recruit did not so much join the Army as a regimental family.[43] Soldiers had defined themselves by their loyalty to a unit; now they defined themself by their occupational specialty.[44] After the 1940s, soldiers reported feeling a lower sense of duty to their units. With the Army's global commitments, soldiers and their families began to lead lives on the edge of society, regularly moving between overseas locations and remote posts.

When the Army entered the Korean War, it faced a crisis of morale, cohesion, and combat effectiveness.[45] Commentators blamed the individual replacement system for the "general decline of discipline and the production of qualified leaders."[46] The Army commissioned Project Doughboy to analyze infantry performance. Led by S. L. A. Marshall, it noted that individual replacements felt isolated and alone during combat. He recommended increasing their sense of solidarity by employing a team replacement system: "If the four-man teams went through the pipeline as one man now moves, they could be fitted into a front-line organization, ready to fight, without the feeling of being personally lost."[47] He wrote to Gen. Matthew Ridgway, "Rotation, never do it again. This is a certain way to destroy an army."[48]

Come peacetime, the personnel system caused continuous issues. In 1954, the Army warned senior leaders of a retention crisis, and units had to make do with just 50 percent manning of experienced noncommissioned officers.[49] Polls consistently confirmed Lt. Gen. John Dahlquist's conclusion that the Army's "increasing inability to attract and hold high-caliber personnel" stemmed from "the instability of assignments and too frequent overseas tours."[50] Maj. Gen. William Yarborough warned that the frequent rotations and rigid career track created alienated, overworked, and anonymous soldiers. He described how in the old Army, if an officer resigned from his career, it was a community concern. In the contemporary Army, no one cared.[51]

The Army signaled attempts to reform its personnel system, such as Gyroscope in the 1950s and COHORT (Cohesion Operational Readiness and Training) in the 1980s, but senior leaders resisted change.[52] Arguing against reform, Gen. William C. Westmoreland maintained that the centralized system of individual assignments allowed for the orderly progression of personnel through schools, promotions, and assignments. He supported the individual replacement system in Vietnam, regardless of the ensuing personnel and morale problems.[53] The Army lost the war, and for the first time, suicide rates surged during war, which prompted the Army to begin centralized efforts to reduce suicide.[54]

Suicides would surge again during the GWOT as the Army increased the pace of personnel moves. To meet manning requirements for the Army Force Generation model to provide ready forces, the Army assigned soldiers to fill units to deploy. The Army leaned further into such scientific management in the 2010s under the heading of "Talent Management."[55]



The US Army Garrison-Kwajalein Atoll community stepped out at sunrise on 27 September 2025 for a 6.4-mile suicide awareness ruck march. The group stopped at various locations along the way to hear personal stories and guidance from the garrison command team about seeing the signs of suicidal thoughts. Before the march, the Kwajalein behavioral health specialist discussed some of the tools for helping someone experiencing thoughts of suicide. (Photo courtesy of US Army Garrison-Kwajalein Atoll)

Whereas before, officer careers had been tightly managed, but enlisted had more flexibility, with some staying at the same installation for a decade, now enlisted were assigned Year-Month Available to move to ensure Human Resources Command could efficiently move them between assignments. At my first unit, a first sergeant had spent his entire career in the battalion. He provided the unit's soul. Such a soul would not exist now.

Other Approaches Are Possible

I had the opportunity to experience a radically different personnel system when I embedded with the Bulgarian Army as an advisor. Their enlisted soldiers only moved if they requested it. Their officers served much of their careers in the same unit. I remember a major being annoyed that, after a decade in the same brigade, he would have to finally leave his unit for a job with Land Forces Command in Sofia, seventy miles away. He was worried about the stress it would put on his family. In the US Army, such a move would be seen as a minor inconvenience, and he would have already moved at least five times during his career. His concerns illuminated the abnormality of the US Army personnel system. Soldiers and families are moved like automatons with no grounding in a sense of place.

Leaders in the Bulgarian Army told me that suicide is incredibly rare, so rare that it is difficult to find statistics on, and they do not feel the need to conduct suicide awareness training. It is infrequent not only because they rarely move, but also because they are integrated into their society. They regularly participated in ceremonies in towns across Bulgaria that brought together the state, religion, history, military, and community. Even as an outsider with a poor grasp of their language, I felt integrated into their society.

Their installations are situated in the middle of cities. No one lives on post; they live in town and walk to work. US Army installations are generally detached from the local community, with separate housing and services. In the early 1900s, a British Army observer praised the Bulgarian Army for integrating installations into their towns. He wrote, "The Bulgarian does not believe in the English plan of building barracks in the most horribly hideous out-of-the-way places that it is possible to find."[56]

The Decline of the Army Social Life

The US Army combines regularly displacing soldiers with installations that are disconnected from local communities, and, within those isolated units, social life has decayed. Once, posts maintained officer and NCO clubs that were frequented after work to create bonds. Now, they are mostly closed. When I was at Fort Campbell, Kentucky, I could not find anywhere on post for my unit to conduct informal after-work socializing. It seemed like a radical concept to have a drink together after work. When I first joined the Army, units still tried to conduct monthly stable calls.[57] Now, such events are the first thing to be removed from training calendars.

A unit reveals much about its level of cohesion by how quickly its members disperse at the end of the presentations at its social events. If soldiers do not

want to socialize with each other more than strictly mandated, the unit has a cohesion problem. Often, soldiers cynically title unit events "mandatory fun days," which highlights the sense that social bonds are a chore. Soldiers who feel forced to be together likely do not feel a deep commitment to live for each other.

When I spent time with a British battalion, I witnessed a different mentality. Officers ate lunch together in the mess, and they regularly had an after-work drink together. The value of the solidarity found in the British regimental system appears in their low suicide rates. Their army also has a lower suicide rate than British society, with the notable exception of soldiers under twenty. Having been uprooted from their homes, young soldiers have not yet had time to integrate into their new regimental family.[58]

High Operational Tempos Shatter Social Bonds

Of course, many soldiers still find a way to build bonds, but during the last two decades, the pace of the Army's operational tempo (OPTEMPO) has made it more difficult to create those bonds. The GWOT increased OPTEMPO, but even as the GWOT ended, the Army's OPTEMPO continued apace. While combat deployments decreased, the Army came to rely on rotational deployments after deactivating many forward-deployed units.

These deployments uproot soldiers from their families and communities. They limit the already scant opportunities soldiers have to integrate into society. When combined with training exercises, many soldiers spend less time in garrison than in the field or deployed. While deployed, they have little opportunity to integrate into the local society. Rotational troops often live in isolated camps and face restrictions on how freely they can interact with the local community. They end up spending much of their time alone in barracks online.

The social isolation of such deployments undermines psychological resiliency; the culprit is not just the fact that a soldier is abroad. The Army unit with the lowest suicide rate was the 2nd Cavalry Regiment, which is stationed in Germany and does not have to contend with long-term rotational deployments.[59]

Displaying the psychological damage of the Army's OPTEMPO, armor brigades, which have been in high demand for deployments, have experienced a significantly higher suicide rate than other units. While in garrison, they spend weeks in the field meeting requirements to deploy. A depressing example can be found in 1st Battalion, 66th Armor Regiment, which between 2019 and 2021, had the highest suicide rate in the Army. Over fourteen months, it suffered six suicides.[60] It had deployments every other year to Kuwait, Korea, and Europe. One disenchanted captain questioned, "What is the mission that we're killing ourselves for? The [armor brigades] weren't at war. They were training in Europe, and they were training in Korea, and they were training in Kuwait, and they were babysitting [bases] in Iraq."[61] No amount of suicide reduction programs can overcome the disruptive effects of such an OPTEMPO.

Misdiagnosing the Problem

In 2022, Secretary of Defense Lloyd Austin made suicide prevention a top priority.[62] He appointed the Department of Defense Suicide Prevention and Response Independent Review Committee, which published 127 recommendations to combat suicides.[63] Meanwhile, at Fort Bliss, Texas, a panel made sixty-one recommendations to address its suicide crisis.[64] While such efforts should be applauded, the avalanche of recommendations risks burying the problem and offering a façade of progress. Leaders can end up focusing on individual stressors and examining subordinate leader interventions while not addressing the wider social environment.

No survey, no high-risk screening, and no counseling can truly assess if a soldier feels disconnected from society. In 2024, at Fort Bliss, zero percent of suicides were previously identified as "high-risk." The installation had initiated screening of its new arrivals in 2022, but by 2024, of the eight suicides among new arrivals, only one had indicated suicidal ideation.[65]

Last year, I attended another ceremony honoring a senior captain who died by suicide in my previous brigade. No one considered him high-risk. He had a fiancée, a supportive family, and was doing well in his career. He killed himself a couple of weeks after we received the unexpected news that our unit would be inactivated, and we would have to scramble for new assignments. I cannot conclude that the news of the inactivation led to his suicide, but the abruptness of the inactivation announcement was alienating.

A Way Forward

The Army's approach to suicide reduction has not stemmed the tide. It needs to look at the problem differently. Just like Durkheim, the Army needs to take a sociological approach to understanding why it is an outlier compared to similar groups.

Fortunately, the Army has initiated policies to increase unit cohesion. The Department of War has incentivized the reduction of moves by cutting the budget for discretionary moves in half by 2030.[66] Already in 2025, 6,200 soldiers chose the Stabilization Retention Option to remain at their duty station. The Army has aimed to reduce relocations by 12,000 in fiscal year 2026 and 13,600 in fiscal year 2027.[67] It has offered pilot programs like bonuses for tankers to remain at their current duty station and the Company Grade Retention and Incentive Program for officers to stabilize at their first installation.[68] The Army is also reviewing professional military education to allow soldiers to attend training while remaining at their current duty stations.[69]

These measures may produce reductions in suicide rates along with other benefits. The Army can already look to the lower suicide rates in units like the 75th Ranger Regiment and Special Forces groups, which allow their members to stabilize.[70] Of course, the cohesion of those units also contributes to their combat effectiveness. In addition to reducing suicides, the Army may find that strengthening social cohesion yields increased lethality and retention.[71] Supporting recent policies to reduce moves, the senior Army career counselor, Sgt. Maj. Enrique Rose, stated, "The goal of our strategy is to solve one of the biggest challenges to readiness and retention, which is instability."[72]

These policies will need to go further because the Army will have to counter increasing social alienation as artificial intelligence provides another means of isolation from society. As an alternative to wider societal atomization, the Army should offer the occupational community that Durkheim recommended. It should serve as a sanctuary from the loneliness epidemic. The Army should provide a band of brothers for soldiers to live for. ■

Notes

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A soldier assigned to Blackfoot Company, 1st Battalion, 4th Infantry Regiment, talks on the radio during a live-fire range in Grafenwoehr, Germany, 16 June 2026. Defaulting to voice radio during fire missions often occurs when digitally transmitted targeting updates are delayed. (Photo by Sgt. Collin Mackall, US Army)

When Delivery Success Fails

Why Time Must Define Authority in Military Networks

Lt. Col. Hyunkeun Shin, Republic of Korea Army, Retired

In Ukraine, intelligence, surveillance, and reconnaissance platforms tracked targets and transmitted data successfully, but by the time fire missions reached shooters, targets had already moved.[1] In Iraq, digitally transmitted targeting updates arrived after pilots had already acted on voice instructions.[2] In both cases, the network performed; the operation did not.

In each case, commanders defaulted to voice radio—not because the digital network had broken down, but because they judged that the digital data had already lost its right to influence the current action. These were not workarounds for a broken network. They were compensation for a network that succeeded at the wrong thing.

This article argues that these failures share a common structural cause: Military networks have no mechanism to determine whether information retains the authority to influence action at the moment of arrival. This condition—structural time failure (STF)—is not a delay problem. It is an architectural absence. The time-gated transmission architecture (TGTA) addresses it through preadmission refusal based on commander-defined reaction windows, with direct implications for joint all-domain command-and-control (JADC2) acquisition, operational testing, and command doctrine.

The Problem Is Not Staleness

The instinctive diagnosis for the aforementioned failures is staleness: The data was old, so it was wrong. That diagnosis misses the structural problem. Staleness is about age. This is about authorization. A three-second-old radar track is not simply "stale"; it is no longer cleared for engagement because the commander's authorization window for that track has closed. A staleness problem is solved by transmitting faster. An authorization problem requires a mechanism that checks, before data enters the network pipeline, whether it is still cleared for operational action—regardless of how quickly it was delivered.

The voice radio fallback illustrates what is missing. A human operator judged that the digital information no longer had the right to influence the current action and refused to use it. That judgment was correct. The digital network had no equivalent structure. TGTA is the systematic replacement for that improvised human check.

Every data type has a commander-defined window during which it remains authorized for operational action—a reaction window. Outside that window, the data is not degraded, it is unauthorized. Military networks currently enforce no such boundary. Every successfully delivered packet is treated as cleared for use, regardless of whether its authorization window has already closed. In a JADC2 environment, where machine-speed systems act on incoming data without any temporal check, this absence does not merely degrade performance, it propagates unauthorized inputs at machine speed.[3] This article names that structural absence "structural time failure" and proposes a solution: the time-gated transmission architecture.

What Makes STF Structural

Tactical networks are built on a reliability principle: If a message fails to get through, send it again. This design made sense when data volume was low and a human reviewed each message before acting on it. In a high-frequency sensor environment—drones, electro-optical/infrared arrays, ground robots generating continuous data streams—this resend-until-delivered approach creates a paradox. In small-scale exercises with few terminals, this paradox remains latent: Collisions are rare, retransmissions are limited, and most data arrives within its authorization window. The failure only becomes visible when subscriber count surges, as it does at the onset of combat operations. By then, the system has already passed every test.

Under network congestion, retransmission queues fill with both cleared and unauthorized data competing for the same bandwidth. The network does not distinguish between them. Unauthorized data, having been waiting longest, holds queue priority and is resent first. The result is a systematic inversion: The reliability mechanism designed to ensure delivery becomes the mechanism that ensures unauthorized information reaches the decision space before cleared information does. Higher reliability produces lower operational validity. This is why STF worsens, not improves, as transmission speed increases. Faster delivery of unauthorized inputs accelerates the problem rather than resolving it.

This is not a contingent failure. It is a structural one. As long as a network lacks a mechanism to stop unauthorized data before it enters the pipeline, there is no speed improvement, no priority scheme, and no scheduling algorithm under which unauthorized data can be reliably excluded from the decision space. The

Table 1. Reaction Window Classification

Class	Reaction Window (Authorization Boundary)	Example Data	Cost When Unenforced
0	No expiration—permanently authorized	Enemy doctrine, terrain overlay	None
1	Hours	Route status, logistics	Unauthorized route data delays resupply
2	Minutes	Troop positions, close air support coordination	Fratricide risk, missed engagement clearance
3	Seconds (3–15)	Radar track, time-sensitive target	Fires on unauthorized track—mission failure

(Table by author)

damage occurs at the moment of entry. Once inside, it cannot be undone.

Empirical simulation using real-world sensor data confirms this inversion; under high-load conditions representative of wartime subscriber density, packet delivery rate remained above 82 percent while the time-valid delivery ratio for time-sensitive targeting data collapsed to zero—a network passing every current test standard while delivering nothing operationally usable.[4]

Time as Authorization, Not Performance

The standard engineering view treats time as a performance variable, something to optimize. The argument here is different: Time defines the authorization boundary for operational action. Networks measure success through delivery rate, latency, and throughput. Commanders measure success by a single question: Did this information arrive in time to act on it?[5] Col. John Boyd's decision-cycle model captures the consequence: A force that cannot act on information within the relevant window loses the ability to influence the unfolding situation.[6] Beyond that window, information may remain accurate and successfully delivered, but it is no longer cleared for use.

Table 1 defines four reaction window classes, each representing the period during which a data type remains authorized for operational action. The classification does not assert that older information is worthless—Class 0 data retains its authorization indefinitely. It asserts that the authorization boundary for each class must be defined by commanders and enforced by the system. STF occurs when the system cannot distinguish between them.

A reaction window is the period during which information remains authorized for operational action. Doctrine does not yet define such a category, and that absence is precisely the gap this article identifies. When that window closes, the information is not degraded, it is unauthorized. It is a command decision, defined by commanders based on the physics of the engagement, the speed of the threat, and the tempo of the operation. Adjusting a reaction window is not a technical setting; it redefines the commander's authorization boundary for that data type in the same way that assigning fires clearance authority defines who may authorize lethal action. STF occurs when this authorization boundary is not enforced. The information may still exist in the system. It is no longer cleared for operational action.

Why Existing Approaches Cannot Solve This

Quality-of-service prioritization, deadline-aware scheduling, and age-of-information metrics each improve some aspects of timeliness. None enforces authorization boundaries. The structural reason is simple; all three act on data after it has already entered the network pipeline. Once data is inside the pipeline, it remains treated as cleared for use regardless of whether its authorization window has closed. Nothing that happens after data enters the pipeline can undo the transmission capacity consumed, the queue space occupied, or the processing time spent on unauthorized data. The

Table 2. Structural Comparison of Time-Management Approaches

Feature	Quality of Service / Scheduling	Age of Information	Time-Gated Transmission Architecture (Proposed)
Acts on data	After data enters pipeline	After data enters pipeline	Before data enters pipeline
Authorization source	System-internal	System-internal	Commander-defined
Can prevent structural time failure?	No	No	Yes—by construction
Refusal logging	No	No	Yes—audit trail

(Table by author)

damage occurs the moment it enters. Once inside, it cannot be undone. The only point at which STF can be prevented is before data enters the pipeline—and that is precisely where all existing mechanisms are silent.

The decisive difference is the authorization source. Existing approaches derive their decision criteria from within the network—traffic load, available capacity, observed delay. TGTA's criterion comes from outside the network: the reaction window is set by commanders at mission planning, not computed from network conditions. A network with perfect traffic management can simultaneously be routing unauthorized targeting data directly to the fires cell. Table 2 captures this distinction.

STF is not a performance anomaly, it is an architectural absence. As long as a network lacks a mechanism to enforce authorization boundaries before data enters the pipeline, no combination of speed improvements, priority schemes, or scheduling algorithms can reliably exclude unauthorized data from the decision space. Without such a mechanism, STF is not a risk, it is a structural inevitability.

The Time-Gated Transmission Architecture

TGTA enforces one rule at every relay point in the network: Before a message enters the pipeline, ask whether it will reach its destination while still within its authorization window. If yes, send it. If no, stop it—and log the decision.

The logic is binary. The decision is independent of the message's accuracy, priority, or network conditions. A Class 3 radar track with a three-second authorization window traveling through a three-relay chain is either still cleared for fires when it arrives, or it is not. If it cannot arrive within its window, sending it wastes transmission capacity, delays cleared data behind it, and delivers an unauthorized track to the fires cell. Stopping it keeps the decision space clean.

Authorization windows are assigned by commanders at mission planning, not derived by the network. TGTA does not require new hardware or changes to existing communication protocols; it operates as a software check at each relay point within current tactical communication standards.[7] This means TGTA can be fielded as an overlay on existing systems without hardware replacement.

The appropriate measure of success is the time-valid delivery ratio (TVDR): the percentage of messages that arrive while still within their authorization window.[8] A network reporting 95 percent delivery and 40 percent TVDR is efficiently routing unauthorized information to decision-makers. Without TVDR, that failure is

TGTA in Practice

A radar track with a three-second authorization window travels through three relay nodes. At each relay, one question is asked: Will this track reach the C2 node while still cleared for engagement?

If no—regardless of priority—it is stopped and logged. The fires cell never receives an unauthorized track. The commander's display shows only what is cleared for action.

The network did not get faster. It became honest.

invisible. Every stopped message is logged with a time-stamp and message class, creating an after-action record that makes STF measurable for the first time.

Implications for JADC2 Acquisition and Doctrine

Treating time as an authorization boundary rather than a performance variable has direct consequences for acquisition, testing, and command doctrine. Three actions follow.

First, the Joint Staff should direct that JADC2 operational testing include TVDR as a coequal success criterion alongside delivery rate. A network that scores well on delivery while failing TVDR is routing unauthorized information to commanders at machine speed—and calling it success.

Second, the under secretary of war for acquisition and sustainment should require all future tacti-cal-communication requests for proposal to specify configurable authorization windows and TVDR thresholds by message class. Requirements documents written today without this language will encode the assumption that all delivered data is cleared for use into fielded systems for a generation.

Third, commanders must assign reaction windows as deliberately as they assign frequency plans, call signs, or fires clearance authority. Authorization windows for each data type belong in communications annexes by mission phase, reviewed at the appropriate command level. The absence of this practice is not a technical limitation; it is a doctrinal gap no different from failing to assign fires clearance authority before an operation.

A 2025 Government Accountability Office assessment identifies JADC2 data-sharing integration as still operating at the "swivel-chair" level, with latency and data-sharing as core bottlenecks.[9] TGTA directly addresses this structural gap. Connectivity that arrives outside the authorization window—regardless of its technical quality—is not decision advantage. It is unauthorized input at machine speed.

Conclusion

Military networks are technologically mature but structurally incomplete. They were designed to deliver information. They were never designed to enforce the authorization boundaries under which information is cleared for operational action.

The instinct is to call this a staleness problem and prescribe faster networks. It is not. Staleness is a symptom. The structural cause is the absence of a mechanism that treats time as an authorization boundary—differentiated by class, defined by commanders, and enforced before data enters the pipeline. No amount of speed closes an authorization gap. A faster network that cannot distinguish cleared from unauthorized data simply delivers the wrong picture more efficiently.

TGTA provides that mechanism. The reaction window classification ensures the argument is precise: Class 0 data remains authorized indefinitely, while Class 3 data loses its clearance for engagement in seconds. The system must know the difference. Currently, it does not. ■

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Army Special Forces and soldiers assigned to 2nd Battalion, 2nd Aviation Regiment, 2nd Combat Aviation Brigade, 2nd Infantry Division, conduct combined air assault training at an undisclosed location on 4 February 2026 in South Korea. The training included fast rope insertions and air assault operations to enhance interoperability, mission execution, and rapid force projection capabilities in an urban environment. This training was conducted in coordination with local authorities at a designated and approved training site to ensure safety, strengthen civil-military partnerships, and maintain readiness for operations in densely populated terrain. (Photo by Cpl. GeonWoo Park, US Army)

The Shape of the Fight

Using LiDAR and Fractal Geometry to Quantify Urban Complexity

Lt. Col. Caleb Dexter, US Army Reserve

The twenty-first century is the urban century. This century has seen, for the first time in human history, the majority of the world's population residing in cities.[1] Nor does this trend show any sign of slowing down.[2] This continuing global phenomenon of people concentrating in dense nodes of economic activity and physical infrastructure—many of which now qualify as "megacities" with populations exceeding ten million—has fundamentally reshaped the strategic landscape across all four instruments of national power: diplomatic, informational, economic, and military.[3] Of the four, it is perhaps the military that is the *most* affected by this global trend of urbanization. Consequently, success in warfare in the land domain, whether great power competition, counterinsurgency, or stability operations, is likely to be decided not on the rolling plains of the Fulda Gap or undulating deserts of the Arabian Peninsula that dominated late twentieth-century and early twenty-first-century military thought but rather in the dense, complex, and unforgiving urban terrain of the world's cities.[4]

Fighting in urban terrain is not a new phenomenon, but the scale, scope, and intensity of recent urban combat is.[5] From the bitter house-to-house fighting in Stalingrad that broke the back of the German Sixth Army to the brutal suppression of Chechen separatists in the 1994–1995 Battle of Grozny that leveled a city and exposed the vulnerabilities of a conventional army and to the labyrinthine alleys of Mogadishu that neutralized US technological overmatch, urban terrain has proven itself again and again to be the costliest terrain in which to conduct military operations.[6]

The US Army has already recognized this reality, stating in its doctrine that "growing urbanization throughout the world raises the possibility of future military operations taking place in urban environments."[7] Recent events have underlined this point with brutal clarity—the grinding, monthslong clearance of Islamic State jihadists from Mosul and Raqqa by coalition-supported forces, the fierce urban combat in Marawi against entrenched militants, and the attritional "meat grinder" of Bakhmut—all have demonstrated that even a nonstate actor or a less-than-near-peer adversary can leverage the urban environment to inflict disproportionate casualties and stall a technologically superior force.[8] The city itself—its verticality, its subterranean voids, its structural mass, and its ability to channelize movement and conceal threats—becomes the primary weapon in the defender's hands and a weapon with which the Army's traditional analytical tools are ill-suited to deal.

This article argues for a paradigm shift in how we quantify and understand this environment. It posits that, to secure victory and protect warfighters in the urban battles of tomorrow, we must augment our doctrinal frameworks with a new, objective, and mathematically rigorous method for quantifying urban complexity. This article introduces such a method: the Urban Density Score, or U Score.

The Limits of Qualitative Analysis

Current US Army and joint doctrine provides a highly structured and detailed intellectual framework for thinking about urban terrain. Field Manual 3-06, *Urban Operations*, and its more recent successor, Army Techniques Publication 3-06, *Urban Operations*, lay out the fundamental principles for analyzing these complex environments.[9] Central to this doctrine is the use of analytical frameworks to help commanders and staffs understand the operational environment. The primary tool is the timeless OAKOC (observation and fields of fire, avenues of approach, key and decisive terrain, obstacles, and cover and concealment) acronym, which is further detailed for urban scenarios by ASCOPE (area, structures, capabilities, organizations, people, and events).[10]

This doctrine has proven its utility over the years and has provided countless Army staffs a lingua franca for planning operations in urban terrain. Under the "structures" consideration of ASCOPE, for example, a planner is correctly directed to ask a series of critical questions: What is the building composition? Are they wood-framed or reinforced concrete? Are there extensive subterranean levels beneath?[11] This qualitative analysis is the current foundation of our understanding of urban terrain. An intelligence officer (S-2), for example, examining satellite imagery and other intelligence sources can assess that a district is composed of prewar European reinforced concrete apartments. This, in turn, leads to the assessment that the district will be difficult to breach, highly defensible, and likely to create significant rubble when targeted with kinetic fires. While such an assessment is qualitative in nature, it nevertheless does help the commander and his or her staff plan and prepare for operations in that terrain.

However, this is where a doctrinal gap appears. As assessments like that above are entirely subjective, they

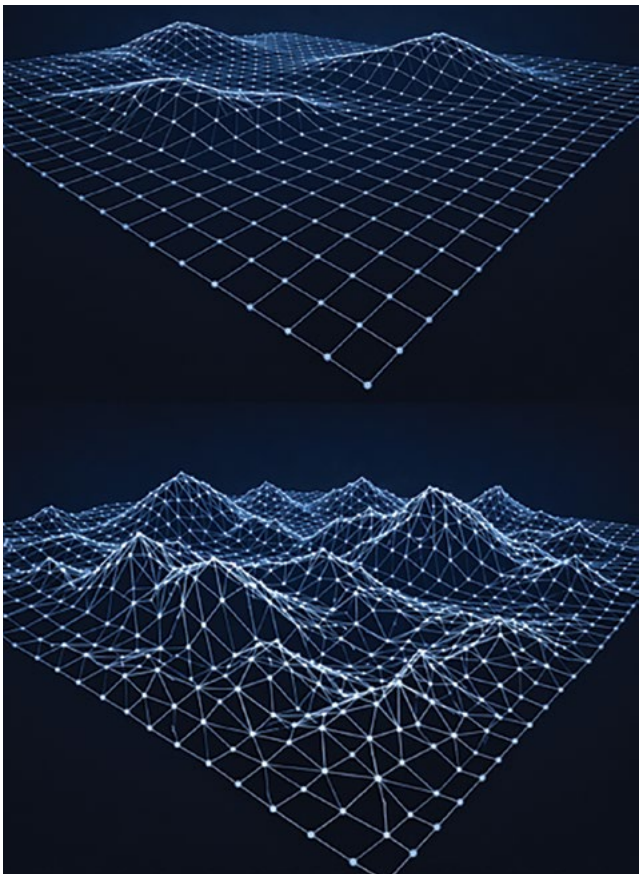
are reliant on individual planner's experiences and mental libraries of past examples. How much more difficult is "difficult to breach" than a district assessed as "moderately defensible"? Is it twice as hard? Ten times as hard? What one battalion S-2, whose formative experience was in the dense urban blocks of Sadr City, considers "dense," another S-2 with experience in the rural villages of Afghanistan might see as "impenetrable." This subjectivity introduces dangerous variance and unquantified risk into the planning process. The lack of a common, objective metric places an impediment on commanders and their staffs' ability to plan for operations in dense urban terrain. They must weigh these unquantified, descriptive assessments against hard numbers for personnel, supplies, and time, attempting to bridge the gap between qualitative art and quantitative science.

The problem becomes particularly acute when planning force ratios. Army Techniques Publication 3-06 provides a table of recommended force ratios based on a collection of historical case studies, suggesting, for instance, an 8:1-to-3:5 ratio for a hasty attack on a city block defended by a platoon.[12] But this guidance, while a valuable starting point, is a historical composite that fails to take into account critical variables (such as the aforementioned building composition). For example, is the "city block" from the Battle of Aachen in 1944, built of brick and stone, the same tactical problem as a modern city block in Seoul, built of steel-reinforced concrete and glass? The doctrine cannot, in its current form, account for this granularity. The U Score is designed to close this gap, not by replacing doctrine but instead by enhancing it with an objective layer of data.

Quantifying Complexity: LiDAR, Fractals, and the U Score

The U Score has been designed to provide this objective measure by leveraging two powerful technologies, light detection and ranging (LiDAR) and fractal analysis.

First, we must have a means of capturing the battlespace in its full three-dimensional reality. Fortunately, such a means exists. LiDAR is a remote sensing method that uses pulsed laser light to measure ranges to the Earth. A complete airborne LiDAR system consists of a laser scanner, a Global Positioning System receiver, and an inertial measurement unit.[13] The inertial measurement unit tracks the precise orientation of the aircraft,

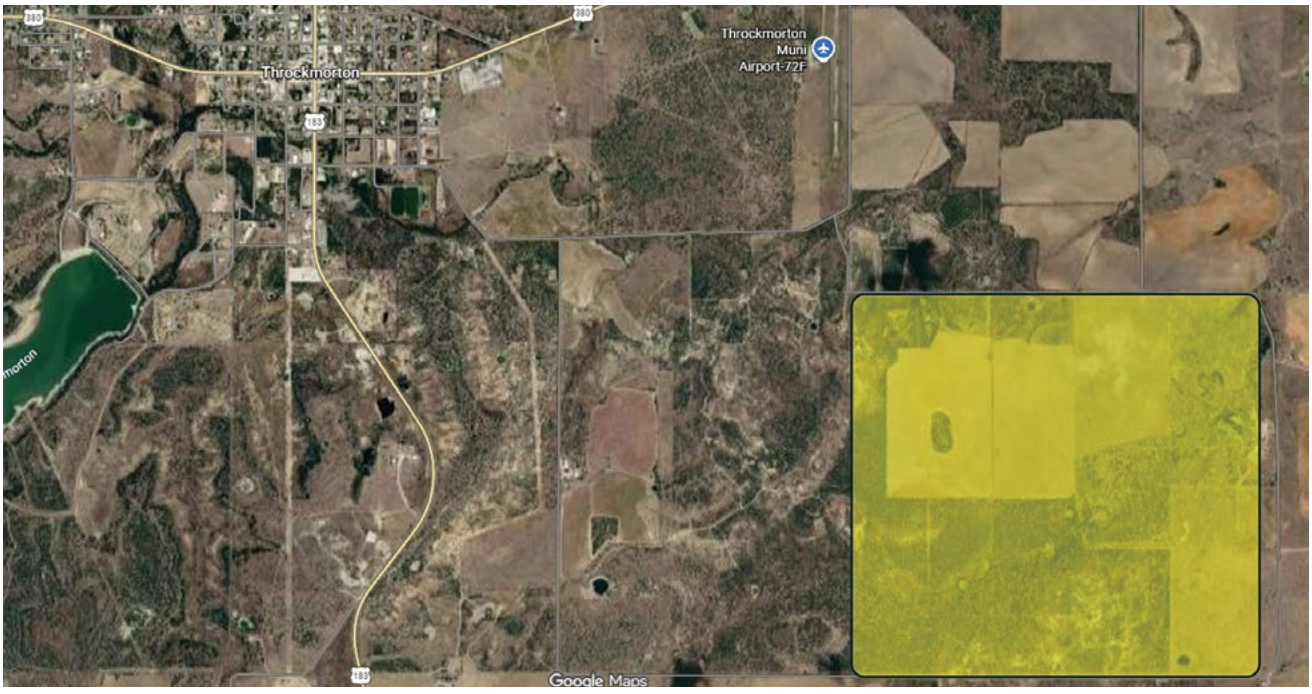


(Figure by author)

Figure. Comparison of Fractals

while the Global Positioning System tracks its location. By combining this data with the laser's range measurements, the system can calculate a precise X, Y, and Z coordinate for every point the laser strikes. When flown over a city by an aircraft, drone, or satellite, it produces a "point cloud" that can contain billions of individual coordinates.[14] The result is a digital twin of the urban landscape. Advanced processing algorithms can then automatically classify this point cloud, distinguishing points that reflect off the "bare earth" from those that reflect off man-made structures or vegetation.[15] This LiDAR point cloud data is widely available free of charge on the internet, with the US Geological Survey and National Oceanic and Atmospheric Administration providing such data in the case of the United States.[16]

Once the cityscape is captured as a point cloud, the challenge is to distill its complexity into a single, meaningful value. The solution lies in the mathematical field of *fractal geometry*. A fractal is a geometric object that



(Map from Google Maps; data derived from National Oceanic and Atmospheric Administration, "2017 - 2018 USGS FEMA LiDAR: Red River, TX," NOAA Office for Coastal Management, 2018, https://coast.noaa.gov/dataviewer/#/lidar/search/-11037253.450607048,3915499.6005857885,-11034860.055183545,3917715.4639311754/details/10027)

Map of Throckmorton, Texas

exhibits self-similarity, meaning it displays the same or similar structural patterns at different scales of magnification. The "complexity" or "jaggedness" of a surface can be quantified by its fractal dimension. A smooth, straight line has a dimension of exactly 1. A perfectly flat, solid plane has a dimension of exactly 2. But what about complex surfaces that are neither perfectly flat nor fully solid?

Consider the two surfaces depicted in the figure. The top image shows a relatively smooth plane with gentle undulations. Because it deviates only slightly from a perfect two-dimensional plane, its fractal dimension is very close to 2.0. The bottom image, however, shows a plane that has been intensely crumpled and folded. While it is still technically a two-dimensional surface, its jagged, mountainous structure is so complex and space-filling that it begins to take on the characteristics of a three-dimensional object. Therefore, its fractal dimension is higher—somewhere between a perfect plane (2.0) and a solid cube (3.0), perhaps around 2.15 (this is a hypothetical number for illustrative purposes only). This single number, 2.15, quantitatively captures the increase in complexity of the crumpled surface.

The U Score methodology applies this principle to isolate the complexity added *by human construction*. Using GenAI.mil to generate code to create a "box-counting" algorithm—that repeatedly lays a 3D grid of shrinking boxes over the point cloud and counts how many are occupied—we compute two values:[17]

The ground dimension (D_ground). This is the fractal dimension of the "bare earth" points, representing the complexity of the natural landscape.

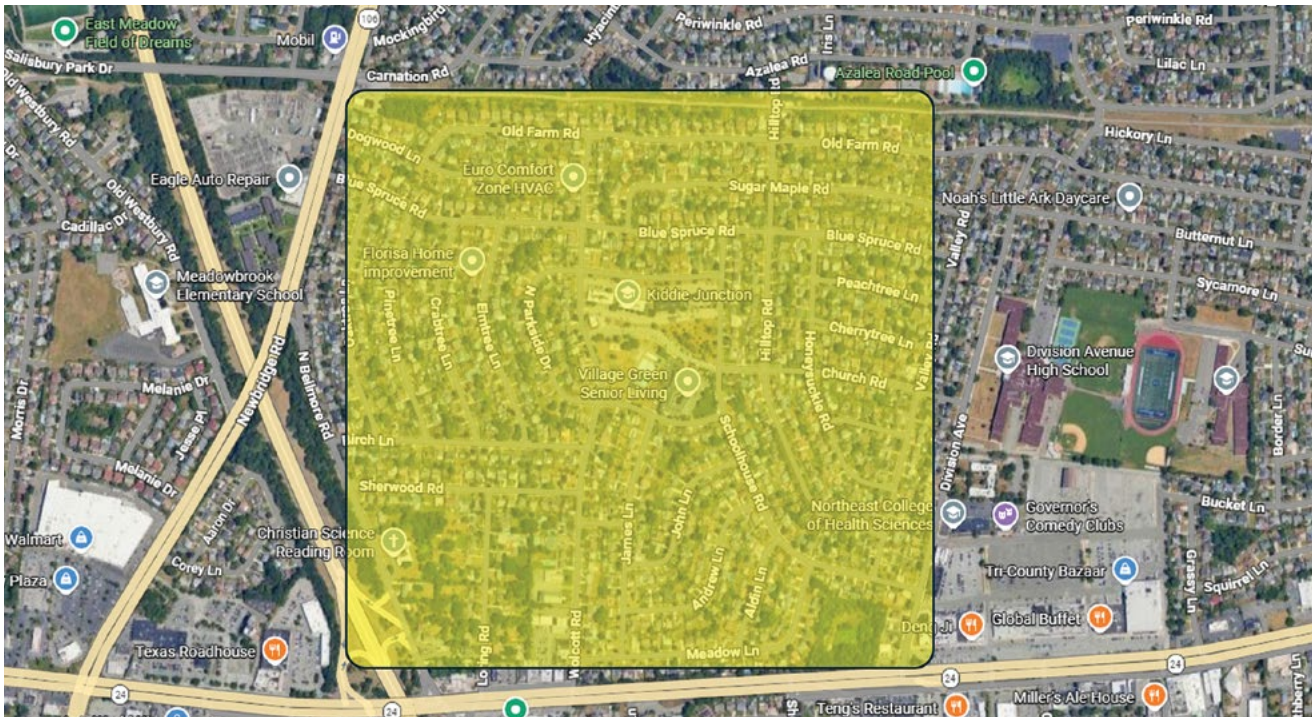
The total dimension (D_total). This is the fractal dimension of the complete environment, including the ground and all man-made superstructures upon it.

The U Score is then the simple subtraction of the two: *U Score = D_total - D_ground*. This simple calculation isolates the variable of interest: a pure, objective measure of the complexity added by human construction, independent of the underlying natural terrain.

A Tale of Three Terrains: A Comparative Analysis

To illustrate the implications of this methodology, let us compare the U Scores of three different pieces of terrain: flat empty terrain, cluttered suburban terrain, and dense urban terrain.

Throckmorton, Texas (flat empty terrain). The area just southeast of Throckmorton, Texas, is almost



(Map from Google Maps; data derived from National Oceanic and Atmospheric Administration, "2014 USGS CMGP LiDAR: Post Sandy (Long Island, NY)," NOAA Office for Coastal Management, 2014, https://coast.noaa.gov/dataviewer/#/lidar/search/-8186302.734583888,4971866.593399718,-8185091.3230212005,4973135.691227296)

Map of Levittown, New York

completely flat and undeveloped. As such, it presents a good example of terrain with a low expected U Score. When the U Score for the piece of terrain highlighted in the Throckmorton map is calculated, it comes to 0.01—relatively close to the value of 0 we would expect for a completely two-dimensional object.

- Ground dimension (D_ground): 1.98
- Total dimension (D_total): 2.08
- Resulting U Score: 0.10

Levittown, New York (cluttered suburban terrain). Levittown, lying about twenty-five miles east of downtown Manhattan, New York City, rose to fame in the middle of the last century as America's first suburb and its level of urban density—closely packed but relatively low single-family homes—is a good overall proxy for what suburban development looks like in terms of building heights, lane widths, and so forth.[18] When its U Score is computed, it comes out to 0.18—about 80 percent denser than the prairie of the Throckmorton area.

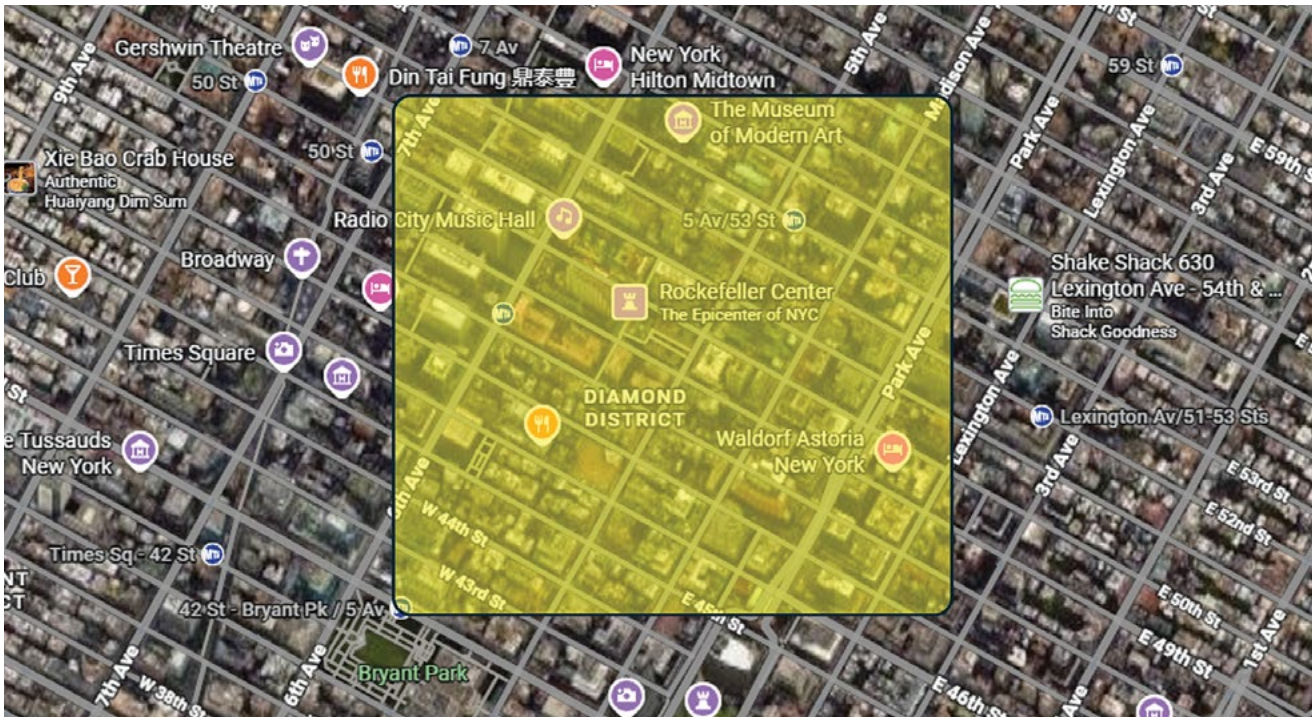
- Ground dimension (D_ground): 1.95
- Total dimension (D_total): 2.13
- Resulting U Score: 0.18

Midtown Manhattan, New York (dense urban terrain). Midtown Manhattan is some of the densest urban terrain in the United States. Roadways are narrow, open space is scarce, and many buildings are in excess of forty stories.[19] It is perhaps the densest urban terrain we have in the United States. Its U Score comes out to 0.38—about twice as dense as Levittown and about 400 percent denser than the Throckmorton piece of terrain.

- Ground dimension (D_ground): 1.84
- Total dimension (D_total): 2.21
- Resulting U Score: 0.38

From Data to Decision: Operationalizing the U Score

The ultimate value of the U Score lies in its applicability to the art of military decision-making. Using the U Score, it would be possible to create a standard digital overlay on our command-and-control (C2) platforms and common operational pictures. Much like a map showing elevation or vegetation, this layer would provide an immediate, visual, and objective understanding of the structural complexity of the urban



(Map from Google Maps; data derived from National Oceanic and Atmospheric Administration, "2017 NYC Topobathy LiDAR: New York," NOAA Office for Coastal Management, 2017, https://coast.noaa.gov/dataviewer/#/lidar/search/-8235826.399421787,4976221.877270037,-8234592.630060924,4977345.256740718)

Map of Midtown Manhattan, New York

battlespace. A commander could glance at his or her screen and see, in a clear color-coded gradient, green (low U Score) to red (high U Score), where the most formidable urban terrain lies.

This data layer could furthermore be integrated with a U Score database, a living repository of combat data that correlates the U Scores of specific historical engagement areas with the actual outcomes: attacker and defender force numbers, casualty rates, rates of advance, ammunition expenditures, and types of units involved. By applying machine learning and statistical analysis to this database, we could move beyond historical analogy to predictive, data-driven planning for operations in urban terrain. A planner could query the database: "Show me the expected casualty rate for a mechanized infantry company attacking a district of this size, defended by this many enemy, with this U Score." The system could then provide a statistically grounded estimate, enabling a far more accurate assessment of risk and the true cost of an operation.

On the C2 map, the commander could click on any grid square and instantly see not only its U Score but also a statistically informed forecast of the resources required to seize or defend it. This transforms the art of command using the science of data, allowing for more accurate predictions of the forces and logistical support required to seize, hold, or defend specific urban objectives. It allows for better war-gaming, more realistic training scenarios at our combat training centers, and more informed procurement decisions about the types of equipment needed for the future urban fight.

Counterarguments

Any proposal to quantify battlefield terrain will invariably draw skepticism from those who view warfare as a fundamentally human art. The U Score is no exception, so I endeavor to address the most likely objections to the U Score here.

First is the timeless argument that warfighting is an art, not a science. This is, of course, somewhat true; it is actually a combination of both. The U Score is not an attempt to create a mathematical formula for victory, nor can it ever replace a commander's intuitive grasp of the battlefield, the commander's coup

d'œil. Rather, its purpose is to sharpen that intuition with objective data and quantitative measures. The modern ground commander already has access to a wide variety of quantitative information, from enemy disposition to logistical statuses. The U Score simply adds the physical environment to that list of knowns and removes the burden of qualitatively estimating urban complexity. By providing a rapid, data-driven assessment of the terrain, it frees the commander to focus on the true art of war: understanding the enemy's will, gauging the population's sentiment, and leading soldiers.

A second critique will likely focus on logistics, arguing that the required LiDAR data is too expensive or difficult to obtain. This perspective, however, is outdated. The proliferation of commercial satellite and aerial LiDAR services, coupled with the miniaturization of sensors for unmanned platforms, makes accessing and collecting such data more feasible than ever. When weighed against the cost in blood, treasure, and national prestige of a failed urban battle, the expense of obtaining LiDAR data of a critical objective becomes a prudent and necessary investment in mission success.

Finally, some may argue that the U Score is flawed because it is static and fails to account for the dynamic nature of battle damage. This is a valid observation, but it mistakes a key feature for a bug. A pristine building has one U Score; the rubble pile it becomes after being struck by artillery has another. One facet of the U Score's utility lies in its ability to measure this change. By conducting repeat scans during an operation, commanders can objectively track the evolving complexity of the urban terrain. This allows them to assess the actual effects of preparatory fires and, crucially, understand the new, rubble-strewn landscape their soldiers must now traverse. The fractal dimension of rubble is not a flaw in the model; it is a critical tactical variable that can assist in assessing the need for assets such as engineers or using alternate routes for both vehicles and dismounted troops.

Conclusion: A Call for a New Cartography

We can continue to rely on twentieth-century methods of qualitative assessment to analyze twenty-first-century battlefields, or we can embrace the technologies that allow us to see and quantify the urban environment with unprecedented clarity. The Urban Density Score offers a path forward, a way to move beyond guesswork and apply data science to one of the most complex problems in modern warfare. It provides a means to predict the cost of urban combat, to allocate forces intelligently, and to give US warfighters a tactical edge where they're most likely to need it. The implementation of the U Score would require a dedicated, multidisciplinary effort. It would require data scientists, geospatial intelligence professionals, and doctrinal writers to work in concert to create a U Score-enabled data layer in our C2 architecture. The tools, data, and methods to do so are all in place. Now it's up to us to put them together and more fully learn from the lessons bought with blood, sweat, and tears of past urban battles. Those lessons teach us that the city itself is a weapon. It is time we learned how to *measure* it so we can better use it to support the warfighter. The development and adoption of the U Score gives us a chance to do exactly that. ■

This article is a product of personal research and does not represent the official position of the Department of the Army, the Department of War, or any US government agency.

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Combat Studies Institute Staff Ride



The Combat Studies Institute Staff Ride Team develops and conducts live and virtual staff rides as an educational tool for the US Army. By focusing on the timeless and universal aspects of warfighting, staff rides provide important insights into military operations, concepts of leadership, and the face of battle, through vignettes and topics for discussion. The Staff Ride Team also provides information and guidance to US Army organizations on how to conduct staff rides for the purpose of educating leaders. For more information, including how to request staff rides, visit https://www.armyupress.army.mil/Staff-Rides/Virtual-Staff-Ride/.

Left: A live staff ride at the Little Bighorn Battlefield, Montana. ***Bottom***: A Normandy invasion virtual staff ride at Fort Leavenworth, Kansas. (Photos courtesy of CSI Staff Ride)





(Photo from Adobe Stock)

The Assurance War

How Saudi Arabia, Qatar, and the United States Are Facing Digital Insurgency

Hamzeh Abu Nowar

In late 2025, something subtle yet dangerous began unfolding in the Arabian Gulf. Aviation and maritime operators near major ports and shipping lanes began reporting erratic satellite navigation data, positioning signals jumping unpredictably, and guidance systems conflicting. Crews were forced to revert to manual and backup procedures.[1] By October, Qatari authorities temporarily restricted maritime traffic, warning that the interference could compromise vessel safety.[2] Though no networks had been breached and no actor claimed responsibility, shipping slowed anyway because the systems that make modern transport safe could no longer verify their own reliability. One of the world's most critical energy

corridors had been constrained without a single missile being fired.[3]

These incidents reveal a form of strategic competition operating beneath traditional armed conflict and gray-zone thresholds. Rather than destroying infrastructure or penetrating networks, adversaries can generate operational effects by manipulating the verification and assurance systems that determine whether critical activities are allowed to continue.

Disruptions targeting digital verification systems present a distinct operational challenge for the United States and its Gulf partners. Existing approaches to cyber defense and gray-zone competition do not fully address the risks posed by interference with the systems that authorize movement, trade, and infrastructure use. For US and partner forces operating in strategically dense regions such as the Arabian Gulf, the ability to deploy, sustain, and maneuver increasingly depends on not only defeating adversary forces but also on maintaining confidence in the civilian assurance systems that permit military movement in the first place.[4]

In a crisis, even temporary degradation of navigation assurance in key transit corridors could delay sealift staging, disrupt air tasking cycles, and prompt marine underwriters to reassess the risks associated with the voyage.[5] Where electronic interference materially increases navigational uncertainty, including the risk of collision, grounding, sanctions violations, or operations within a designated Additional Premium Area or other war-risk area, underwriters may require advance notification, additional war-risk premiums, revised policy terms, or agreement on revised terms before coverage can be continued or reinstated.[6] If adequate insurance cannot be obtained or maintained, shipowners may defer or decline the voyage, while ports, charterers, and contracting authorities may withhold access or authorization, delaying contracted logistics.[7] Within tightly synchronized deployment and sustainment plans, delays measured in hours or days can produce strategic consequences disproportionate to the initial disruption. Early indicators of this dynamic have already appeared in safety-governed industrial systems. In 2017, the malware Triton triggered a safety shutdown at a Saudi petrochemical facility.[8] While no explosion occurred, the plant halted operations as its safeguards could no longer verify safe conditions.

Precautionary suspension is not new. Nuclear facilities during the Cold War and industrial control systems for decades have been designed to halt when reliability cannot be confirmed. What has changed is the ability of adversaries to deliberately induce uncertainty in those systems through digital manipulation. Disruption no longer requires physical destruction; it just requires the introduction of doubt into systems designed to halt under uncertainty.

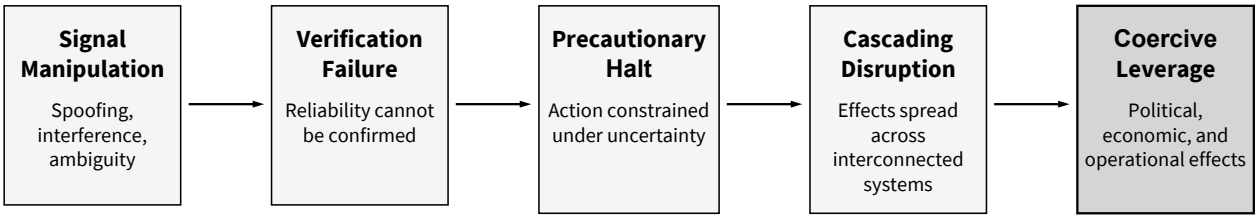
What is unfolding in the Gulf may represent an emerging form of strategic competition—one in which rivals can erode US, Saudi, and Qatari power by manipulating the systems that energy markets, governance, and shipping depend on. The interference is aimed not at physical assets but at the mechanisms that authorize their use.

The Logic of Precautionary Shutdown

In safety-critical industries, unverified data is treated as danger. If operators cannot confirm that signals, controls, or safeguards are reliable, then systems are designed to halt—not because failure has occurred but because continued operation could be catastrophic.[9] This logic is embedded across engineering standards, compliance regimes, and insurance rules. It saves lives, but it also introduces a new strategic vulnerability. This vulnerability is not purely digital but structural. Predating the cyber era, safety-critical industries have long been designed to halt operations when reliability cannot be confirmed. During the Three Mile Island accident, precautionary shutdown procedures were triggered amid uncertainty rather than confirmed destruction.[10]

An attacker no longer needs to penetrate networks or destroy physical assets. They only need to introduce enough ambiguity—spoofed signals, synthetic warnings, or data that appear official but cannot be quickly verified. When verification fails, systems do exactly what they are designed to do: stop. The causal loop is simple: signal manipulation → verification failure → precautionary halt → cascading disruption → coercive leverage (see figure).

These are not fragile systems, they are obedient. The same safety reflexes that prevent catastrophe can now be turned into instruments of coercion—long before any military threshold is crossed. The disruption exploits the precautionary logic embedded in modern



(Figure by author)

Figure. Mechanism of Assurance Disruption

Adversaries exploit the precautionary logic embedded in assurance systems. Rather than attacking infrastructure directly, they manipulate verification to trigger precautionary restrictions, cascading disruptions, and ultimately coercive leverage.

safety and verification systems, creating a distinct strategic risk for the United States and its Gulf partners.

Limits of Existing Threat Frameworks

The dynamics described here intersect with established concepts such as hybrid warfare, gray-zone competition, and cyber operations. Each captures important dimensions of contemporary conflict; yet, disruptions targeting assurance systems reveal a pattern not fully explained by any single framework.

Hybrid warfare describes coordinated state campaigns that integrate military, irregular, informational, and economic tools within a coherent strategic design under centralized political direction.[11] Assurance disruption differs in a critical respect: strategic effects can emerge without centralized direction, unified campaign architecture, or synchronized political signaling. Rather than an integrated strategy executed across domains, dispersed actors operating within digital ecosystems exploit institutional precautions to produce cumulative disruption.

Gray-zone competition refers to calibrated coercion and influence below the threshold of armed conflict, premised on intentional signaling and escalating pressure among political actors.[12] In assurance disruption, however, leverage arises not from strategic coercion or signaling but from institutional hesitation triggered by structural exploitation.

Traditional cyber frameworks emphasize operations that target networks, data, access, or the integrity and availability of systems, including actions that compromise data confidentiality, disrupt services, or degrade infrastructures.[13] Assurance disruption operates differently. Systems may remain technically intact, yet manipulation of signal integrity or verification confidence triggers precautionary shutdowns embedded within modern safety architectures. The attack is not on infrastructure but on trust.

Modern infrastructure is designed to halt when reliability cannot be confirmed. Aviation routing, maritime transit, energy production, and financial authorization depend on automated verification systems built to prevent catastrophic failure. When those systems are contested, even briefly, they default to restriction rather than risk.[14] Strategic leverage therefore emerges from inducing institutional hesitation, not destroying infrastructure.

These dynamics point to a broader environment this article describes as digital insurgency: a decentralized ecosystem in which state-linked, proxy, ideological, and illicit actors operate across encrypted and attribution-resistant digital infrastructures. Rather than unified command or shared ideology, the environment is defined by dispersed actors employing similar fieldcraft and adaptive practices within stateless digital environments.[15] Within safety-governed infrastructures, such activity converts signal manipulation and verification uncertainty into precautionary delays that slow movement and compress operational tempo. Assurance disruption represents one operational manifestation of this environment, translating digital manipulation into precautionary constraint within safety-governed infrastructures.

Recent assessments of transnational cybercrime networks and documented global navigation satellite



King Fahad Industrial Port in Jubail, Saudi Arabia, is the largest industrial port in the Middle East. The targeting of digital systems, such as the 2017 safety shutdown of a Saudi petrochemical facility by the Triton malware, present a challenge that existing approaches do not fully address. Any delay measured in hours or days could produce strategic consequences disproportionate to the initial disruption. (Photo from Adobe Stock)

system (GNSS) interference incidents illustrate how loosely aligned digital actors and signal manipulation can generate cascading operational effects without centralized command or overt escalation.[16] This ecosystem is susceptible to exploitation by state-linked services, proxy networks, and illicit digital operators targeting the decision systems on which governments, regulators, and corporations rely rather than the machines themselves. Nowhere are these dynamics more visible than in the Arabian Gulf, where digital assurance systems intersect with one of the world's most critical energy and logistics corridors.

The Gulf's Invisible Battlespace

The Arabian Gulf is one of the world's most critical energy and logistics corridors. Oil, liquefied natural gas, aviation, shipping, finance, and governance all flow through it at speed; however, the Gulf is structurally exposed.[17] It is where physical chokepoints intersect with digital assurance systems—and where localized uncertainty can generate global consequences. When assurance mechanisms falter in this environment, disruptions propagate far beyond the region.[18]

Satellite navigation guides tankers through narrow channels while marine insurance markets influence whether voyages are commercially feasible by determining the availability and terms of hull, protection and indemnity, and war-risk coverage.[19] Compliance regimes decide which cargoes can move, and undersea cables carry the financial and logistical data that sustain global trade.[20] When navigational interference or geopolitical conditions materially increase operational risk, marine underwriters may require advance notification, charge additional war-risk premiums, revise policy terms, or, where the assessed risk becomes unacceptable, decline to provide war-risk cover. Because charter parties, ports, and many commercial contracts require adequate insurance, these underwriting decisions can delay, reroute, or prevent voyages from proceeding.[21]

None of these systems fail loudly; rather, they fail cautiously. When signals become unreliable or confidence mechanisms break down, vessels are delayed, terminals slow operations, insurers reassess coverage,

and regulators intervene. Energy exports need not cease entirely to generate strategic effects. Small, distributed pauses across a tightly optimized system can shift markets and alter political calculations.

Within this environment, two of the region's most strategically integrated states—Saudi Arabia and Qatar—face distinct but related forms of structural exposure.

Saudi Arabia's Strategic Transformation in a Contested Assurance Environment

Saudi Arabia's Vision 2030 is built on speed, scale, and integration.[22] Projects such as new Red Sea ports and digitally managed energy infrastructure are designed to move people, power, and capital with minimal friction.[23] That efficiency creates structural dependencies. As these systems become more tightly optimized, they rely increasingly on uninterrupted assurance mechanisms: satellite positioning, automated safety checks, insurance clearance, regulatory approval, and cross-border data flows. In such an environment, even small disruptions propagate quickly—a delayed signal, a disputed sensor reading, or a challenged compliance report can ripple through logistics chains, construction schedules, and energy exports.

Saudi Arabia does not need to be attacked directly to be constrained. It only needs to face conditions in which its own safeguards hesitate to authorize continued activity. Vision 2030 is producing an economy in which government, finance, logistics, and major projects depend on shared digital coordination layers.[24] While this integration enhances efficiency, it also concentrates risk. Growth-driving systems can slow or stall projects nationwide when those assurance layers are contested.

Qatar's Strategic Embeddedness and Exposure

Qatar occupies a distinct position within the Gulf's strategic architecture as its influence derives from deep integration into global energy, financial, and diplomatic systems.[25] As one of the world's largest exporters of liquefied natural gas and an active mediator in regional conflicts, Qatar functions as both an energy hub and a diplomatic actor.[26] That role depends on continuous external validation. That same integration creates structural sensitivity to disruption. Qatar operates within a multilayered international system in which energy contracts, banking relationships, airspace access, and diplomatic initiatives are mediated through foreign regulators, insurers, investors, and compliance bodies.[27]

Pressure does not need to disrupt Qatar directly. It needs to generate doubt—through headlines, investor anxiety, and coordinated influence operations including disinformation—that triggers regulatory reviews, procedural delays, and institutional slowdowns. The 2022 World Cup illustrates this dynamic as sustained narrative pressure generated regulatory scrutiny, commercial caution, and reputational friction without formal sanctions or legal findings.[28]

Together, Saudi Arabia and Qatar are central to the Gulf's security, energy, and diplomatic architecture that supports US deterrence in the region—making both structurally exposed to assurance-based pressure.[29] These dynamics illustrate how assurance disruption in the Gulf carries direct implications for US military operations dependent on regional stability and uninterrupted mobility.

Strategic Leverage Without Escalation

For decades, US military presence anchored stability in the Gulf.[30] American power now depends in part on commercial logistics networks, insurance markets, regulatory regimes, and navigation systems beyond direct military control—creating opportunities for competitors to impose costs without overt confrontation.[31] US leverage can erode in real time when those systems freeze.

This dynamic helps explain why assurance-based coercion is so attractive to major competitors such as Russia and Iran and why it aligns with the counterspace and electronic warfare investments pursued by China—it is cheap, deniable, and difficult to attribute or punish. By leveraging the same digital insurgency networks used for cybercrime and influence operations, it can generate economic and political disruption without triggering military retaliation.[32] This pattern is already visible beyond the Gulf. Russian satellite interference in European airspace, which has forced civilian aircraft into degraded navigation modes, illustrates how this logic operates beyond the Gulf.[33]

Open-source aviation safety reporting and regional navigation advisories between 2022 and 2025



An aerial view of Qatar's Ras Laffan Industrial City, one of the world's largest liquefied natural gas production and export hubs. As one of the world's largest exporters of liquefied natural gas and a diplomatic actor in the Gulf region, Qatar's integration within a multilayered international system creates structural sensitivity to disruption. As the author states, "Pressure does not need to disrupt Qatar directly. It needs to generate doubt." (Photo from Adobe Stock)

documented repeated GNSS interference affecting civilian aircraft operating across eastern European and eastern Mediterranean air corridors.[34] These incidents forced pilots and air traffic authorities to implement contingency procedures, demonstrating how disruption below the threshold of armed conflict can impose operational constraints without formal escalation.[35]

In this new environment, power is exercised not only through force but also through the ability to shape whether ports operate, planes fly, cargo moves, insurance is extended, and contracts are enforced. When those systems slow or suspend, strategic influence shifts—without military escalation.

Implications for US and Joint Force Operations

The US Army's multidomain operations concept emphasizes contesting freedom of maneuver, sustainment, and decision advantage across domains rather than focusing solely on force-on-force engagement.[36] Assurance disruption operates within this logic by targeting the civilian and commercial systems that enable operational movement rather than military forces themselves, introducing operational friction without direct kinetic engagement.[37] Operational reach and tempo depend on sustained access, basing, and distribution networks across domains. When civilian verification systems hesitate, they compress operational reach and slow tempo without a single shot being fired.

The emergence of digitally mediated assurance disruption presents a growing operational challenge for the United States and partner forces, particularly in regions where commercial infrastructure and military mobility operate within tightly integrated systems. Digital insurgency does not primarily target military forces as it exploits the civilian verification systems—navigation validation, port certification, insurance authorization, airspace clearance, and logistics coordination—on which modern military operations depend.[38]

While no publicly reported US deployment has yet been halted solely due to contested assurance conditions, increasing reliance on commercial ports, contracted shipping, satellite navigation services, and multinational sustainment networks create clear pathways for such constraints in future crises.

Mobility under contested assurance. US joint force operations depend extensively on civilian and



(Photo courtesy of Army Cyber Command)

commercial systems. Access to ports, shipping insurance, satellite-based navigation, airspace approval, and digitally coordinated logistics networks all require continuous verification. Confidence is high when these systems function efficiently, but they default to caution when uncertainty is introduced.

Navigation interference illustrates this vulnerability. Persistent degradation of satellite positioning—even without a confirmed attack—can force aviation authorities, maritime operators, and port regulators to impose precautionary restrictions. Aircraft shift to degraded navigation modes, shipping lanes slow, and insurers reassess exposure. None of these actions constitute hostile engagement, yet their cumulative effect can slow force projection, complicate sustainment, and reduce operational tempo.

For combatant commands operating in the Arabian Gulf—where access to host-nation ports, commercial shipping, and multinational sustainment agreements form the backbone of deployment architecture—this dynamic introduces a new form of operational friction, a reality increasingly visible in Indo-Pacific maritime corridors and European airspace as well. Commanders may retain full combat capability while facing mobility constraints imposed by civilian safety and regulatory systems outside direct military control.

Operational freedom increasingly depends not only on defeating adversary forces but also on whether surrounding assurance systems continue to validate safe movement.

Planning for continuity, not just protection. The operational challenge is not solely protection against attack, it is continuity of operations amid contested verification. This shift requires reframing resilience not as recovery after attack but as sustained authorization under degraded assurance. Resilience must extend beyond recovery after disruption. It must include the ability to operate safely and confidently while assurance systems are degraded or questioned. Planning assumptions that rely on uninterrupted digital validation may unintentionally create exploitable vulnerabilities within otherwise capable forces.

As military and civilian infrastructures converge, battlespace expands beyond networks and territory into the systems that determine whether activity is permitted to continue. Operational freedom may

depend as much on maintaining assurance confidence as on defeating adversary forces. For practitioners, assurance disruption functions less as a conventional attack than as a constraint on mobility and sustainment. It compresses decision timelines, shifts authority toward civilian regulatory and insurance institutions, and generates cascading delays across access, basing, contracting, and movement control. The decisive question for commanders is increasingly not whether forces possess combat capability but whether the systems that authorize movement continue to permit operations. Sustaining tempo when verification systems hesitate is now a core operational requirement.

Operational Adaptation to the Assurance War

For operational commanders, assurance disruption is unlikely to appear first as a cyber incident. It will appear as operational friction: delayed clearances, degraded navigation confidence, halted commercial logistics, or regulatory hesitation affecting mobility corridors. The challenge is maintaining operational tempo when civilian verification systems begin restricting movement despite the absence of confirmed attack. Preparing for this environment requires treating contested assurance conditions as routine planning factors rather than exceptional crises.

For Army commanders, this means planning for contested assurance conditions the same way they plan for degraded communications or contested airspace—anticipating friction in the systems that authorize movement, not just in the domains where forces fight. Operational plans should incorporate degraded-assurance scenarios into exercises and contingency development. Ports may remain physically accessible while commercial certification, navigation validation, or insurance authorization temporarily restrict movement. Training forces operating under verification constraints reduces decision paralysis during real disruptions.

Combatant commands should incorporate contested assurance into major multinational exercises. These scenarios should test degraded navigation confidence, temporary port-access restrictions, and commercial certification delays under realistic operating conditions. Headquarters should establish standing coordination mechanisms with civilian assurance actors, including port authorities, navigation providers, insurers, and aviation regulators, whose risk decisions shape military freedom of maneuver. Early information sharing can prevent localized anomalies from cascading into precautionary shutdowns that delay deployment. Units operating in contested regions at the force level should develop redundant validation pathways for navigation, logistics authorization, and infrastructure access. Multisource verification reduces the risk that limited interference produces disproportionate delay. At the senior level, commanders will increasingly face uncertainty rather than confirmed failure. Doctrine and operational guidance must define acceptable risk thresholds for continued operations under degraded-assurance conditions. Without predefined decision frameworks, organizations default to suspension—granting adversaries leverage without escalation.

Deterrence must also adapt. Adversaries exploit ambiguity when response thresholds are unclear. Demonstrating the ability to sustain operations despite verification disruption denies the leverage adversaries seek through uncertainty. The objective is not to eliminate uncertainty, it is to prevent uncertainty from translating into paralysis. Forces capable of operating confidently within contested assurance environments retain strategic freedom of action. Operational adaptation strengthens resilience, but if civilian assurance systems remain vulnerable to manipulation, freedom of maneuver will remain contingent. Sustaining operational momentum ultimately requires broader institutional resilience across the verification networks on which modern military operations depend.

How to Fight the Assurance War

Operational adaptation addresses only part of the assurance war. Military forces do not control the civilian verification architecture on which modern mobility depends. Energy markets, shipping corridors, insurers, regulators, and navigation authorities collectively determine whether activity continues under contested conditions. If those systems remain vulnerable to manipulation, military adaptation will remain reactive.

Fighting the assurance war therefore requires strengthening assurance itself. Five shifts are essential: building deterrent resilience, engineering hostile-resilient verification, establishing joint validation networks, institutionalizing continuity under uncertainty, and

reforming liability frameworks to protect prudent continuation during disruption.

Deterrent resilience. Assurance systems must demonstrate credible continuity under contestation. Ports, airlines, and energy infrastructure should be structured to continue operating when signals are degraded or disputed rather than defaulting immediately to suspension. The objective is not to eliminate interference but to deny adversaries the expectation that manipulation will automatically halt activity. When precautionary shutdown becomes predictable, it becomes exploitable.

Hostile-resilient verification. Most assurance architectures are designed to manage technical malfunction, not deliberate manipulation. Verification systems must be built with adversarial intent in mind. Redundant validation pathways, anomaly discrimination tools, and cross-domain corroboration mechanisms reduce the likelihood that limited interference produces systemic hesitation. Systems that assume benign error remain structurally vulnerable to strategic exploitation.

Joint validation networks. No single regulator, insurer, or navigation authority should carry the burden of validation alone. Shared data environments and standing coordination mechanisms reduce the risk that localized anomalies cascade into systemic paralysis. Collective verification strengthens confidence and reduces both overreaction and delay.

Continuity under uncertainty. Institutions must define degraded-mode operations in advance. Clear thresholds for continued activity under contested conditions prevent precautionary suspension from becoming the default response. Predefined operating envelopes preserve tempo while maintaining safety margins. The goal is controlled continuation not reckless persistence. When organizations know how to operate under degraded assurance, hesitation loses its coercive power.

Resilient liability frameworks. Legal and regulatory structures must protect operators who continue functioning cautiously during signal manipulation or contested verification events. If decision-makers face punitive exposure for acting amid uncertainty, the rational response will always be suspension. Liability frameworks should incentivize prudent continuation rather than institutional paralysis. The objective of these reforms is not perfect information but functional confidence.

The assurance war is not won by eliminating uncertainty but by preventing uncertainty from translating into systemic hesitation. Its strategic value declines when verification manipulation fails to halt movement, suspend trade, or trigger cascading shutdowns. It is decided not solely by force but decided by whether institutions continue to function under pressure.

What Comes Next

The next Gulf crisis may not begin with military action; it may in fact begin when automated safeguards start making decisions before governments do. In that moment, leaders will confront constraints imposed not by adversary force but by the precautionary systems embedded in their own infrastructure. The strategic question will not be whether power exists but whether it can move. Resilience functions as a form of deterrence in environments where uncertainty triggers hesitation. The challenge for military and civilian leaders is not only to prevent disruption but to ensure that institutional caution also does not replace strategic choice. The assurance war is not coming; it is already here. ■

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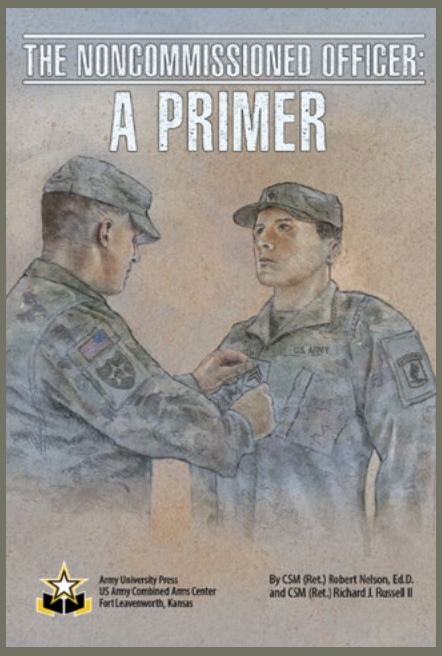
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From the introduction

"In 1999, I was a young enlisted soldier and a combat medic stationed with the 296th FSB, 3 BDE, 2 ID, at (then) Fort Lewis, Washington. Staff Sergeant Rob Nelson came into my organization when I was a Private First Class. It was before 9/11. We were serving a peacetime US Army, and SSG Nelson taught me through his actions and words what a good NCO ought to bring to the human dimension of leading soldiers …

"Over twenty years later, far removed from my active-duty days, I had used my educational benefits to earn a PhD in American history … and in 2020, I found my dream job, amplifying and publishing the scholarship of soldiers, for soldiers. As the managing editor for the Research & Books team for Army University Press (AUP) located in Fort Leavenworth, Kansas, my colleagues include *The NCO Journal*. In 2022, the managing editor of *The NCO Journal* briefed a pending publication, a co-authored essay entitled 'The Understated Art of Knowing When to ST*U' by Sgt. Maj. Robert Nelson & Retired Command Sgt. Maj. Gabriel Arnold. That headline was something that could only have come from my former platoon sergeant. …

"The US Army is a small world and the longer you continue to serve in it or adjacent to it meaningful connections will inevitably resurface in sometimes delightful and serendipitous ways. I asked him to write a book for the noncommissioned officer, a primer, that uses effective narration and storytelling informed by professional experience. I asked him to lift the veil and demystify the contributions NCOs make to mission command and military successes."

To read ***The Noncommissioned Officer: A Primer***, scan the QR code or visit https://www.armyupress.army.mil/Books/.



(AI image generated by Gerardo Mena, Army University Press)

Artificial Intelligence and Professional Authorship

Ownership of Thought When the Machine Is Off

Sgt. Maj. Dustin Shepherd, US Army, Retired

In a recent editorial published in *Military Review*, Col. Andrew Steadman raised an increasingly relevant set of questions for the profession regarding the use of artificial intelligence (AI) in professional writing. As AI tools become more accessible and capable, Steadman noted that three simple but consequential questions emerge: "How prevalent is AI in professional writing today? Where, if anywhere, should AI be prohibited versus disclosed? And what intellectual and institutional risks do we incur if AI becomes a substitute for human cognition and creativity?"[1]

These questions arrive at a moment when AI tools are rapidly entering professional environments across government, academia, and industry. For military professionals, the implications are particularly significant. Writing within the profession is not merely an exercise in communication; it is a mechanism through which leaders clarify thinking, test ideas, share lessons learned, and contribute to the development of doctrine and professional knowledge. Staff estimates, operational plans, professional articles, and doctrinal discussions all depend on the credibility of the reasoning behind the words on the page.

AI can assist this process in meaningful ways. AI tools can help organize information, clarify language, and accelerate early drafts of written work. Used carefully, such tools may improve efficiency and help professionals communicate their ideas more clearly. Yet the increasing capability of these systems raises an important concern: At what point does assistance begin to replace the thinking that professional writing is intended to capture?

The distinction matters because credibility in professional discourse depends on intellectual ownership. When a military professional publishes an article, drafts a staff product, or presents an argument for consideration, the institution assumes that the reasoning behind those ideas belongs to the author. That assumption underpins the trust that allows ideas to be debated, refined, and applied within the profession.

As AI tools become more integrated into writing processes, the profession must focus less on whether these tools exist and more on how they are used. The central challenge is not technological but professional: ensuring that AI remains a tool that supports judgment rather than a substitute for it. AI will inevitably become part of professional writing, but the credibility of military discourse will depend on whether authors retain ownership of the thinking behind the ideas they publish.

The following discussion examines this challenge by exploring the role of AI as a professional reference tool, the distinction between assistance and substitution in professional thinking, and the importance of preserving intellectual ownership in military discourse.

AI as a Professional Reference Tool

Before the widespread digitization of medical information, physicians routinely relied on a large reference volume known as the *Physicians' Desk Reference* (*PDR*), the predecessor to today's digital Prescriber's Digital Reference.[2] The book was a comprehensive compilation of pharmaceutical information that documented drug interactions, recommended dosages, contraindications, and treatment considerations. It represented an enormous body of accumulated medical knowledge organized into a single reference source that physicians could consult when evaluating treatment options.

Importantly, the *PDR* was never intended to replace the physician's judgment. It served as a repository of patterns derived from clinical research and pharmaceutical testing. Physicians consulted the reference to inform their decisions, but diagnosis and treatment remained the responsibility of the practitioner. The value of the *PDR* depended entirely on the physician's ability to interpret the information in the context of a specific patient, weigh competing considerations, and apply professional judgment.

Military professionals rely on similar reference systems. Field manuals, doctrinal publications, and professional journals collectively function as repositories of institutional knowledge derived from decades of operational experience, experimentation, and analysis. These publications document patterns of observed military behavior—how units maneuver, how staffs plan, how operations are synchronized, and how leaders manage complex environments. They provide leaders with a body of professional knowledge that informs decision-making across a wide range of operational contexts.

Like the *PDR*, however, military publications are not substitutes for professional judgment. Doctrine provides a framework for thinking about military problems, but leaders must interpret and adapt that framework to the conditions they face. Commanders and staff officers routinely consult doctrine while planning operations, but they also recognize that no publication can account for every operational variable. The responsibility for

decision-making ultimately rests with the leader applying the guidance, not with the publication itself.

AI tools can be understood in a similar way. Modern AI systems are capable of synthesizing vast quantities of information and identifying patterns within that information at remarkable speed. In many respects, these systems function as dynamic repositories of patterns drawn from large bodies of text and data. Used responsibly, they can assist professionals by organizing information, suggesting alternative ways to frame a problem, or helping clarify written arguments. As with any reference tool, however, the value of AI depends on the professional judgment of the individual using it.

The critical distinction, therefore, is not whether AI tools are consulted during the writing process, but whether the professional retains ownership of the thinking that follows. Like doctrine or medical reference manuals, AI can support professional reasoning. It cannot replace the responsibility of the practitioner to understand, defend, and apply the ideas being presented.

Assistance and Substitution in Professional Thinking

Understanding the role of AI in professional writing requires distinguishing between two very different uses of the technology: assistance and substitution. When used appropriately, AI can assist professionals in organizing ideas, identifying gaps in an argument, or refining language for clarity. In this role, the technology functions much like other professional reference tools, supporting the author's thinking while leaving responsibility for the ideas firmly with the writer.

Many professionals already use AI in this way. A leader drafting an article or staff paper may consult an AI tool to explore alternative ways of framing an argument or to clarify complex sentences that obscure the intended message. Used responsibly, the technology can improve the efficiency and readability of professional writing without altering the fundamental ownership of the ideas being expressed.

Concern arises when the relationship between the professional and the tool begins to shift. AI systems are capable of generating complete passages of text that appear coherent and persuasive, often within seconds. While this capability can be useful for brainstorming or exploring unfamiliar topics, it also creates the possibility that the technology may begin to substitute for the author's reasoning rather than support it.

In professional military discourse, this distinction carries significant implications. Written products within the profession often represent more than personal expression; they contribute to the collective body of knowledge from which leaders draw insight when confronting complex operational problems. Staff estimates, doctrinal discussions, and professional articles all form part of a broader intellectual process through which the institution evaluates ideas and refines its understanding of warfare and leadership. The credibility of that process depends on the expectation that the ideas presented reflect the considered judgment of the author.

Professional military education has already encountered early versions of this challenge. During my time as a teaching fellow at the US Army Sergeants Major Academy from 2021 to 2023, faculty initially struggled to interpret the use of digital writing tools such as Grammarly, which can automatically suggest revisions to grammar, structure, and tone. Early reactions often treated the tool as a form of academic dishonesty because instructors had not yet developed a shared understanding of how the technology functioned. Over time, however, it became clear that Grammarly operated less like a ghostwriter and more like an editing assistant. By the time I departed the academy, no formal policy prohibited its use; instead, instructors relied on professional judgment when evaluating whether a student's work reflected their own thinking.

This experience illustrates the broader challenge AI now presents to professional writing. When technology assists the author's thinking, the resulting work remains an extension of the author's reasoning. When technology substitutes for the author's thinking, the product may appear polished but lacks the intellectual foundation that gives professional writing its credibility.

Recognizing the difference between these two uses of technology is therefore essential for maintaining the integrity of professional discourse. The challenge for the profession is not simply determining whether AI tools are used in the writing process, but ensuring that responsibility for thinking, reasoning, and defending ideas remains with the author.

> Coherent logic, consistency of argument, accurate representation of facts, and the integration of verifiable knowledge all serve as indicators that the ideas presented reflect the considered judgment of the author.

A Practical Standard for Intellectual Ownership

As writing technologies become more capable, the profession will need practical ways to evaluate whether professional writing still reflects the judgment of the author. Policies and disclosure requirements may play a role, but they cannot fully resolve the underlying issue. Ultimately, the credibility of professional writing rests on whether the individual presenting the ideas understands and accepts responsibility for them.

In practice, this question often becomes clear when the written work is discussed rather than simply read. Military professionals are accustomed to this dynamic. Plans are briefed, staff estimates are questioned, and professional arguments are debated among peers and leaders. In these settings, written products rarely stand alone; they are part of an ongoing conversation in which the author is expected to explain and defend the reasoning behind the work.

It is in these moments that the difference between technological assistance and intellectual substitution becomes visible. When a professional has used technology to clarify or refine their thinking, they can usually explain how the argument developed, describe why certain conclusions were reached, and expand upon the ideas beyond what appears on the page. The written product serves as a reflection of reasoning that continues even when the document is closed.

When technology substitutes for the author's thinking, however, the conversation often stops at the text itself. The individual may be able to restate what is written but struggles to explain the logic behind the argument or extend the discussion beyond the words that appear in the document. The ideas exist on the page, but they are not fully owned by the person presenting them.

A useful way to think about this distinction is through what might be called the "machine turned off test." If the computer is turned off and the tool removed, can the author still explain, defend, and extend the reasoning behind the written product? If so, the technology likely served as an aid to the author's thinking. If not, the technology may have begun to substitute for the intellectual work that professional writing is intended to capture.

This standard does not require the profession to reject emerging technologies, nor does it assume that the use of AI tools is inherently problematic. Instead, it reinforces a long-standing expectation within the military profession that individuals who present ideas must be prepared to explain and defend them. Technology may assist in expressing those ideas more clearly, but it cannot assume responsibility for the reasoning behind them.

While this standard is effective in settings where ideas can be discussed and challenged directly, it does not fully address how intellectual ownership is assessed in published work. A related challenge emerges when considering how intellectual ownership can be established in published work where direct engagement with the author is unlikely. In staff settings or classroom environments, understanding can be tested through discussion, questioning, and refinement of ideas. Published writing, however, does not offer readers the same opportunity. Once an article is in circulation, the audience must assess it without the benefit of dialogue with the author. This raises a legitimate concern in an era of AI: If the process by which a document is produced is not visible, on what basis can intellectual ownership be trusted?

In practice, professional military writing has never relied on direct verification of authorship at the level of composition. Editors and readers do not observe how an article is written, nor do they evaluate each step of its development. Instead, credibility is inferred from the internal quality of the work itself. Coherent logic, consistency of argument, accurate representation of facts, and the integration of verifiable knowledge all serve as indicators that the ideas presented reflect the considered judgment of the author. In evaluating

written discourse, these indicators are often visible through the clarity of the argument, the alignment of ideas across sections, the depth of understanding demonstrated, and the relevance of the content to the problem at hand. These same standards are routinely applied to assess staff products, professional publications, and doctrinal contributions across the force.

This approach reflects a broader characteristic within military writing. Institutional publications such as doctrine, policy, and regulations derive authority from the organizations that produce them. Trust in those documents is granted in advance based on the credibility of the institution. Professional discourse, by contrast, operates differently. Articles and essays do not carry the same presumption of authority and must instead earn credibility through the strength of their reasoning. Readers evaluate clarity, relevance, and consistency with known facts to determine whether the argument is sound. In this context, intellectual ownership is not confirmed through the visibility of the writing process but through the quality and integrity of the finished work.

AI complicates the question of authorship by obscuring how text may have been generated or refined. It does not, however, change the fundamental standard by which professional writing is judged. The profession has long accepted that absolute certainty regarding the process of composition is neither practical nor necessary. What remains essential is accountability. The author whose name appears on a document is responsible for the ideas it contains, regardless of the tools used in its preparation. When a written product demonstrates clear reasoning, aligns with verifiable knowledge, and withstands critical scrutiny, it provides sufficient basis for the profession to attribute intellectual ownership to the author and to engage the ideas on their merits.

Beyond Prohibition: Developing AI Literacy in the Profession

Building on the questions raised by Steadman, discussions surrounding AI have increasingly focused on where the technology should be restricted, disclosed, or integrated within professional military writing.[3] While concerns about misuse are understandable, framing the issue primarily as a matter of prohibition risks overlooking the more important challenge facing the profession. AI is advancing rapidly and becoming increasingly accessible across both civilian and military environments. Attempting to prevent its use entirely is unlikely to succeed and may ultimately leave professionals less prepared to understand and govern the technology effectively.

A more productive approach is to recognize that AI will inevitably become part of the professional landscape and prepare leaders accordingly. Just as the military has historically incorporated new technologies into training and doctrine, the profession must now develop a baseline level of AI literacy across the force.[4] Leaders should understand what these systems can do, where their limitations lie, and how their outputs should be evaluated critically.

Developing this literacy is not simply a matter of technical familiarity; it is equally a matter of professional responsibility. Military professionals are expected to exercise judgment when interpreting doctrine, evaluating intelligence, and making operational decisions. AI should be approached in the same way—as a tool that may assist analysis and communication but does not replace the responsibility of the leader to think critically about the information being presented.

The military has repeatedly encountered similar moments as emerging technologies moved from experimental capabilities to routine tools used by operational units. During my first deployment following the attacks of 11 September 2001, I was assigned to an operating base that housed one of the early Predator unmanned aircraft systems. At the time, such platforms represented a relatively new capability concentrated at higher echelons of command. Within a decade, however, small unmanned aircraft systems such as the Raven had been fielded to infantry platoons, providing small-unit leaders with their own organic reconnaissance capability.[5] What began as a specialized technology quickly became a routine component of tactical operations.

AI appears to be following a similar trajectory. Many commercially available unmanned systems now incorporate AI-enabled software to assist with navigation, targeting, and data processing. As these capabilities continue to evolve, it is reasonable to expect that AI-assisted systems will increasingly appear across military platforms and decision-support tools.

For this reason, the profession must focus not only on regulating emerging technologies but also on preparing leaders to understand and evaluate them.

Just as commanders and staff officers learned to integrate unmanned systems into operational planning, the profession must now ensure that leaders possess the literacy required to use AI responsibly. Doing so will help ensure that AI remains a tool that enhances professional capability rather than one that undermines the intellectual foundations of military decision-making.

Conclusion

The questions raised by Steadman reflect a broader conversation now unfolding across the profession. AI is entering environments in which military professionals think, write, plan, and communicate. Like other technological innovations that preceded it, AI will likely become increasingly integrated into the tools leaders use to analyze problems and express ideas. The challenge facing the profession is therefore not simply whether these tools exist, but how they are used.

Throughout its history, the military has adapted to new technologies by pairing technical capability with professional judgment. Radios expanded command and control but did not replace leadership. Unmanned systems extended battlefield awareness but did not eliminate the need for commanders to interpret the information they produced. AI should be approached in the same way. It may assist professionals in organizing information, refining arguments, and improving the clarity of written communication, but it cannot assume responsibility for the thinking behind those ideas.

Professional credibility ultimately depends on intellectual ownership. When leaders publish an article, present a staff estimate, or contribute to doctrinal discussion, the institution assumes that the reasoning behind those ideas belongs to the individual presenting them. Maintaining that expectation will remain essential as writing technologies become more capable.

For this reason, the profession must focus on developing AI literacy while reinforcing the enduring standards of professional responsibility. Leaders should understand how these systems function, recognize their limitations, and evaluate their outputs critically. At the same time, the expectation remains unchanged: those who present ideas must be prepared to explain and defend the reasoning behind them.

AI may increase the efficiency with which ideas are written and shared, but credibility in professional discourse will always depend on the human judgment behind those ideas. The profession therefore faces a familiar task—integrating new technology while preserving the intellectual responsibility that defines professional military writing.

In the end, the simplest measure of intellectual ownership remains straightforward: when the machine is turned off, the thinking should remain. ■

Notes

1. Andrew Steadman, "Letter from the Editor," *Military Review* 106, no. 1 (January-February 2026): 1, https://www.armyupress.army.mil/Journals/Military-Review/English-Edition-Archives/January-February-2026/Letter-from-the-Editor/.

2. *Physicians' Desk Reference*, 71st ed. (Physicians' Desk Reference Inc., 2017); "Prescriber's Digital Reference," ConnectiveRx, accessed 30 June3 2026, https://www.pdr.net/.

3. US Department of Defense (DOD), *Summary of the 2018 Department of Defense Artificial Intelligence Strategy* (DOD, 2018), 7, https://media.defense.gov/2019/feb/12/2002088963/-1/-1/1/summary-of-dod-ai-strategy.pdf.

4. DOD, *Summary of the 2018 Department of Defense Artificial Intelligence Strategy*.

5. "Small Unmanned Aircraft Systems Raven," Program Executive Office Aviation and UAS Project Office, accessed 28 May 2026, https://safety.army.mil/Portals/0/Documents/ON-DUTY/AVIATION/RAVEN/Standard/Raven_9June2011.pdf.

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Understanding Death Is the Best Way to Prevent It

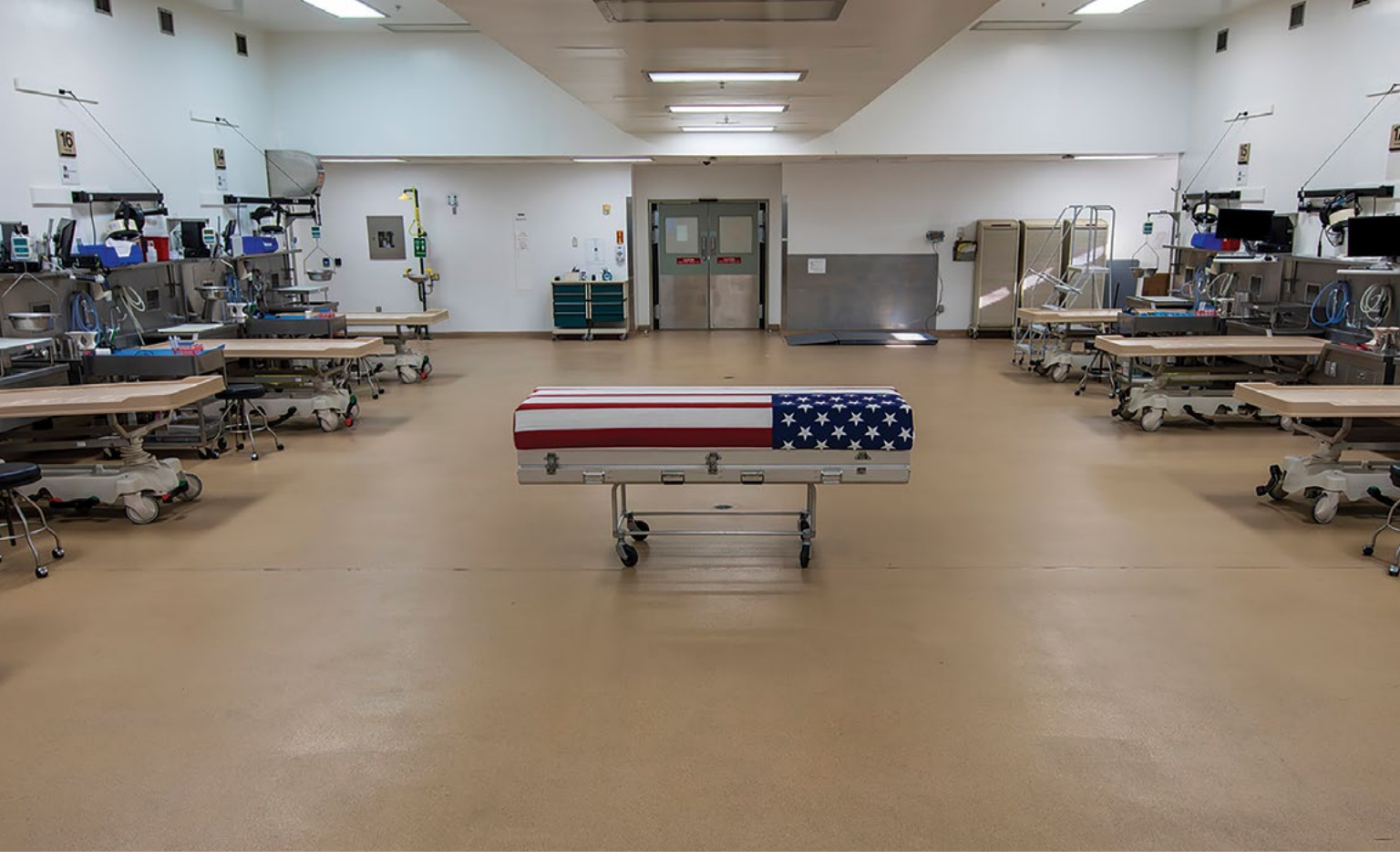
Postmortem Data Is a Strategic Enabler for the Joint Force

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Understanding how and why service members die is not just a medical necessity, it is a multidomain strategic imperative. In recent conflicts, detailed postmortem analysis of combat deaths uncovered systemic gaps in equipment, tactics, and medical care that would have otherwise gone unrecognized. The postmortem data collected from both individual cases and epidemiologic studies led to insights that directly drove improvements in warfighter survivability.[1]

One of the most powerful academic articles published during the Global War on Terrorism was aptly titled "Death on the Battlefield."[2] The authors analyzed autopsy reports from the Armed Forces Medical Examiner System (AFMES) and demonstrated that 87 percent of deaths on the battlefield occurred in the prehospital setting. Additionally, 24 percent of these prehospital deaths were from potentially survivable injuries; hemorrhage was the primary physiologic mechanism of death in 91 percent of this subset. This pioneering article incited change across the entire DOTMLPF-P (doctrine, organization, training, materiel, leadership and education, personnel, facilities, and policy) spectrum and contributed to the United States eventually achieving the lowest case fatality rate in its history.[3]

The current system for collecting and analyzing fatality data in the Department of War (DoW) was built on experiences from the conflicts in Afghanistan and Iraq. Postmortem efforts were shaped primarily by counterinsurgency operations, low-intensity conflict, and an operational fatality management construct that allowed rapid repatriation of the deceased to the Dover Port Mortuary for postmortem examination. However, there is risk in the current system impacting the clinical, operational, and strategic domains. The fundamental risk is that the DoW lacks a policy for postmortem data collection, resulting in an absence of requirement and ultimately a lack in capability.



Shown here is the autopsy suite of the Armed Forces Medical Examiner System laboratory, where deceased US service members undergo full autopsy examinations. (Photo by Staff Sgt. Noah D. Coger, US Air Force)

Other considerations when caring for fallen service members are the expectation of the US public regarding the DoW's moral obligation and the effect of casualties on US morale. Military recruitment and reenlistment numbers may be impacted by the perception of casualties not receiving dignified care of their remains. Public support for the human toll of large-scale combat operations (LSCO) will wane if families of the fallen are not provided timely information as to how their loved ones died or if they do not receive a death certificate to close out estates and initiate benefits. Death certification and family correspondence may seem like administrative functions that do not directly support a warfighting effort, but their importance cannot be undervalued when considering the general public's support of combat activities. Prompt postmortem data collection can help address these sensitive future challenges.

Regardless of the nature of the next conflict, fundamental questions from commanders, families, and the public at large will remain unchanged. What happened? How did the service member die? Did they suffer? How can we prevent other deaths? The inability to answer these questions will increase risk to service members and negatively impact survivability, effectiveness of operations, and the support of the US public. If the joint force is to be data informed, postmortem data collection and analysis should be prioritized, resourced, and institutionalized.

Survivorship Bias Risk

In an era of peer competition with distributed operations in potentially contested environments, not only will management of the fallen be challenging, but the subsequent Military Trauma Mortality Reviews (MTMR) may not be feasible without complete postmortem data to inform decision-making. Without postmortem data, the risk to the DoW is clear: succumbing to survivorship bias.

"Survivorship bias is a cognitive shortcut that occurs when a successful subgroup is mistaken as the entire group, due to the invisibility of the failure subgroup."[4] A well-known operational example of survivorship bias is from World War II, where unsustainable numbers of bomber aircraft were being shot down. Those that

A Brigade Commander's Perspective

Col. Scott Wence

With several friends in Arlington National Cemetery, the thought that their deaths could have been prevented by information we already had is a heavy burden. As a commander, I was unaware that detailed postmortem data from our casualties even existed—data that could have exposed vulnerabilities in our tactics, medical processes, and equipment in near real-time.

This isn't just a documentation failure; it's an operational one. Every casualty provides a lesson, and postmortem data is the textbook. It's the linchpin of adaptive leadership, transforming raw loss into actionable intelligence that saves lives and preserves combat power. When frontline leaders don't have this critical analysis, we are destined to repeat mistakes.

We owe it to every soldier, past and present, to close this gap. We must ensure every sacrifice serves to protect the living by directly informing our decisions on unit training, equipment, and tactical planning.

returned from missions were riddled with bullet holes in the wings and fuselage. To improve survivability, the solution was to bolster the armor of those bullet-laden areas, a classic example of survivorship bias. Statistician Abraham Wald posited that the damage to the wing and fuselage was not the problem, but the true vulnerabilities were in the areas of the returning aircraft without damage. The cockpit and engines needed to be bolstered with additional armor to improve aircraft survivability.[5]

Survivorship bias in combat deaths can lead to wrong conclusions, clinically and operationally. A clinical example is seen in the data coming out of Ukraine, which indicates a greater than 95 percent survivability rate of combat casualties; however, the data is only from hospitals and does not take into consideration prehospital deaths.[6] Without full-spectrum trauma data, especially postmortem data, the trauma system cannot evolve to improve aspects of prehospital care. Missing data risks incorrect solutions and misapplication of precious resources to solve the wrong problems. Since the DoW is not postured to maintain postmortem data collection and analysis in LSCO, battlefield survivorship bias will exist across multiple domains, with no means to understand the true opportunities for improvement. As a quote commonly attributed to philosopher and educator John Dewey states, "We do not learn from experience … we learn from reflecting on experience."[7]

The "Legacy" AFMES Model for Postmortem Data Collection

AFMES was established in 1988 and thus did not exist during previous large-scale conflicts.[8] During combat operations from 2001 to 2025, AFMES developed a system capable of performing a complete postmortem examination of every combat fatality. Upon death, the fallen were rapidly repatriated to Dover Air Force Base, Delaware. Each fatality was then identified scientifically (using fingerprints, dental records, or DNA methods) and underwent postmortem full-body computed tomography (CT) scan, a complete autopsy by a forensic pathologist with photographic documentation, and toxicology testing. Forensic evidence (ballistic material, blast fragments, and personal protective equipment) was collected and provided to intelligence and investigative authorities for analysis. In many cases, explosions or high-energy events resulted in fragmented remains that required DNA testing for both identification and reassociation of remains. Finally, an autopsy report was generated that documented injuries and evidence of medical interventions and provided a medical examiner's (for the purposes of this article, synonymous with forensic pathologist) determination of the cause and manner of death.

This AFMES legacy system exists today and remains capable of collecting a gold-standard postmortem data set on combat fatalities but is entirely dependent on relatively low case volume. The legacy model cannot be scaled to meet the high-volume demand of potential future operating environments in which vast distances and contested logistics will hinder centralized postmortem analysis. Without rapid repatriation of remains to Dover Air Force Base, the AFMES will be unable to adequately perform their unique mission of generating postmortem data; therefore, a vital data set within the DoW will be diminished or lost.

The Battlefield Learning Health System

Postmortem data feeds a critical DoW asset that directly improves warfighter survivability. The Joint Trauma System (JTS) is a continuous battlefield learning health system that uses clinical and postmortem data to identify and close gaps in DOTMLPF-P decisions across the joint force.[9] Through this threat-informed and data-driven process of medical performance optimization, the JTS captures and transforms clinical data into actionable insights to improve all aspects of the battlefield trauma system. Examples of medical performance optimization in action include the Golden Hour Policy, Tactical Combat Casualty Care (TCCC) Guidelines, prehospital blood, battlefield walking blood banks, burn care management, and hypothermia prevention kits.[10] The JTS did not exist when the United States entered the Global War on Terrorism. Thus, the early years of the conflict had unnecessary fatalities from potentially survivable injuries as there was no trauma system in place to analyze and implement medical processes and countermeasures. The joint force did not go to war with a functioning JTS, and much like the legacy AFMES model, the JTS grew out of necessity and need.

Together the JTS and AFMES have aligned forensic postmortem data with combat-casualty-care performance measures, establishing the MTMR as a gold standard of evidence-based practice and data-driven medical performance optimization. Without an updated, modern, and distributed solution for postmortem data collection, the battlefield learning health system will lose access to one of its most critical data streams.

Mortality Reviews: JTS-AFMES Partnership as a Model for Combat Support

Building on the model of a battlefield learning health system, the JTS and the AFMES have developed ad hoc MTMRs to better synchronize and integrate clinical and forensic data into the DoW trauma registry. By combining forensic autopsy data, postmortem radiographic imaging, and clinical case reviews with a multidisciplinary panel of medical experts (e.g., trauma surgeons, forensic pathologists, prehospital specialists), the MTMR improves the completeness and fidelity of causes and mechanisms of battlefield deaths. Clinical care is examined to ensure it met standards set by the JTS TCCC Guidelines and JTS Clinical Practice Guidelines.[11] Injury survivability is determined, and the death is deemed preventable or nonpreventable. While injuries are survivable or nonsurvivable, death preventability includes the tactical and operational conditions of the battlefield that may limit medical care. So, there are times that a survivable injury results in a nonpreventable death secondary to battlefield conditions.

However, the success of MTMRs is constrained by the same challenge facing broader postmortem data efforts: there is no overarching policy or programming for mortality reviews. MTMRs were done by JTS and AFMES because it was needed to improve battlefield survivability. Without a program of record, postmortem data analysis and the determinations of survivability and preventability of battlefield deaths rely on personality-driven coordination between JTS and AFMES physicians. Volunteerism should not be the engine for this critical (yet unofficial) function.

Leveraging the Intelligence Community: JTAPIC and Beyond

The Joint Trauma Analysis and Prevention of Injury in Combat (JTAPIC) program further illustrates the power of postmortem data analysis and cross-domain integration. JTAPIC was created to bring together the medical, logistics, and intelligence communities to analyze injury mechanisms and combat incidents in a holistic manner.[12] JTAPIC has provided timely and actionable incident reports, which combine operational, medical, and forensic intelligence (including ballistic evidence) to inform personal protective equipment development, vehicle design, and operational tactics. For example, analysis of fragment patterns and injury severity helped drive both helmet redesigns and optimized body armor plate coverage. The JTAPIC data and this multidisciplinary and cross-domain approach ensures that battlefield deaths are not merely cataloged but actively analyzed for data-driven lessons and actionable process improvement.[13]

From Data to Action: Tourniquets and V-Hulls

The value of postmortem data collection and analysis by AFMES, JTS, and JTAPIC can be illustrated through

A Prehospital Provider's Perspective

Maj. Michael Broussard, DMS, PA-C

As a prehospital provider, I have treated casualties under night vision and under fire, making irreversible decisions in seconds with incomplete information. In 2017, I lost two service members. Once they were packaged and moved off the objective, they entered a formal chain of custody. I never saw them again. I never saw them under white-light conditions, nor did I fully understand the precise mechanisms that ended their lives. What remained were impressions formed in chaos and the profound uncertainty that follows loss at the point of injury.

Several weeks later, I received the call from the Joint Trauma System and the medical examiner. The process began with accountability: I was asked to describe what I saw and what I did, the injuries as I perceived them, the interventions performed, the timelines, the constraints. Only then were the findings delivered. The report was clinical and precise. Assumption was finally replaced with fact. The full extent of injuries was realized. Survivability was clarified. Whether confirming nonsurvivable injury or identifying gaps, the information resolved doubt and provided direction.

MTMR is not retrospective critique; it is what responsible institutional learning looks like. We have an imperative duty to ensure that every fallen service member's sacrifice contributes to protecting those who follow, a duty owed not only to the service member but also to their family and to their teammates who continue the mission. Postmortem data transforms loss into doctrinal training, materiel, and policy improvement. That responsibility is collective. It is enduring. And it is not optional.

lifesaving changes impacting the entire joint force. In the early years of the Afghanistan and Iraq conflicts, some individuals died from injuries limited to the extremities. It was only through this MTMR process that patterns of potentially survivable injury from extremity hemorrhage became apparent. This led directly to the widespread fielding of modern combat application tourniquets and their inclusion as a pillar of TCCC doctrine. With the proper tools and training, troops on the ground were empowered to rapidly control hemorrhage in the first minutes after injury. The results were dramatic, and tourniquet use significantly reduced mortality from extremity hemorrhage when applied early and effectively.[14] Subsequently, among US special operations military personnel injured in combat, no fatality has died exclusively from extremity hemorrhage since 2006; this powerful force-maintaining intervention was made possible by postmortem data and analysis.[15]

Postmortem data also influenced survivability efforts beyond the medical domain. The evolution of Mine-Resistant Ambush Protected vehicles offers a clear example. Early designs provided increased protection over up-armored HMMWVs, yet fatalities from underbody blasts persisted. Autopsy examination and postmortem data analysis of this cohort revealed common patterns of pelvic and lower extremity injuries. These findings, in conjunction with operational and test data, informed the decision to adopt V-hull configurations that were better able to deflect blast energy. The design change significantly improved occupant survivability and remains a foundational feature of many modern military vehicles.[16] These force-enabling outcomes were not achieved by accident. They were the result of deliberate processes and multidomain, data-driven analysis of combat deaths.

Challenges in Sustaining the MTMR Process: No Policy = No Scalable Capability

The ability to adapt postmortem data collection and analysis to future large-scale and distributed operations will need deliberate planning for implementation.

No policy. There is no policy for postmortem data collection and analysis across the DoW. Part of the challenge is delineating responsibility outside of the centralized, legacy AFMES model. AFMES falls under the control of the Defense Health Agency, but during military operations fatality management is a military service responsibility. Postmortem data collection is currently a unique AFMES capability. Overarching policy and designation of a lead organization for postmortem data collection must bridge the historical gap between the Defense Health Agency's combat support functions and the services' operational responsibilities. The value and utilization of postmortem data extends across multiple domains (medical, logistics, intelligence, and operations), and policy must provide clear ownership to drive the requirements needed to enable sustained success and avoid structural fragmentation leading to missed opportunities, duplicated efforts, and data silos.



A Mine Resistant Ambush Protected (MRAP) vehicle lies in disarray after being struck by an improvised explosive device while returning from a humanitarian mission to give medical attention to women and children at a small Afghan village on 12 August 2009. Airman 1st Class Destiney Dowdy, a medic with Provincial Reconstruction Team Nuristan, was in the MRAP when it was hit and was able to perform life-saving aid to injured soldiers who were also on the convoy mission. (Photo by Airman 1st Class Destiney Dowdy, US Air Force)

No scalable capability. Without guiding policy to drive requirements, the DoW will continue to depend on the legacy AFMES model, which assumes low-volume casework, permissive airspace, and robust transport infrastructure to enable rapid repatriation of remains. In any future conflict, fatality evacuation timelines may be prolonged, and centralized autopsy facilities may be unreachable and/or demand may quickly exceed capacity. Novel approaches for forward-capable postmortem data collection solutions should be investigated and developed for implementation by the services when the legacy model is overwhelmed or unable to function due to logistical constraints.

Proposed Solutions to Enable Future Postmortem Data Collection

To meet the challenges of postmortem data collection in LSCO or major combat operations, efforts should be established to accomplish the following:

- Improve awareness and establish formal policy for postmortem data, to include capabilities for data collection (in both operational and training environments), the MTMR process, and other stakeholder analysis.
- Designate a lead organization to synchronize policy, doctrine, and data integration with a focus on accessibility for both medical and operational lines of effort.
- Research, develop, and invest in forward-capable systems that can operate at or near Role 2 and Role 3 locations, mortuary affairs collection points, or theater mortuary evacuation points to enable early collection of perishable postmortem data and to ease contested logistical burdens. For example, a ruggedized, high-throughput, deployable CT scan system with artificial intelligence-assisted image analysis would allow for scalable, rapid collection of postmortem data.

Conclusion: Postmortem Data and the MTMR are Enablers of the Future Fight

Understanding mortality is not solely a medical responsibility; it must be a total force endeavor. The systematic examination of combat deaths provides

critical insight into how and why service members die, transforming individual loss into institutional learning resulting in improved warfighter survivability. Detailed postmortem analyses and the MTMR have historically revealed deficiencies across the battlefield continuum of care, from point of injury to evacuation and definitive treatment. These analyses have uncovered systemic gaps not only in medical capability but also in equipment design, operational tactics, and command decision-making. When properly analyzed and integrated, postmortem data becomes a type of operational intelligence that informs force development, enhances medical readiness, strengthens combat effectiveness, and increases warfighter survivability.

As the joint force continues to prepare for LSCO, leaders will assume higher risk if they fail to understand combat death. Investments in maximizing lethality and survivability must be unbiased and data driven. Avoiding survivorship bias cannot be accomplished without postmortem data and MTMRs. There is an unbroken continuum among mortality, survivability, and lethality. Reducing mortality increases survivability, and improved survivability directly enhances lethality and the capacity of the joint force to fight, endure, and prevail.

The United States has led the world in combat casualty care due in no small part to its willingness to confront hard truths and learn from loss. That legacy must continue. Policy for postmortem data collection and analysis, integrated into a threat-informed battlefield trauma system, is essential to preserving and expanding our historical advantage on the battlefield, because if we do not understand death, we cannot prevent it. And if we cannot prevent death, we will lose more than lives—we will lose a strategic advantage in the future fight. ■

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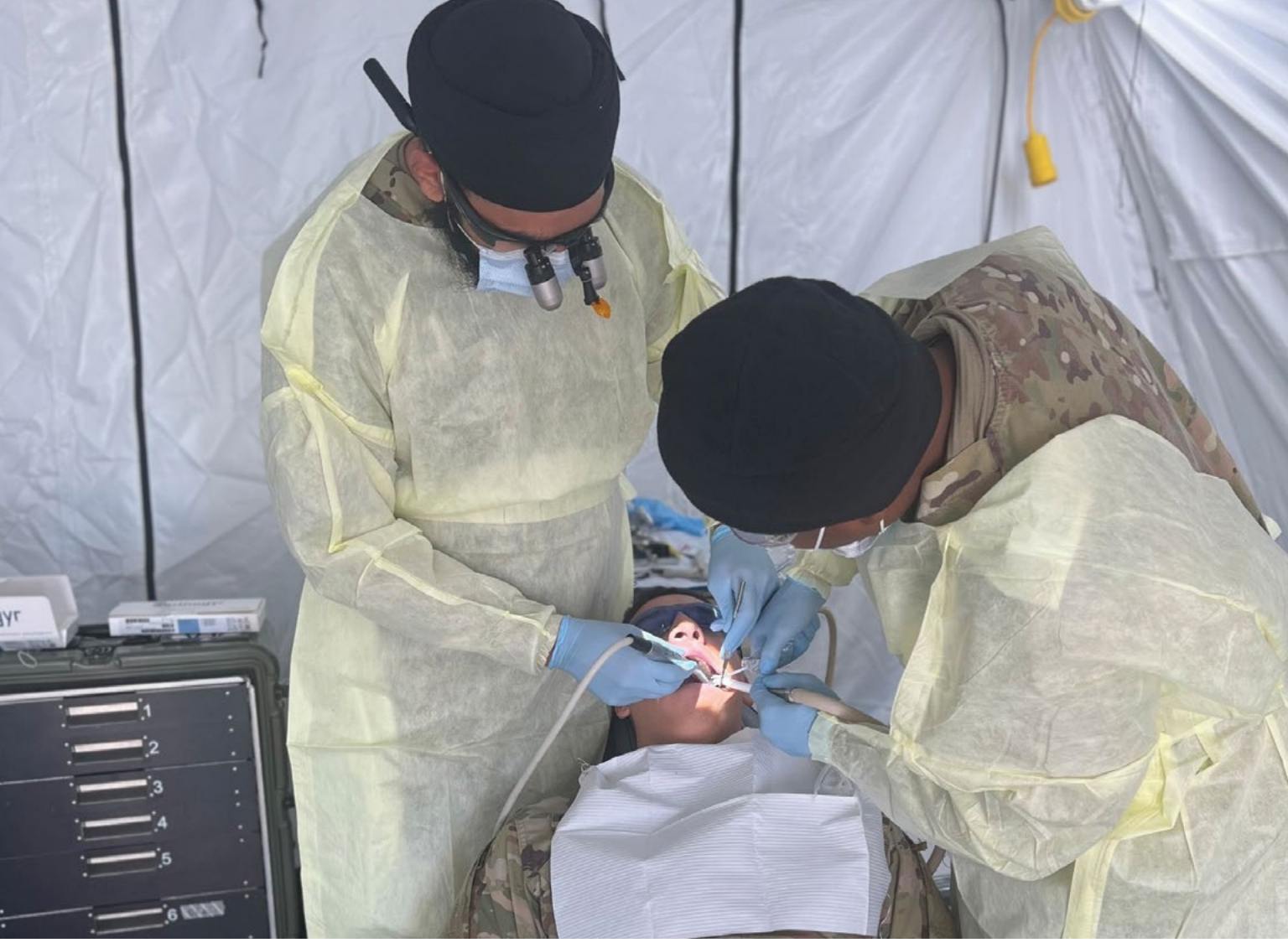
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The Hidden Casualty Load

Integrating Dental Emergencies into Large-Scale Combat Operations Planning

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In March 2025, retired Air Force Maj. Gen. Paul Friedrichs warned the US Senate Committee on Armed Services about the growing number of soldiers "pulled off the line" for dental emergencies—an issue he identified as especially acute within the National Guard and Army Reserve.[1] Dental readiness has long challenged modern militaries, but its consequences are magnified in large-scale combat operations (LSCO). Disease and nonbattle injuries (DNBI) have compromised combat effectiveness throughout military history. Modern medicine and public health practices like food and water sanitation have minimized the threat to life, but DNBI remains a significant cause of morbidity. Dental emergencies are a subset of DNBI in a deployed theater of operations that can degrade the combat strength of brigades and divisions. Dental teams play a critical role in sustaining the force by treating emergencies swiftly and returning soldiers to duty, thereby preserving combat power and operational momentum. Yet, despite this essential contribution, deployable dental capabilities continue to decline. This trend underscores the urgent need for accurate dental emergency modeling in war-gaming to emphasize the importance of DNBI and dental emergencies. To give



A patient receives dental care at a field dental clinic during a January 2023 field training exercise at Fort Bragg, North Carolina. (Photo by Lt. Col. Gamal Baker, US Army)

commanders a true picture of medical risk in future conflicts, Warfighter exercises (WFX) must incorporate historically grounded casualty estimates that fully account for DNBI.

The Operational Challenge for Commanders

In contested logistics environments characteristic of LSCO, commanders face stark choices about scarce evacuation assets. A dental emergency—while not life-threatening—creates the same immediate non-mission-capable status as a combat injury. A dental emergency is any acute oral condition impairing a service member's mission readiness, operational performance, or ability to consume normal daily rations, which increases the risk of mission failure. Severe dental pain renders a soldier unable to operate weapons, make tactical decisions, or sustain physical exertion. Oral conditions that are expected to result in a dental emergency within twelve months are designated as Dental Readiness Category 3 (DRC 3).[2] Per Department of Defense Instruction 1332.45, *Retention Determinations for Non-Deployable Service Members*, DRC 3 is now an individual medical readiness deficit rather than a barrier to deployment; thereby, allowing commanders to deploy their soldiers in this status.[3] Consequently, medical operations planners must recognize the increased likelihood for dental emergencies in theater and ensure robust, forward-deployed dental support to prevent mission degradation. Unfortunately, current WFX models don't force commanders to make triage decisions

involving dental casualties because these casualties aren't included in scenarios.

Unrealistic expectations regarding casualty estimates for wounded in action (WIA) and DNBI were discussed during the 2024 LSCO Symposium. While acknowledging the severity of projections, senior leaders admitted training objectives must be balanced with the need to train fires and maneuver.[4] But the methodology and design of WFXs must include all types of casualties and with the appropriate proportion. Experiments that solely focus on the number of WIA will invariably produce flawed results since they only consider a portion of the total expected casualties. What distinguishes dental casualties is their return-to-duty (RTD) potential. When dental capability exists, 96 percent of dental casualties RTD within one to two visits.[5] This contrasts sharply with many surgical casualties requiring prolonged recovery. In LSCO's attritional environment, the ability to restore combat power within hours rather than days has strategic value—but only if planning accounts for this requirement.

Dental Emergencies: The Persistent Problem

The history of warfare has shown that armies incur more casualties from DNBI than from direct combat.[6] The burden of dental emergencies on combat effectiveness must be historically understood within the overall context of medical readiness. During World War II from 1942 to 1945, the US Army Dental Service performed approximately 12.6 million fillings and 2.5 million extractions while overseas.[7] During the Korean War in 1952, the 7th Infantry Division dental surgeon reported that 23 percent of their entire division required emergency dental treatment.[8] In 1968, during the Vietnam War, field commanders reported that dental emergencies incapacitated their soldiers for up to seven days, with an annual rate of 142 dental emergencies per one thousand soldiers.[9] During Operation Iraqi Freedom, dental emergencies accounted for 19 percent of all treatment at a US Role 2 medical treatment facility, far exceeding the number of WIAs.[10] In 2016, a Canadian-led Role 2 medical treatment facility deployed to Iraq and reported that dental emergencies comprised 46 percent of all patient encounters.[11] James A. Nicholson et al. estimated that dental emergencies will account for 17 percent of all casualties during prolonged field care scenarios.[12]

The war in Ukraine has shown the gritty reality of large-scale combat with a near-peer adversary. The over seven hundred attacks on Ukrainian hospitals reflect a targeted destruction of civilian medical infrastructure.[13] These attacks include 181 ancillary facilities like pharmacies and dental clinics, which often serve as casualty collection points.[14] The degradation of civilian dental facilities has allowed dental disease to progress for years, resulting in severe treatment needs and compromised readiness among the Ukrainian military.[15] Even potential adversaries acknowledge challenges—Russian military medical literature explicitly identifies dental casualty management as an operational planning factor requiring dedicated resources and capability.[16] The Russian military has documented its own struggle addressing dental emergencies on the battlefield. Dense artillery fire across the entire operational depth endangers Russia's ability to deploy standard dental equipment. These reports emphasize dental's strain on the ability to clear future battlefields and how the movement of dental patients can limit a commander's freedom of movement.

Evacuation of Dental Emergencies

The Russian and Ukrainian experiences are not unique. French forces deployed to Mali reported that 16 percent of all medical evacuations were for dental emergencies.[17] For this reason, North Atlantic Treaty Organization (NATO) allies, including Canada and the United Kingdom, maintain dental capabilities at Role 2 facilities, recognizing that forward treatment minimizes evacuation burden. A review from 2009 to 2023 shows that within the US Central Command (USCENTCOM), there were 34,318 service members, civilians, and contractors who required evacuation to a dental treatment facility.[18] The staggering volume of intratheater patient movement for dental needs peaked during narrow windows of combat operations. In Afghanistan, 14,514 (90 percent) of these evacuations occurred from 2010 to 2014.[19] In Iraq, 13,660 (96 percent) occurred from 2010 to 2011, with an average of 723 evacuations per month.[20] Unfortunately, USCENTCOM's reported dental evacuations underrepresents the true burden throughout the Global War on Terrorism. The system tracking these metrics, implemented in 2009, lacks data from earlier periods.[21] Given the implementation



Soldiers from 3rd Brigade Combat Team, 101st Airborne Division, converse ouside the entrance of a Role 2 medical facility on 3 June 2024 at the Joint Multinational Readiness Center, Hohenfels Training Area, Germany. (Photo by Capt. Roland Alonzo, US Army)

timeline, data from the 2003–2008 surge periods remain undocumented, indicating an even greater historical burden. Escalating the problem of patient movement, 58 percent of the dental emergency evacuations within USCENTCOM occurred by ground vehicle, predominately casualty evacuation.[22]

In LSCO, the absence of air superiority renders air medical evacuations infeasible. Consequently, ground vehicles will serve as the primary means for patient movement. The avenues of egress away from the forward line of troops will likely be congested with materiel and personnel. Furthermore, service members will continuously experience dental emergencies outside the context of prolonged conflict and combat operations. In 2016, the 101st Airborne Division deployed in support of Operation Inherent Resolve and did not forward deploy their dental officer. Without dedicated organic dental treatment options available in theater, patients were evacuated to Kuwait for dental emergencies. This became so problematic that the division surgeon cell tasked a dental officer in Kuwait to conduct battlefield circulation in theater.

Medical planners and combatant commanders must anticipate that dental emergencies, as shown, will account for a significant burden on movement and present from outside their formations. Among NATO allies, dental standards and capabilities vary significantly, with some nations prioritizing predeployment screenings due to minimal deployable dental assets. This disparity can complicate multinational operations, making interoperability in dental care critical to maximize RTDs. Additionally, the demand for dental care extends to detainees, who often present untreated chronic conditions, further straining limited evacuation assets and requiring dedicated protocols for humane treatment without compromising force readiness. To offset the rearward movement of thousands of dental casualties, the forward deployment of small expeditionary dental teams is necessary. Unique to the profession of dentistry, dental teams can diagnose and treat most service members' emergencies within the same day, maximizing RTD. Recently, US dental teams supported a population of partner forces and 96 percent were successfully RTD with one or two visits.[23] Expeditionary dental teams

conducting battlefield circulation during LSCO can move along supply routes from Role 2 and Role 3 units to provide care. This approach ensures each soldier maintains optimal performance while minimizing evacuation delays on patient movement platforms.

Degradation of Role 2 and Role 3 Dental Support

Ongoing cuts to operational dental billets threaten expeditionary support and capabilities, further exacerbating the burden of dental emergencies. The Army Structure Memorandum for Fiscal Years 2025–2029 will inactivate a Role 3 dental asset from the Army's formation and significantly reduce the number of active-duty dental company (area support) (DCAS) units providing area support for the division and corps.[24] In addition to increasing the risk of untreated dental emergencies in the division area of operations, this DCAS inactivation will limit access to care and reduce training opportunities for dental teams. Reductions to Role 3 dental support will create greater demand at Role 2 units, namely the brigade support medical company (BSMC) and medical company area support.

Experiments removing organic Role 2 dental support from the BSMC and medical company area support, especially without accurate dental casualties, will inevitably degrade access to care. Eliminating organic brigade dental surgeons from the BSMC will delete an entire medical function from US Army divisions and disregard the increased reliance on the BSMC following DCAS cuts. This substantially reduces the division's ability for prolonged care of dental patients, as there are few available providers on the battlefield to fill this gap. After dental officers, the next most formally trained providers to manage dental emergencies are the 18D Special Forces medical sergeant who receives sixteen hours of dental training and the 65D physician assistant with ten hours of dental training. A 68W combat medic receives zero dental training in the Advanced Individual Training course.

In addition to their primary mission of providing operational dental care, dental officers serve as supplementary medical manpower during critical moments. The Dental Corps' Individual Critical Task Lists prepare dental officers to augment medical personnel during mass casualty (MASCAL) events. These skills can be expanded through Army dental residency programs offering courses, such as Pre-Hospital Trauma Life Support and Combat Casualty Care Course, to further prepare dentists for medical emergencies. Moreover, dentists' skills with soft tissue management, conscious sedation, and triage protocols allow them to assist in times when medical providers will be overwhelmed with wounded patients. A cut to US Role 2 dental support would result in the same capability gap observed in Ukrainian Armed Forces, who place dentists at Role 3 or 4. Consequently, Ukrainian medics report sending critical dental treatment to hospitals.[25] The elimination of dental teams from the modified table of organization and equipment creates the conditions for dental emergencies to accumulate near the forward line of troops, relying on fewer and smaller DCASs as the contingency.

Casualty Estimates in LSCO Must Include DNBI

To accurately strain and test the medical system prior to conflict, WFXs must include dental emergencies and all forms of DNBI among MASCAL events. Doing so may reveal that the medical system could also be overwhelmed with a MASCAL of DNBI patients. Matthew Fandre reported that during the Mission Command Training Program WFXs, a force of one hundred thousand soldiers will incur up to fifty-five thousand casualties.[26] To accurately model dental casualties, WFXs should incorporate a volume that 15–20 percent of DNBI casualties are dental, comprising 8,250–11,000 for a fifty-five-thousand-casualty scenario. The distribution of dental casualties means a steady accumulation of twelve to fifteen cases per brigade per day rather than spikes during engagements. If there is forward dental capability, then there would be a 90 percent plus RTD within twenty-four hours, and without a forward dental capability, RTD would take three to seven days. The evacuation mode would be primarily ground transportation with implications for convoy composition and protected persons requirements. It is noteworthy that eleven thousand dental casualties would represent nearly 30 percent of the real-world intratheater dental evacuations that occurred in USCENTCOM from 2009 to 2023. WFXs must include a volume of dental emergencies to accurately test the medical treatment and evacuation system. These parameters would allow commanders and staff to test medical task organization, evacuation priorities,

and resource allocation against the complete casualty burden rather than a subset.

Furthermore, the Military Health System (MHS) is obligated to care for soldiers after a war has ended. At the 2024 MHS Research Symposium, the chief surgeon for the Ukrainian Armed Forces outlined ten tactical medical priorities. Among the challenges he discussed was the need for "prosthetics with modern dentures."[27] Because routine dental decay had progressed for years without dental care, over half of Ukrainian dental emergencies could only be treated by extractions.[28] This observation validates the Ukrainian chief surgeon's assertion of the need for dentures among Russo-Ukrainian War veterans. If tooth extraction is the only option for soldiers engaged in real-world large-scale combat, then there will be a significant need to replace missing teeth with dentures. These circumstances necessitate planning for the dental treatment needs a military population will require after a war ends and as soldiers reintegrate back into civilian life.

The long-term costs extend beyond individual health. Service members requiring extensive prosthetics in their twenties and thirties will need ongoing care throughout their lives. The lifetime cost of prosthetic rehabilitation ($5,000–$50,000 per case with ongoing maintenance) dramatically exceeds the cost of early preventive and restorative care ($200–$2,000). Multiply these costs by thousands of service members experiencing progressive disease during LSCO, and the long-term financial burden to the MHS and Veterans Affairs becomes substantial—likely exceeding the cost of properly resourcing forward dental capability during operations. Beyond economics, allowing preventable dental disease to progress due to inadequate planning creates service-connected disabilities that could have been avoided through proper force health protection measures.

Recommendations

For WFX planners—include dental casualties at historically accurate rates (15–20 percent of DNBI) with appropriate distribution patterns, treatment timelines, and evacuation modes. Track dental RTD rates as a measure of medical system effectiveness.

For medical planners—preserve organic Role 2 dental capability pending operational analysis that includes dental casualty rates and evacuation burden. Reconsider active-duty DCAS inactivation or identify alternative capability to maintain sufficient deployable dental assets for multidivision operations.

For operational commanders—include dental emergency projections in medical estimates alongside battle injuries and other DNBI. Task organize dental capability based on anticipated casualty rates and evacuation constraints. Establish dental readiness metrics as a component of unit readiness reporting.

For the Defense Health Agency and services—consider developing dental emergency predictive models that holistically assess overall health and oral health to better address the needs of the most vulnerable warfighters before deployment. Consider updating the dental readiness classification system to quantify risk for a dental emergency based on patient dental history and oral health risk factors. Update Department of Defense Instruction 1332.45 to reflect DRC 3 nondeployable to reduce dental emergencies in theater.

Conclusion

Sean O'Keefe, deputy under secretary of war for personnel and readiness, echoes Friedrich's warning, stating, "Dental health issues are often the largest cause of nondeployability within a military unit."[29] Excluding dental-related DNBI from casualty estimates in large-scale combat WFXs creates a dangerously incomplete picture of operational risk. Planning to react after capabilities degrade is planning for failure—at the expense of both combat effectiveness and the long-term health of the force. Real-world experience shows that dental emergencies place significant strain on Role 2 medical treatment facilities and evacuation assets; WFXs must therefore incorporate historically accurate DNBI data to reflect these demands. Only by accurately simulating the full medical burden can exercises reveal the true value of forward-deployed dental teams. Future large-scale combat will likely feature brief, intense MASCAL events separated by extended periods of disease, injury, and attrition. To sustain lethality through these cycles, the MHS must prioritize rapid RTD capabilities—delivered by expeditionary dental teams positioned at the point of need—to include Role 2 BSMC. ■

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A Brazilian Army barge transports cavalry troops and equipment on a fifteen-day journey along the Amazon River from Belém to Manaus, the initial staging area for Operation Roraima, in December 2023. (Photo courtesy of Brazilian Army Logistics Command)

Operational Art in LSCO

Forging Deterrence by Denial on the Amazon Frontier

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In late 2023, rising tensions between Venezuela and Guyana over the disputed Essequibo region (see figure 1) created a significant risk of regional instability, threatening Brazilian interests. In response, Brazil implemented Operation Roraima to strengthen its military presence along the border with the two countries.[1] This article examines that operation, arguing that effectively applying operational art within a potential large-scale conventional combat scenario established a credible strategic deterrent, thereby safeguarding Brazil's national interests and influencing regional actors' considerations. Ultimately, no Venezuelan forces traversed Brazilian territory to conduct overland operations against Guyana during the period of heightened tension.[2] This outcome suggests that Brazil's enhanced forward posture may have materially constrained the most practical overland option, contributing to a posture consistent with deterrence by denial and the protection of national sovereignty.

The strategic logic for Brazil's actions is rooted in geography: the only practical overland routes from Venezuela to the disputed region pass through Brazilian territory.[3] Consequently, any Venezuelan military incursion into Guyana would violate Brazilian sovereignty. This reality shaped Brazil's policy, as clearly stated by the minister of defense, who firmly declared that Brazil would not permit Venezuelan forces to cross its borders to attack Guyana, thereby reinforcing a deterrence strategy based on denial.[4]

Although extensive literature exists on the theoretical foundations of deterrence and the logistical challenges in complex environments, this case study also offers a unique contribution. It provides a different perspective of the maritime-centric modern reception, staging, onward movement, and integration (RSOI) doctrine by systematically examining how operational art—particularly through the RSOI process—helps achieve strategic deterrence by denial. Operation Roraima serves as a case study of a road/river-centric model of continental power projection, providing insights into modern nonkinetic settings while also acknowledging the inherent limitations and ongoing debates within deterrence theory, especially regarding the challenge of definitively proving deterrent effects in a multicausal international landscape.

To support the thesis, this article will provide an overview of deterrence theory. It will then use Operation Roraima as a case study, analyzing its operational components. Finally, the conclusion will demonstrate how these measures achieved strategic deterrent outcomes, thereby safeguarding Brazilian sovereignty and fostering regional stability.

Deterrence and Operational Art in History

The idea of deterrence is ancient, and its failure is clearly demonstrated in Thucydides' Melian Dialogue. In this account, the Melians' attempt to prevent an Athenian siege failed because their threats lacked credibility; they didn't have the military capability to resist, and they did not have a guaranteed alliance with Sparta. The Athenians dismissed the Melians' appeals for justice with the harsh saying, "The strong do what they have the power to do and the weak accept what they have to accept."[5] The Athenian strategy then shifted to using pure force—pushing for surrender—while the destruction of Melos later served as a strong deterrence, discouraging other states from challenging Athenian power.

Sun Tzu highlights deterrence by stating, "To subdue the enemy without fighting is the acme of skill."[6] This reflects a focus on influencing the enemy's decisions to avoid conflict and achieve objectives. Moreover, Carl von Clausewitz sees war as an act of force used "to compel our enemy to do our will."[7] From this perspective, deterrence is a method of achieving political goals by influencing the opponent's decision-making process. Collectively, these classical theorists demonstrate that deterrence is not a modern concept, but a traditional strategy rooted in the enduring dynamics of honor, fear, and interest.

Although its philosophical roots are ancient, the formal study of deterrence theory emerged with the development of nuclear weapons during the Cold War. The arrival of the atomic bomb transformed the strategic landscape, shifting the main challenge for great powers from winning conflicts to preventing them. Thinkers like Thomas Schelling, Herman Kahn, and Patrick Morgan became the founders of modern deterrence, creating the language needed to analyze a world dominated by the potential for mutually assured destruction.[8] They explored the complex psychological factors of coercion, credibility, and escalation, moving strategy from physical battlefields into the minds



(Figure by author)

Figure 1. The Essequibo Region

of decision-makers. Their work established the three fundamental principles of modern deterrence theory: for a deterrent threat to be effective, a state must have the ability to carry it out, the enemy must trust in the state's credibility and willingness to act, and this ability and intent must be clearly communicated.[9]

A major breakthrough in this field was the discovery of two main approaches to deterrence. Glenn Snyder, in his influential 1959 monograph *Deterrence by Denial and Punishment*, explained the key difference between denial and punishment.[10] He, along with later scholars, described deterrence by punishment as the threat of causing unacceptable costs to an adversary after hostile acts—such as the concept of massive nuclear retaliation. In contrast, deterrence by denial aims to stop aggression by making success unlikely or impossible, thus eroding an aggressor's confidence in achieving their goals.[11] Snyder argued that deterrence by denial, which focuses on an adversary's perceived capabilities, is often more credible and stable than deterrence by punishment, which relies on the willingness to escalate if deterrence fails. His analytical framework is central to this article's argument. Operation Roraima should be viewed not as a punitive threat but as a strong example of deterrence by denial.

In the twenty-first century, especially with the renewed emphasis on great-power competition, deterrence by denial has been increasingly acknowledged by the strategic community as essential to avoiding conventional conflict. As John Mearsheimer points out, "Conventional deterrence is all about persuading an adversary not to initiate a war because the expected costs and risks outweigh the anticipated benefits."[12] Modern analysts like Michael Mazarr argue that denial strategies are considered "inherently more reliable" because they rely on concrete defensive measures instead of uncertain threats of future punitive actions.[13] In the context of "gray zone" competition—marked by efforts to achieve objectives below the threshold of major conflict—the ability to physically prevent aggression stands as the strongest form of deterrence.[14] This requires combat-credible, forward-deployed forces capable of stopping an adversary from rapidly and cheaply achieving their goals. This article argues that

Operation Roraima demonstrates this modern strategic need. Through operational planning and execution, Brazil created a credible, conventional deterrent that effectively prevented a regional actor from pursuing a certain course of action, thereby protecting Brazil's national interests and supporting regional stability.

Operational Art as the Mechanism for Deterrence by Denial

A nation achieves strategic deterrence not through abstract threats but through credible demonstrations of capability and will. In the conventional realm, the primary method for this demonstration is operational art, which is the key process through which the theory of deterrence by denial is applied.[15] The arrangement of military forces in time, space, and purpose—the core of operational art—is what generates clear signals of capability and credibility that influence an adversary's strategic decision-making. Operation Roraima, therefore, should be viewed not just as a simple deployment but as a deliberate application of operational art aimed at producing a specific deterrent effect.

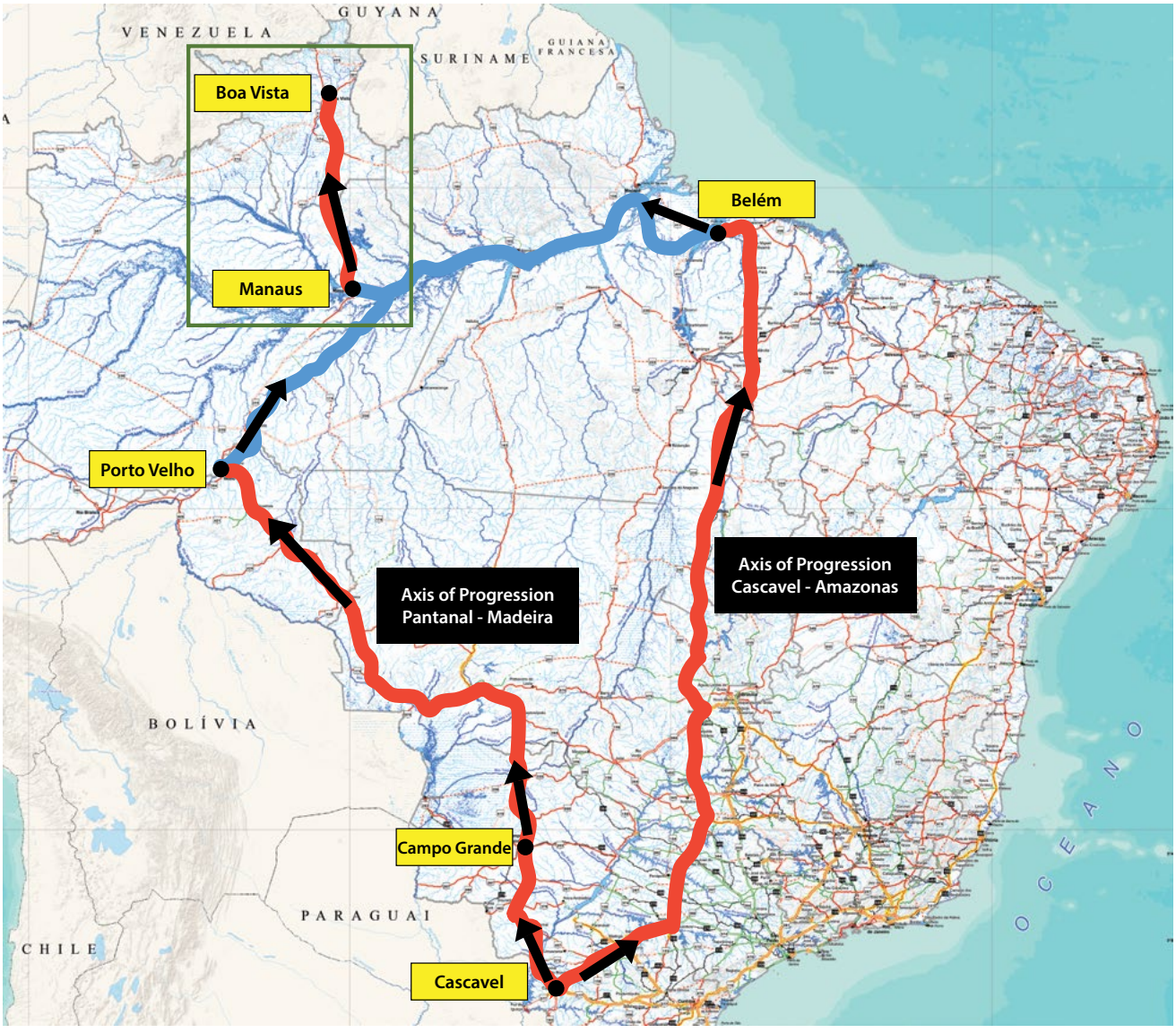
Operational art translates strategic objectives into tactical actions.[16] While force projection connects all three levels of war, it is at the operational level where this link is most actively planned and coordinated. The doctrinal process of RSOI illustrates this. RSOI begins with a strategic decision to deploy forces to meet a national objective, and its success hinges on the tactical readiness of the integrated units. However, the primary focus of planning, synchronization, and management stays at the operational level. According to Joint Publication 3-35, *Joint Deployment and Redeployment Operations*, the combatant commander is responsible for the entire RSOI process within the theater, but may delegate some of those responsibilities to the joint task force commander.[17] Coordinating diverse capabilities—logistics, force flow, security, etc.—over time and space is a core function of operational art. Successfully orchestrating this sequence is more than administrative; it is an art that requires the "creative and skillful exercise of authority" by a commander to handle these complex actions.[18] Therefore, analyzing how a commander executes RSOI shows their practical application of operational art.

The following analysis of Operation Roraima examines three critical elements of operational art that played a key role in establishing a deterrent effect: the arrangement of operations, tempo, and, most importantly, logistics and operational reach. The integration of these elements, demonstrated through the RSOI process, allowed Brazil to influence perceptions of a regional adversary by effectively showing both capability and credibility. In terms of capability, decision-makers in Venezuela needed to see that any potential military action would be unfeasible. Brazil's successful implementation of the complex RSOI into the northern Amazon, along with its improved forward posture, aimed to prevent any land incursions through its territory. This operational reach directly signaled capability. As for credibility, it was crucial for Venezuela to understand Brazil's determination to act. The operation's deliberate pace and phased approach—not just a single maneuver but a continuous buildup leading to a permanent stance—were meant to make Brazil's resolve to defend its sovereignty unequivocally clear.

Case Study: The Execution of Operation Roraima

Before addressing the specifics of Operation Roraima, it is important to note that Brazil maintains friendly relations with both Venezuela and Guyana, with diplomatic staffs in Caracas and Georgetown. Worries in Brasília were therefore not about direct aggression from either of these friendly countries but rather that a conflict between them could spill over into Brazilian territory, threatening its neutrality.

The deterrent posture adopted by Brazil was a direct response to a clear and growing conventional military buildup by Venezuela. In late 2023, the Maduro government moved about five thousand troops to the border area with Guyana.[19] This was not just a political act; satellite images from early 2024 confirmed a major expansion of Venezuela's Anacoco Island military base, opposite the contested territory, with the garrison likely increasing from a company to a battalion. The buildup involved the forward deployment of a mixed group of armored fighting vehicles, including Scorpion light tanks and EE-11 Urutu amphibious armored personnel carriers. This activity was coordinated under Venezuela's new "Guyana Esequiba Integral Defense Operations Zone," indicating a clear and organized military focus on the region.[20] It was this demonstrated conventional capability, not just political rhetoric, that Operation Roraima aimed to deter.



(Figure by author; map from the National Department of Transport Infrastructure)

Figure 2. Movement of Assets From Brazil to Boa Vista

The heart of the Brazilian Army's forward presence in the northern Amazon is the *1ª Brigada de Infantaria de Selva* (1st Jungle Infantry Brigade), the region's main tactical unit, which serves as the primary defender of national sovereignty in Roraima.[21] This unit plays a key role in the Army's *"Estratégia da Presença"* (strategy of presence) for the Amazon region. The brigade operates as a permanent regional defense force—a first response unit—tasked with maintaining a continuous presence, defending the territory, and responding quickly to various threats, including illegal activities and low-intensity conflicts within its area of responsibility.[22] The increasing regional tensions in late 2023 prompted a rapid shift in the brigade's operational focus, moving from standard presence missions to developing a credible conventional deterrence capability. This transition involved support from units associated with Brazil's strategic response forces. These units, similar to national-level global response or rapid deployment teams, maintain higher readiness levels and have specialized skills such as operating with armored vehicles, conducting airborne assaults, providing long-range fires, and contributing operational logistics. Their main role is to quickly project force across the country during major

Table. Command Structure

Brazilian Organization	Closest US Doctrinal Equivalent	Role in Operation Roraima
Comando Logìstico	Army Materiel Command	Strategic-level logistics planning and coordination
Comando Militar da Amazônia	Army Service Component Command	Theater-level command and RSOI execution
1ª Brigada de Infantaria de Selva (1ª Bda Inf Sl)	Light Infantry Brigade	A tactical-level combined arms army combat unit that is specifically trained for jungle warfare
1º Batalhão Logístico de Selva (1º B Log Sl)	Brigade Support Battalion	Tactical-level logistics and maintenance execution

(Table by author)

emergencies or to reinforce first response units during crises (see figure 2).

With Operation Roraima, Brazil's strategic goal was to prevent any potential adversary from using its territory.[23] Additionally, other government efforts, such as Operation Catrimani II, aimed to increase state presence along the northern border.[24] As a result, the measures discussed in this article were specifically designed to strengthen the organic capabilities of the 1st Jungle Infantry Brigade, providing greater mobility, firepower, and shock effect essential for countering conventional military threats. The operational response to the growing crisis required a complex, multiphase strategy, implementing the principles of RSOI within the challenging Amazon environment. This initiative relied on the coordinated efforts of all three branches of the Brazilian Armed Forces, although most tactical actions were carried out by the Army.

While the entire process was successful, this analysis argues that the most important doctrinal adaptations and operational innovations occurred during the reception and staging phases. It was here that the armed forces addressed the unique challenge of continental power projection in an environment lacking infrastructure. This case is especially instructive because it offers a strong contrast to the maritime-focused RSOI models typical of US doctrine, instead showcasing a road-river-centric approach to projecting power across a landmass. The operation was coordinated through a two-tiered command structure, with the national Logistics Command providing strategic guidance and the Amazon Military Command (CMA) overseeing the in-theater execution (see the table).[25]

Reception. The operation's reception phase was the first critical act of operational art, centered on organizing military operations in space to manage the flow of combat power into the theater. The operation aligns with the doctrinal definition of *reception*—"the process of receiving, off-loading, marshalling, and transporting of personnel, equipment, and materiel from strategic and/or intratheater deployment phase to a sea, air, or surface transportation POD [port of debarkation] to the marshalling area."[26] This was achieved by both air and via a network of inland hubs instead of traditional coastal seaports.

The airports in Manaus and Boa Vista served as the primary air ports of debarkation, receiving a total of 122.3 tons of critical supplies via Brazilian Air Force aircraft. This airlifted cargo was diverse and essential for enhancing forward posture; it included military personnel; ammunition and explosives; antitank missile systems; communications modules; and different classes of supplies such as general goods, lubricants, and vehicle spare parts. The use of both military and contracted civilian aircraft for personnel highlights the adaptable approach of the reception plan.[27]

For the heavy cargo, reception relied on a river corridor that reinforced Manaus's role as the central hub. The main road movements, organized along two distinct axes of progression, began in the Brazilian southernmost regions and ended at the cities of Belém and Porto Velho, which served as reception and



High-explosive ammunition is unloaded from a Brazilian Air Force KC-390 Millennium aircraft at Boa Vista Air Base on 1 December 2023 to reinforce Class V stocks for the 1st Jungle Infantry Brigade. (Photo by author)

intermodal transfer points. Here, the main combat platforms—including more than forty armored combat vehicles—along with specialized nonarmored vehicles and a logistics module, were received from the road network and transferred onto river barges. These barges then navigated the rivers to the primary riverine port of debarkation at the Army's Center for Riverine Craft in Manaus.[28] This made Manaus the focal point for the entire surface deployment, where the majority of the combat power was received in the theater.

The execution of these reception activities faced unique constraints that tested the flexibility of the planners. One key challenge was Boa Vista's runway's inability to support the larger A330 Multi-Role Tanker Transport aircraft, which forced cargo to be rerouted through Manaus for a subsequent 780-kilometer road transfer. At the river ports, the reception process encountered environmental issues; transloading vehicles in Belém took a full day, hampered by the river's significant tidal changes and infrastructure limitations that created a small window for safe embarkation. To address these challenges, the Army used specialized units like the *Pelotão de Operação de Terminais de Cargas* (Cargo Terminal Operation Platoon) to manage the flow of materiel and personnel at transfer hubs. These obstacles highlight that in any RSOI operation, the official plan must continually adapt to the constraints of the physical environment.[29]

Staging. The staging phase of Operation Roraima was carried out as a complex, multiechelon staging pipeline, an application of operational art that managed tempo to steadily enhance combat power without the significant operational pauses that a single staging base would require. Instead of using a single, large intermediate staging base, which is common in US operations, the operation employed a layered system that started with support points along deployment routes, followed by an initial theater staging phase in Manaus, and concluded with a more advanced staging and integration phase in Boa Vista.[30]

To alleviate the immense strain on vehicles and personnel during intertheater movement, the staging pipeline included logistical detachments set up by the Brazilian Army along the main ground routes of advance. These detachments accomplished essential preparatory tasks including performing corrective maintenance and providing sustainment. The

effectiveness of this ongoing focus on maintenance is notable because there were no vehicle breakdowns during the strategic move despite the extreme distances involved.[31] Upon entering the Amazonian theater, Manaus served as the initial staging area. Here, forces were received, assembled, and then tactically organized into separate march columns.

The final stage of the pipeline occurred in Boa Vista, where disparate elements were unified and prepared for integration. In this forward staging area, the 18th Mechanized Cavalry Regiment was officially activated under the purview of the Amazon Army Command. A key activity in Boa Vista was advanced, system-specific maintenance. The introduction of high-technology platforms—such as armored personnel carriers, light tactical vehicles, wheeled armored reconnaissance vehicles, and remote weapon stations—created a specific demand for updating and resizing the maintenance structure. The construction of new maintenance pavilions in Boa Vista and the deployment of maintenance personnel, specifically to provide second echelon support for these new assets, was a key action in this phase, setting the conditions for the force to be made fully mission capable.[32]

The staging pipeline model was shaped by constraints that required an innovative application of doctrine. The severe infrastructure limitations of the region meant that no single location before the final destination could receive, house, maintain, and prepare the entire force at once. Additionally, the challenge of sustainment included the need to improve the distribution of strategic supply depots in the Amazon.[33] Resupplying the force during staging was a major constraint that demanded careful, long-range planning from the Logistics Command and precise execution by the CMA to address.

Onward movement. Onward movement clearly demonstrated phasing and operational reach as the force moved from its initial staging areas to its final tactical positions.[34] This was a multilevel process, dividing the movement into two specific stages: an operational move to position the force in the area of operations followed by a final tactical deployment.

The first stage of this phase was the operational movement from the initial staging area in Manaus to the final staging area in Boa Vista. This major ground movement covered about five hundred miles along the BR-174 federal highway. Under the control of the CMA, forces organized into march units during staging in Manaus to execute this move.[35] This journey served as the critical link that transported the prepared force from the theater's main reception and staging hub to the forward staging area where it would conduct its final tactical deployment. The last echelon involved the tactical movement from the staging area in Boa Vista to the force's assigned positions in northern Roraima near the Venezuelan and Guyana borders. This crucial final step was executed by the CMA using its organic theater assets. This included, but was not limited to, logistical elements from the 1st Jungle Infantry Brigade such as the 1st Jungle Logistics Battalion, the brigade's organic support unit responsible for transportation, maintenance, and delivery of supplies.[36]

The deployment of the enhanced 1st Jungle Infantry Brigade in northern Roraima marked the end of the movement phase and established the conditions for final integration. The main obstacle remained the harsh environment and limited road network of Roraima. This operational challenge was confirmed by convoy commanders on the ground, who identified the "Amazonian climate and the logistical challenges of moving heavy, sensitive materials on a simple BR highway" as the main difficulties of the final tactical move, making each convoy a significant logistical effort that benefited from the local expertise of existing in-theater logistics assets.[37]

Integration. The integration phase was the culmination of the entire operational design, during which the arrangement of forces in time, space, and purpose was finally executed and validated.[38] Operation Roraima's integration phase was uniquely centered on the comprehensive reinforcement of the 1st Jungle Infantry Brigade. This was achieved by operationally integrating the newly activated 18th Mechanized Cavalry Regiment as a subordinate unit and simultaneously enhancing existing brigade units with new equipment and supplies. A foundational element of this integration was the final, advanced maintenance required to transform both newly arrived and preexisting equipment into a fully mission-capable force. This task fell to in-theater assets, primarily the 1st Jungle Logistics Battalion. This battalion performed crucial preventive and corrective maintenance to bring the brigade's key organic assets to peak operational status.[39]



Vehicles temporarily stage at Campo Grande on the Pantanal-Madeira axis of progression to Manaus in December 2023. (Photo courtesy of Brazilian Army Logistics Command)

With the equipment prepared, the next step was the formal integration of command and control and new capabilities into the theater commander's order of battle. This action brought multiple capabilities into the brigade: the new cavalry regiment provided armored mobility, while the brigade's organic antitank defense was simultaneously strengthened by integrating the new MSS 1.2 AC antitank guided missile systems into the 7th Jungle Infantry Battalion.[40]

The final validation of this complex integration was accomplished through a series of exercises and demonstrations that verified the brigade's new capabilities. At the tactical level, integrating expertise was a key focus, with personnel from the initial convoys conducting the first training sessions for the crews receiving the new vehicles. Readiness was then confirmed by the strategic airlift of live ammunition directly to the 1st Jungle Logistics Battalion, clearly indicating that the force was not just present but fully armed and mission ready.[41] High-level inspections further verified this, with the Brazilian Army's commanding general observing live demonstrations of the new laser-guided missiles. The final mission rehearsal for the entire reinforced brigade was Operation Atlas, the largest joint military exercise of 2025, designed to test the unit's readiness and joint interoperability in the field.[42] This division-level exercise involved elements of the Brazilian Army, Navy, and Air Force.

Conclusion: Deterrence, Denial, and Lessons in Operational Art

Operation Roraima stands as a compelling case study in the successful application of operational art to achieve strategic deterrence. The Brazilian Army's planning and execution of its RSOI processes showed its capability and built the credibility needed to send a powerful message of deterrence through denial. This was demonstrated by deploying advanced armored vehicles and maintaining the airlift of critical supplies, which displayed Brazil's materiel and organizational strength. Credibility was established not through words but through actions; the physical deployment and sustainment of the force, reinforced by the long-term commitment to establish the 18th Mechanized Cavalry Regiment, made Brazil's defensive stance unmistakable.

Following the establishment of the full operational capability of the 1st Jungle Infantry Brigade in northern Roraima, the security situation along the border stabilized.[43] Most critically, no Venezuelan forces traversed Brazilian territory to conduct overland operations against Guyana during the period of heightened tension.

While Venezuelan military exercises and public messaging continued within its own territory, the specific threat of an armed incursion utilizing Brazil's road network did not materialize.[44] Venezuela maintained its posture in the Guyana Esequiba Integral Defense Operations Zone, yet its armored and maneuver elements remained contained within Venezuela, never challenging the Brazilian defensive lines. Public reporting described the Brazil–Venezuela border as calm and open during the period following the deployment, while Venezuelan activities remained on their side of the frontier.[45] Taken together, this absence of hostile movement is consistent with a denial effect: the forward posture of Brazilian forces likely raised operational risk and friction associated with any attempt to use Brazilian territory as an overland corridor.

While the lack of any overland incursions through Brazilian territory is consistent with a successful deterrent effect, a key counterargument suggests that Venezuela's internal political and economic turmoil was the main inhibiting factor.[46] However, such a view overlooks how severe domestic crises can paradoxically act as a strong incentive for external military aggression—a "diversionary war" to boost nationalist support. This potential for miscalculation highlights where the decisiveness of Operation Roraima becomes clear. Brazil's strategy was not based on an abstract threat but on changing the physical situation of the conflict zone. By deploying a complex, multimodal force to its northern border, Brazil presented a hard operational reality to Caracas's strategic calculations, blocking a critical invasion route and making sure the reality on the ground overshadowed any chance of strategic misjudgment.

The operation offers lessons for military professionals. First, it emphasizes that doctrinal principles are flexible frameworks, not rigid checklists, as demonstrated by the successful adaptation of the port of debarkation concept to inland riverine hubs. Second, it highlights the effectiveness of a distributed, multiechelon staging pipeline as a viable alternative when a single, consolidated staging base isn't feasible. For a US and allied audience, the operation reveals a "two-speed" logistical model: heavy equipment is moved deliberately via a fifteen-day riverine transit, while the Brazilian Air Force provides a rapid flow of critical items to forward units. This highly visible, slow-paced riverine journey of armored vehicles also serves as a continuous signal of national commitment, showing how the tempo of a deployment can be explicitly linked to deterrent objectives.

For a planner at a combatant command like US Western Hemisphere Command, these lessons are more than just theoretical. They suggest a model for prioritizing strategic lift in a coalition setting, showing how a quick airborne insertion of key enablers can be coordinated with a slower, partner-led movement of heavy forces. This method decreases the immediate strain on limited air and sea lift resources. Additionally, the operation acts as a reminder that the timing and visibility of deployment are strategic tools, essential for influencing an adversary's decision-making during precrisis situations.

Finally, the crisis emphasizes a key lesson in the information domain. While the Venezuelan Armed Forces conducted information operations by releasing well-produced videos of their military activities, Brazil's response was mainly kinetic and logistical, not rhetorical. In a time filled with disinformation, the clear display of combat power and logistical skill acted as a highly visible "ground truth" that was more convincing than any press release or social media campaign. This grounds a strategic narrative in reality, forcing an adversary to respond to facts on the ground. Although maintaining such a heightened presence presents long-term sustainability challenges, Operation Roraima's role in helping prevent a specific course of action offers a valuable model for how military forces can generate strategic effects in gray-zone conflicts without direct combat. It underscores the ongoing need for adaptable military planning in a world where outcomes are seldom linked to a single cause. ■

Notes

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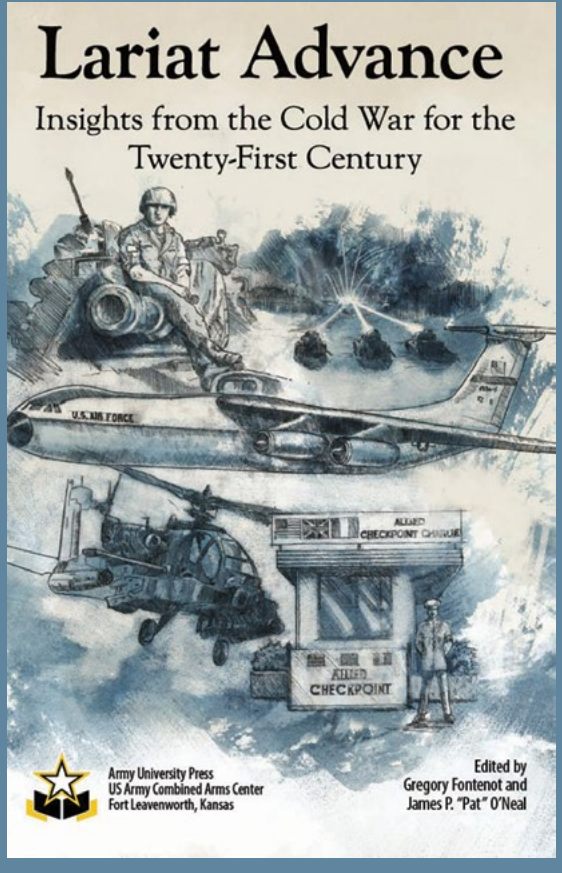
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Lariat Advance: Insights from the Cold War for the Twenty-First Century

Edited by Gregory Fontenot and James P. "Pat" O'Neal

Lariat Advance. Those two words were used to send more than two hundred thousand US troops serving in Europe into alert status during the Cold War. The code words were for a field exercise, but the lessons learned from rigorous practice reflected the US Army's recognition of the need to transform following the Vietnam War. An all-volunteer American fighting force required soldiers committed to practice and study; they were competing against a USSR interested in growing and a Warsaw Pact that demanded coordinated NATO responses. Fortunately, a shooting war never occurred during the Cold War years between American and Soviet forces, but those who served in the Air Force or the Army in Europe took the possibility seriously. ***Lariat Advance*** provides contemporary lessons from influential US Army leaders who helped transform the US Army during the Cold War period. That transformation is not unlike the one that must occur in today's force—experiences that current and future generations of soldiers can learn from as they reshape their Army.

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Tradition and Transformation

Forging Strategically Minded, Globally Competent Officers

Col. William Lyons, JD, US Army, Retired
Travis Morris, PhD
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Often, the character of conflict changes faster than the institutions designed to prepare people for it. Today, strategic competition with peer adversaries, the changing nature of American hegemony, the proliferation of hybrid warfare, the weaponization of information, the accelerating integration of artificial intelligence (AI) into military operations, and the emergence of space as a contested region demand a new kind of officer, one who can think critically across the domains of warfare, collaborate across many cultures, and lead effectively under conditions of perpetual ambiguity.[1]

This article examines how Norwich University has oriented its educational mission toward the areas defining the contemporary security environment: cyber and information warfare, hybrid conflict, and strategic leadership. Using US Armed Forces professional military education (PME) doctrine and precommissioning protocols as analytical lenses and the organizational theory literature on institutional adaptation for methodological framing, this article argues that Norwich's private governance structure generates structural conditions like compressed decision timelines, embedded interagency learning environments, and discretionary authority over curriculum design, which are analytically distinct from those available to federally governed PME institutions. The case is evaluated against organizational theory frameworks: ambidextrous organizations, dynamic capabilities, and adaptive capacity. The article concludes by identifying governance architecture as an underexamined variable in PME literature and proposing it as a productive focus for comparative institutional analysis.

The PME Problem: Continuity versus Adaptation

The twenty-first-century security environment is being defined by relentless strategic competition, rapid technological innovation, and the ever-present prospect of large-scale combat operations.[2] As adversaries use emerging technologies and information operations, the cognitive demands placed on military leaders have grown significantly. Statutory requirements for PME programs must address national military strategy, joint planning across all levels of war, joint doctrine, command and control, ethical



Norwich University's color guard marches in a parade for students and their families on 1 October 2022 during the Parents and Family Weekend at its campus in Vermont. The Parents and Family Weekend provides families an opportunity to connect with the students and experience Norwich's campus and culture. (Photo courtesy of Norwich University)

applications of military power, and joint force development.[3] Some educators would argue the character of contemporary warfare, which is now marked by rapid changes in hybrid threats, cyber, and information operations, requires a much different approach to PME.[4] In 2014, historian and author Williamson Murray argued that the military had become "stagnant" and "self-satisfied," producing officers who were technically competent but strategically underdeveloped.[5] Meanwhile, PME in the United States has faced criticism for insufficient adaptation to new realities. As former Secretary of Defense James Mattis strikingly noted in 2018, PME must evolve to become a strategic asset, one that builds trust and interoperability across the joint force and with allied and partner nations.[6] The *2018 National Defense Strategy* called for PME to prepare officers not only for today's challenges but also for the unpredictable conflicts of tomorrow.[7] The structural factors he identified included accreditation requirements, congressional mandates, and service bureaucracies that can slow and complicate curriculum change.

Scholars of management have made a similar argument about organizations in general: institutions that can reconfigure rapidly in response to environmental shocks tend to be more resilient than those optimized for stability.[8] Today, the pace of technology and innovation requires near constant adaptation and incorporation in the classroom and on the battlefield. Nevertheless, for institutions steeped in tradition and governed by long-standing bureaucratic structures, reconfiguration can be a herculean task. Organizational change requires deliberate institutional choices and the chance that some institutional identity and legacy will shift.

To assess how PME is delivered within a civilian institution operating under conditions of environmental uncertainty, this study draws on several

complementary organizational theories from the management literature. The first, organizational ambidexterity and the exploration-exploitation framework, provides a lens for examining how institutions balance the competing demands of preserving established missions while pursuing innovation.[9] The dynamic capabilities theory extends our analysis by focusing on the processes through which organizations identify emerging opportunities, mobilize resources, and reconfigure structures and capabilities in response to changing conditions.[10] Finally, the concept of "adaptive capacity" further complements these perspectives by emphasizing institutional resilience and the ability to sustain effectiveness through periods of disruption and transformation.[11] These frameworks are particularly well-suited to the study of PME-generating institutions, which must simultaneously maintain legitimacy through continuity while ensuring relevance through adaptation to evolving strategic, technological, and geopolitical shifts.

Norwich University as a Case Study

Norwich University provides a useful case to examine these dynamics because of its historical continuity, evolving educational portfolio, and responsiveness to changing defense and security requirements. Literature on the sociology of civil-military relations often shows that the quality and character of military education is shaped by curriculum content as well as the institutional structures by which that curriculum is produced and governed. These structures determine who can hold authority to define educational priorities, how readily those priorities can be revised, and how much insulation an institution enjoys from short-term political or bureaucratic pressures.[12] In addition to organizational frameworks, an institution's governance architecture is also a constitutive variable shaping the substance and responsiveness of military education.

Norwich's undergraduate and graduate programs seek to address several needs within precommissioning and early career PME continuum. First, the university recognizes that coalition warfare requires cross-cultural competency and actively works toward developing it within the student body. Joint PME (JPME) doctrine describes joint acculturation as a foundational outcome of PME. Norwich's view is that this acculturation can be deliberately and rigorously seeded before an officer's first joint assignment, rather than waiting until JPME Phase I. In addition, Norwich's emphasis on experiential leadership development ensures that cadets exercise leadership under structured pressure before commissioning, rather than encountering it for the first time at the unit level. The dual military-civilian student body serves as an embedded interagency environment, developing the whole-of-government orientation that JPME doctrine identifies as essential. At the undergraduate level, Norwich's cybersecurity curriculum demands that technological and information warfare fluency are treated as foundational competencies rather than specialized supplements.[13] Taken together, these elements reflect an institutional design that continually reinforces precommissioning and early-career PME fundamentals.

An institution's governance architecture is particularly significant when considered alongside the organizational theory literature on adaptation. The concept of "organizational ambidexterity" suggests that ambidextrous organizations are those capable of simultaneously exploiting established competencies while exploring new domains, achieving this through structural arrangements that allow units oriented toward continuity and innovation to operate in productive tension rather than competition.[14] Work on the exploration-exploitation dialectic establishes another theoretical foundation: institutions that overinvest in exploiting existing capabilities risk obsolescence, while those that overinvest in exploration risk incoherence.[15] The ambidextrous organization resolves this tension not by choosing between tradition and adaptation but by creating the structural conditions that allow both to coexist in parallel.

Applied in the context of an established university, where tradition is paramount, both frameworks illuminate a dimension of Norwich's institutional position. As a private institution governed by a board of trustees and subject to regional accreditation standards, Norwich vests curricular decision-making authority in institutional leadership rather than in the layered approval structures that characterize most federally governed PME institutions. The practical consequence is a governance architecture that compresses the timeline between strategic identification of an educational need and operational curricular response. For example, in 2023, Norwich was awarded funding to construct



Cadets assemble by the flagpole in this aerial view of Norwich University. The university is historically known as the birthplace of the Reserve Officers' Training Corps. (Photo courtesy of Norwich University)

the Cyber Fusion Research and Development Center.[16] While the initiative required years of sustained institutional vision, faculty development, and donor and congressional investment, the decision-making pathway followed internal governance processes rather than the multiyear programmatic review cycles that comparable federal initiatives typically require. In terms of organizational theory, the structure of authority shapes not only *what* gets taught and researched but also *how quickly* the institution can determine that something new needs to be taught or researched at all.

Global Partnerships and Overseas Education

The most structurally significant expression of the governance conditions that enable organizational ambidexterity is the university's deliberate, structured integration of international engagement long before an officer's first joint or combined assignment. JPME has long emphasized international engagement and allied interoperability as core competencies for future joint force leaders. Yet for some officers, meaningful cross-cultural military education might not begin until the intermediate-level education phase, often years into a career. Yet, as a private college, Norwich retains the institutional authority and discretion to determine the timing of international programs. Consistent with the ambidextrous organizational model, Norwich embeds international engagement into its educational architecture at both the undergraduate and graduate levels, treating overseas study and foreign military exchange as a structured domain of exploration that operates alongside, rather than in tension with, the institution's core precommissioning mission.[17]

One institutional example of this approach is with funding from the Olmsted Foundation, which has enabled Norwich students to participate in overseas experiential learning. The Olmsted Field Studies in Peace and Conflict immerses students in international environments where they encounter firsthand the local history, religion, politics, civil-military relations, and security conditions of a given region. This experiential model cultivates habits of observation, humility, cultural interpretation, and adaptive leadership that only direct engagement outside the classroom can develop. Since 2005, Norwich students on a commissioning

track through the Reserve Officers' Training Corps have benefited from this funding, which supports two-week spring excursions to destinations across the globe.[18] During these trips, students engage with local government officials and military personnel while participating in field studies and guided tours.

Beyond bilateral and institutional exchanges, Norwich also maintains partnerships with the International Security and Officer Military Academy and the International Forum on Peace, Security, and Prosperity (IFPSP). The IFPSP convenes civilian youth and officer cadets from across allied and partner nations to examine the role of armed forces in peacemaking, peacebuilding, and stabilization activities.[19] Cadets attending the forum engage directly with counterparts and other nations, producing a structured cross-cultural encounter consistent with experiential learning theory. The result is a concrete experience with foreign military peers, followed by deliberate reflective observation in a professionally bounded context. Cadet reflections from the forum document precisely the competencies JPME identifies as foundational to joint and combined operations: allied interoperability, coalition management, and the capacity to recognize how American strategic choices are received beyond domestic institutional frames. Viewed through the lens of ambidextrous organizational theory, IFPSP participation functions as a low-structural-cost exploratory investment to extend international educational reach without displacing core precommissioning requirements.[20] The result is officer candidates who arrive at their first assignment having already navigated the interpersonal and professional complexity of multinational military dialogue.

At the graduate level, Norwich has formalized this international orientation through a partnership with Helmut Schmidt University (HSU) in Hamburg, Germany. HSU is a federal Bundeswehr (German military) university whose student body consists predominantly of officer candidates of the German armed forces. In one instance, a graduate-level course is codelivered by Norwich and HSU faculty, combining ten weeks of virtual instruction with a one-week in-residence seminar held on the HSU campus.[21] The program is structured as a capstone experience that challenges students from both the American and German military education systems to apply strategic frameworks, collaborate across cultural and institutional boundaries, and communicate effectively under sustained pressure using a values-based decision-making process.[22] Viewed through the lens of ambidexterity, the Norwich-HSU partnership exemplifies the exploratory function: It extends the educational reach into allied professional military culture while drawing on Norwich's graduate curriculum as the exploitative foundation. The German-American cohort structure makes the cross-cultural dimension an explicit and assessable feature of the learning design rather than an incidental byproduct of international enrollment.

Norwich also holds graduate-level immersion courses in Berlin with a format that reflects the same governance flexibility that characterizes Norwich's broader curricular approach. The institution can expand delivery modalities in response to identified demand without requiring a programmatic redesign, thereby sustaining the partnership's exploratory reach while lowering structural barriers to participation. For instance, the university developed a graduate-level hybrid warfare course in 2026 that was in direct response to identified operational requirements and practitioner feedback from the field.[23] The course has students arrive in Germany after spending two months developing theoretical and conceptual fluency in hybrid conflict. The Berlin residency centers on an intensive cumulative exercise in which students from the United States and NATO allies develop offensive and defensive hybrid campaign strategies, analyze information and influence operations, and apply structured frameworks for risk and escalation management in a city whose geography and history provide concrete grounding for these concepts.[24] The sequencing reflects a deliberate application of the experiential learning cycle, moving students from abstract conceptualization to active experimentation in an environment that renders tangible the operational stakes of hybrid conflict.

Cybersecurity, Information Warfare, and AI

The ambidextrous organizational logic that characterizes Norwich's international programming is similarly evident in its approach to emerging technologies in cybersecurity, information warfare, and AI. These areas are experiencing a pace of environmental change so rapid that exploitation of existing curricula alone cannot sustain their educational relevance. The



Students attend a leadership workshop on 16 January 2023 at Norwich University. (Photo courtesy of Norwich University)

concept of dynamic capabilities explains that institutions operating in volatile environments must develop competencies to perform current functions as well as the meta-level capacity to sense shifts in the environment, seize emerging opportunities, and reconfigure internal resources accordingly.[25] Norwich's sustained investment in cyber education reflects this kind of dynamic capability development.

The university's cyber and information operations initiatives can also be also understood as manifestations of "adaptive capacity."[26] Since 2001, Norwich has maintained the National Centers of Academic Excellence in Cybersecurity Cyber Defense designation, providing an institutionally validated foundation for its cyber curriculum.[27] Its School of Cybersecurity and Advanced Computing serves as the organizational hub for this work, and the university's DoD Cyber Institute and Cyber Leader Development Program extends that foundation into integrated cyber and AI education.[28] In 2023, a $16.4 million investment in the Multidisciplinary Cyber Fusion Research and Development Center further expanded Norwich's capacity in information operations and AI, representing the kind of deliberate resource reconfiguration that the management literature identifies as central to dynamic capability.[29]

Norwich has also partnered with Seerist Inc., an augmented analytics platform serving intelligence, threat, and security professionals.[30] The partnership enables Norwich students to gain hands-on experience using an industry-relevant tool. Through a multiyear agreement, students gain practical exposure to Seerist's open-source intelligence-based platform, which integrates AI with expert human analysis to forecast and assess global threats, track instability trends, and deliver actionable strategic insights.[31] Students also use Seerist's Foreign Influence Risk Index, a real-time dashboard designed to monitor and analyze global foreign influence activities.[32] Users can make faster, better-informed risk decisions to examine instability indicators, foreign influence activity, narrative trends, and emerging security risks in near-real time. This partnership directly supports a larger effort to produce graduates who are competitive candidates for roles within the US intelligence community, the Department of War, fusion centers, and the private sector. The extent to which exposure to practitioner technologies produces

measurable improvements in graduate performance remains an empirical question; however, the initial longitudinal assessments show a steady upward trajectory.

Finally, Norwich's development of geospatial intelligence (GEOINT) education is another example of how the institution has adapted to contemporary security competition within the framework of dynamic capability development. Today, GEOINT is central to how military and civilian leaders understand exact locations of allies and adversaries, as well as natural and man-made obstacles in contested environments.[33] To meet this need, in 2024, Norwich advanced GEOINT as an applied educational capability.[34] This effort connects geographic reasoning, remote sensing, open-source data, and visualization to real-world problem-solving. This commitment earned formal recognition in 2025, when Norwich's undergraduate GEOINT certificate program received accreditation from the US Geospatial Intelligence Foundation.[35] The accreditation itself functions as an external validity marker, and future GEOINT programs will advance into student research and applied analytical products.

In Summary

Twenty-first-century security competition has exposed tension within PME: the need to preserve the institutional continuity that gives PME its authority while also developing the adaptability that makes it relevant to the battlefield. This tension is present in many institutions, but the pace at which hybrid warfare, cyber operations, information conflict, and AI are reshaping the operational environment has made resolving it much more urgent. Norwich University's case illustrates just one way to manage that tension. The organizational frameworks examined in this piece—ambidexterity, dynamic capabilities, and adaptive capacity—illuminate the structural logic that has allowed a centuries-old institution to respond to operational demand signals, build internationally integrated graduate programs, and develop cutting-edge curricula without abandoning the citizen-soldier identity that has defined it since the early 1800s.

This article does not suggest that Norwich's model is the only viable response to the educational demands of contemporary security competition, nor are its specific programmatic choices directly transferable to other institutional contexts. What this case suggests is that governance architecture matters, and that the structures that determine who can authorize curricular change, on what timeline, and at what cost to institutional coherence can shape educational outcomes in ways the PME literature has not fully examined. As the conversation in this series continues, the relationship between institutional structure and educational responsiveness deserves sustained analytical attention across the full range of institutions preparing individuals for today's highly technical battlefield. ■

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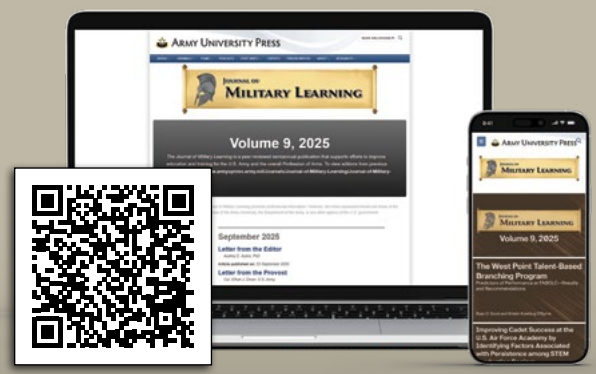
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International officers participate in a training exercise at the Army War College, Carlisle Barracks, Pennsylvania, in March 2022. (Photo by Michael Marra, US Army)

Intellectually Nurturing the Warfighters the Nation Needs

Lt. Col. David P. Oakley, PhD, US Army, Retired

At the Department of War, we have a duty to ensure that our professional military education develops real leaders, warfighters who dominate into the future.

—Secretary of War Pete Hegseth

In March 2026, Secretary of War Pete Hegseth announced the formation of a ninety-day task force to review the curriculum of the US senior service colleges (SSC).[1] This assessment came amidst ongoing educational changes within the Department of War, to include ending relationships with certain civilian universities.[2] Pentagon leadership wants professional military education (PME) to instill the "warrior ethos" and for it to be "grounded in the founding principles and documents of the republic."[3] The critique is pointed, but the description of the type of officer PME should nurture is vague.[4] The desire to develop "critical thinkers" and "warfighters who dominate into the

future" is admirable, but what does that guidance mean for course assessment and curriculum development, or more importantly, how officers are educated throughout their careers?[5]

Hegseth is not the first Pentagon leader to call for reevaluation of SSCs after more than two decades of the military instrument failing to achieve policy ends.[6] And a reassessment of SSCs is welcomed if it helps move the focus of officer education away from credentialing while resolving the "episodic" issue identified by previous military leaders.[7] To be useful though, any review of senior leader PME needs to consider it within a broader assessment of how military officers are intellectually nurtured from commissioning to retirement.

The current officer education progressional approach that aligns the PME "continuum" with the tactical, operational, and strategic levels of war is logical but insufficient for developing the strategically minded national security practitioners the military desires post SSC.[8] This is because the first decade of an officer's PME is largely focused on the officer developing technical skills and "tactical knowledge."[9] Although this tactical focus is needed and should be a priority, it also encourages a rigid approach that is reliant on processes and procedures. This can lead to an "action is achievement" mindset where the purpose of why something is being done is lost and the focus becomes on the activity or process instead of the desired ends.[10]

A 2023 study at the Army War College highlights the "shallow strategic" thinking and reliance on "heuristics" that occurs even after military students spend a year of focused study on strategy.[11] Although some might blame SSCs for the failure to produce strategic thinkers, this is an unfair critique that highlights the military's "just in time" approach to education. Only about 40 percent of officers are selected to attend SSCs, and these individuals are selected based on their potential to serve at higher levels.[12] If these top-performers are deficient in strategic thinking, is it the fault of ten months of SSC or the inadequacy of the previous two decades of intellectual development?[13]

The first decade of an officer's PME journey is largely professional military training, not education. Again, this observation is not intended to denigrate the essential "primary education" an officer receives, but it is important to acknowledge the difference in purpose between military training and education.[14] Training should develop confidence in the officer that they are technically and tactically proficient and capable of doing their job. Education should nurture humility, uncertainty, empathy, and an open mind—all necessary attributes for the practitioner to seek new knowledge, deepen their understanding of the strategic environment, improve planning, and ultimately take more purposeful action. Education helps the officer assess the operational and strategic environments to determine what actions should be taken to achieve the desired conditions aligned with policy objectives, while training ensures they can plan, conduct, and lead operations in pursuit of military objectives. Both are essential for success and complimentary if they are not conflated.

The failure of the PME continuum is not a critique of specific PME programs but rather a critique of how the military approaches education. Instead of viewing education as a distinct experience that an officer goes through every four to six years, the military needs to embrace an ongoing approach that intellectually nurtures officers throughout their career. Despite Chairman of the Joint Chiefs of Staff Instruction 1800.1, *Officer Professional Military Education*, including "thinking critically/strategically" as "desired leader attributes" throughout the PME continuum, the current educational approach is ineffective at nurturing those attributes. To remedy this problem, intellectual development, like leader development, needs to be a priority at the individual, command, and institutional levels.[15]

Any assessment of officer education must begin by asking broader questions: What purpose do military officers serve throughout their career? What is their role in not only waging war but in advising if the military instrument can achieve policy ends and what possible repercussions could arise through military action? Given their role in advising on and carrying out war, what are the intellectual attributes that officers require? Given these required attributes, what are the best ways to nurture them over a career?

What Is a Military Officer's Role? Advise on War, Wage War, or Both?

We see, therefore, that war is not merely an act of policy but a true political instrument, a continuation of political intercourse, carried on with other means.

—Carl Von Clausewitz[16]

The US military has been criticized for not winning wars.[17] This critique often implies the "loss" of a war is a military performance failure like a sporting loss. If the "team" would have fought a little harder or been a little more lethal, then victory would have been attained. Although this notion is comforting because the solution seems simple (just improve performance or increase lethality), it ignores a difficult reality—war is a social-political phenomenon, so failure is not merely a technical performance issue.[18]

While tactical and operational losses occur, what is considered a "military loss" is often a policy failure. This is not to say the military is perfect, but its imperfection frequently occurs in the analyzing, assessing, and advising stages. Failures in Iraq, Afghanistan, or even the recent failure of military force to compel Iran in a short period is not a "lethality" or "warrior ethos" failure but rather a failure to appreciate what policy outcomes the military instrument can achieve and the potential consequences if their analysis proves wrong.[19] This analysis requires a deep appreciation of the adversary and the social-political environment.[20]

While civilian and military leaders are calling for PME to focus on lethality and warfighting, there seems to be little evidence that US military defeats were due to a lack of "capability or capacity to destroy."[21] Rather, failure was largely due to an inability to appreciate the limitation of, or any repercussions from, employing the military instrument. What seems to be a focus on performance at the tactical and operational levels as a solution to stop losing wars is counterintuitive given lessons learned from Vietnam and the post-9/11 wars.[22]

The problem with the "why is the military not winning wars" question is it centers the discussion on tactical and operational performance while neglecting more important policy questions regarding the utility of the military instrument in achieving policy objectives. If it is a failure implementing the military instrument at the tactical or operational level, the solution most likely resides in developing more technically competent officers imbued with the warrior ethos and capable of increasing lethality across the battlefield. If the issue is about the fit of the military instrument in achieving policy aims or the consequences in introducing the military instrument, then the educational solution resides outside military technical expertise, most likely in the social-political space.

When it comes to uniformed military leaders advising civilian policymakers on the efficacy of the military instrument to achieve a policy outcome, what constitutes a "warfighter"? Is the warfighter merely the technician who understands how to array military forces to be the most lethal? The person whose expertise is solely focused on the employment of military systems to close in and destroy the enemy without any consideration of the political, social, or psychological aspects of the adversary or the broader social-political environment in which both ally and adversary exist? Or is the "warfighter" the individual who appreciates both the capability and limitations of the military systems she commands and the potential effectiveness and consequences in employing the military instrument within specific social-political environments? For the military to be purposeful, its warfighting leaders must be the latter.

What Type of Officers Does the US Military Need? What Attributes Do They Require?

The value of a good soldier is increased many fold if he possesses, in addition to his knowledge of military matters, competency in one or more of the humanities and thinks about the contributions these subjects can make to the defense of his country.

—Gen. Dwight D. Eisenhower[23]

The critique that PME should focus on developing warfighters is correct, but effective warfighting is more than a technical expert employing his weapon system or the commander successfully determining operational reach and achieving her tactical or operational objectives. Ultimately, winning wars is not about the level of destruction or degree of lethality, but the ability to assess the adequate amount of force necessary to help achieve policy outcomes without creating negative consequences that would put the United States in a worse strategic position. This demands that an officer not only understands the military instrument but also appreciates how the operational environment will respond to the introduction of the military instrument. This requires appreciation of the social-political environment in which the military operation is occurring and an understanding of the interagency and US domestic political dynamics involved.

If you are trying to coerce an adversary to do something, it helps to have an appreciation of how they see the world, their priorities, and what they value.

In addition to critical thinking, purposeful warfighting requires that practitioners possess attributes such as curiosity, empathy, and humility.[24] The objective of nurturing empathy is not to create woke warriors but to ensure that military officers appreciate the environments they operate within so the plans they develop and the actions they take have a greater likelihood of success. If you are trying to coerce an adversary to do something, it helps to have an appreciation of how they see the world, their priorities, and what they value. Without this awareness, you might underestimate the amount of coercion (deterrence and compellence) or brute force that is required.[25] This is often called "strategic empathy," but it can also be thought of as practical empathy since it is essential to effective planning and purposeful warfighting.[26]

A curious mind drives the officer to ask questions, deepen her understanding, and explore possibilities.[27] It allows the officer to accept uncertainty within the tactical, operational, or strategic environments, while helping them develop the necessary flexibility of mind to deal with uncertainty and adapt to surprise. The final attribute, humility, opens the officer up to the possibility that he or she could be wrong and that he or she should continue to strive for increased understanding. This openness to being wrong protects from fundamental surprise and helps the officer adapt when failure occurs.[28]

Critical thinking, curiosity, empathy, and humility are not developed through military training or by executing processes, procedures, and drills. These attributes are nurtured through an education that challenges an officer's beliefs and forces them to grapple with ideas that might make them uncomfortable. The intent is not to weaken the officer or make them less effective; on the contrary, it is to build an intellectually stronger officer who is firm in their beliefs, committed to their service, and who possesses the flexibility of mind so they can adapt to the challenges they face.

How to Develop the Warfighters America Needs?

We have found that there is no place for the closed mind in the Marine Corps. In order to get the right type of men with the right type of mind we must begin at the beginning.

—A. A. Vandegrift[29]

If the military is committed to developing critical-thinking warfighters, they must move away from the episodic education approach its leaders have criticized.[30] This shift does not require a complete overhaul of the PME continuum but rather an investment in the "career-long development" it identifies but supports mainly through rhetoric.[31] Moving from rhetoric to action will require military leaders to prioritize intellectual growth to the same degree they do physical fitness and unit readiness.[32] This will require leaders to encourage individual intellectual growth opportunities and reward those who seek those opportunities. It will also require commanders to make collective educational opportunities a priority on the unit calendar and identify and seek out the resources necessary to support these opportunities.

Moving from episodic to ongoing does not require a herculean effort or a large investment of resources, but it does require time and commitment. Simple actions like introducing a speaker series or a book club focused on topics related to international affairs, strategy, the US national security enterprise, professionalism, or civil-military affairs, to name just a few, is a great start. These programs can be a mixture of guest speakers who are subject-matter experts on a given topic or warfighters within the unit who are encouraged to research a given topic and facilitate a discussion with her colleagues. These sessions should not be treated as the typical mandatory training PowerPoint briefs but should be more akin to a university seminar where the power of ideas and not the power of rank takes precedence.

> Although recently maligned, civilian universities are essential to America's national security.

Although recently maligned, civilian universities are essential to America's national security.[33] These institutions not only serve as the "precommissioning" educational institutions for the future warfighters but also provide "the talent pipeline" for the other instruments of national power.[34] These same universities provide much of the basic and applied research that helps US national security institutions appreciate and respond to the threats and challenges the Nation faces.[35] It is safe to say that without civilian universities, the US national security institutions could not fulfill their mandates. The same universities that support US national security through research and precommissioning education can also help enable the US military to resolve their episodic education problem. Commanders at all levels should develop collaborative partnerships with their local universities to support their educational plans. Collaboration between military units and local universities will not only provide the necessary resources to enable the unit's educational plans and further the warfighters intellectual growth but will also be valuable in dispelling myths about each other and alleviating mistrust of each other.[36]

Some commanders or leaders might balk at the idea of increasing their role in the intellectual nurturing of their warfighters. They might argue that there is not enough time in the training calendar with all the other requirements thrusted upon them. While time is often in short supply, the choice to invest or not is more about priorities than time available. What is more important, the multiple staff meetings and formations, mandatory training, and quarterly golf tournaments, or the development of warfighters? To date, the answer has been the former. The military will not move beyond the episodic approach to education and improve its strategic education until it is a priority throughout the chain of command.

Conclusion

It is false to suggest a stark trade-off between the tactical and technical competence of officers and the ability of officers to appreciate political impact. Neither alone is satisfactory, and the U.S. military needs both. Twenty-first-century military officers can be excellent tacticians and astute observers of political affairs.

—Suzanne C. Nielsen and Hugh Liebert[37]

The US military needs patriotic officers who live by the warrior ethos—technically proficient officers who understand the tactical and operational levels of war and the importance of managing lethality to accomplish the mission. While these characteristics, along with knowledge of the Constitution, the Federalist Papers, and the US founding are necessary and ideal for military officers, they are insufficient for nurturing warfighters capable of advising policymakers on the strengths, limitations, and risks in employing the military instrument—just as a nationalist historiography approach focused on increasing esprit de corps and patriotism is inadequate to understand historical context or to develop the critical-thinking skills necessary to appreciate the potential consequences of military advice.[38]

The military has been criticized for not winning wars and military actions not resulting in desired policy outcomes. Much of the recent blame for this failure has been placed on PME institutions, particularly senior service colleges, and their faculty, but this is largely undeserved. Although regular assessment of PME to ensure it is doing its part to intellectually develop warfighters is positive, we cannot expect a ten-month program to make up for a nearly two-decade void in strategic education. The military's episodic educational approach is the main culprit because it is ineffective at producing the warfighters the Nation needs. Instead of encouraging intellectual nurturing, it prioritizes credentialing and the acquiring of skill identifiers. Although change is difficult, by prioritizing education at the individual, organizational, and institutional levels, the military should start to see an improvement in its officers' strategic performance.

This will not be due to an increased focus on lethality or the Nation's founding documents, but the result of a deliberate focus on developing warfighters who not only understand how to employ the military instrument but also appreciate its potential and risks in specific social-political environments. ■

Notes

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20. Suzanne C. Nielsen and Hugh Liebert, "The Continuing Relevance of Morris Janowitz's *The Professional Soldier* for the Education of Officers," *Armed Forces & Society* 47, no. 4 (2021): 744, https://doi.org/10.1177/0095327X20960480. This article provides a great perspective on why Morris Janowitz's research is important for military officers. The authors rightfully argue, "While officers must still be highly trained to perform their specialized tasks, that is not enough. To be militarily effective, officers must understand the political significance of military operations."

21. Field Manual 3-0, *Operations* (US GPO, March 2025), 5. The Army's definition, "lethality is the capability and capacity to destroy."

22. See Harry G. Summers, *On Strategy: A Critical Analysis of the Vietnam War* (Presidio Press, 1982), 12; John Q. Bolton, "Military Power Is Insufficient: Learning from Failure in Afghanistan," *Military Review* 103, no. 1 (January-February 2023): 63–75, https://www.armyupress.army.mil/Journals/Military-Review/English-Edition-Archives/January-February-2023/Bolton/. Beyond highlighting

that tactical/operational success does not always result in strategic success, the RAND report, *The U.S. Army and the Battle for Baghdad*, states, "PME can provide future senior Army leaders with the intellectual underpinnings to be able to understand the tactical, operational, and strategic implications of a range of operational environments. This broader approach to PME should also improve the ability of senior Army leaders to advise civilian appointees and senior officers about whether a strategy will achieve policy objectives." See David E. Johnson et al., *The U.S. Army and the Battle for Baghdad: Lessons Learned—And Still to Be Learned* (RAND Corporation, 2019), 231–32, https://www.rand.org/pubs/research_reports/RR3076.html.

23. Dwight D. Eisenhower et al., "Liberal Education in the Military Forces (A Symposium)," *Journal of General Education* 1, no. 1 (October 1946): 35, https://www.jstor.org/stable/27795125.

24. Oakley and Obadal, "Want 'Strategically Minded Warfighters?'"

25. SWJ Staff, "Coercion Theory: A Basic Introduction for Practitioners," Small Wars Journal, 30 July 2025, https://smallwarsjournal.com/2025/07/30/coercion-theory-a-basic-introduction-for-practitioners/. An ongoing recent example of this is Operation Epic Fury, the military action against Iran. Although the secretary of war declared it a "decisive military victory" for the United States, at the time of this writing, more than a month after that proclamation was made, it is still unclear what political outcomes the "decisive" victory achieved. This highlights that the enemy has a vote on what is and what is not "decisive."

26. Stephanie Crider, host, *Conversations on Strategy Podcast*, episode 25, "On Strategic Empathy," US Army War College Strategic Studies Institute, 28 September 2023, https://ssi.armywarcollege.edu/SSI-Media/Recent-Publications/Article/3944764/on-strategic-empathy/.

27. Laurence Thomson, "The Thinking Soldier: Why Intellectual Curiosity Belongs in Your Belt Kit," Wavell Room, 10 December 2025, https://wavellroom.com/2025/12/10/the-thinking-soldier-why-intellectual-curiosity-belongs-in-your-belt-kit/.

28. For more information on fundamental surprise, see Energy Academic Group, "Resilience Corner: Don't Design for Threats, Design for Surprise," Naval Postgraduate School, accessed 22 June 2026, https://nps.edu/web/eag/resilience-corner-don-t-design-for-threats-design-for-surprise..

29. Eisenhower et al., "Liberal Education in the Military Forces," 37.

30. CJCS, *Developing Today's Joint Officers for Tomorrow's Ways of War*.

31. CJCSI 1800.01G, *Officer Professional Military Education Policy*, A-15.

32. Oakley and Obadal, "Want 'Strategically Minded Warfighters?'"

33. Brian L. Heuser, "Let's Talk About Hard Power: America's Universities Are National Security Assets," Inside Higher Ed, 10 November 2025, https://www.insidehighered.com/opinion/views/2025/11/10/us-universities-are-national-security-assets-opinion.

34. Heuser, "Let's Talk About Hard Power."

35. Heuser, "Let's Talk About Hard Power."

36. Recommend Ryan Shaw and Ken Gleiman's article in this same series for a good exploration of the value of partnering with civilian universities. See "From Consumer to Custodian: A New Model for Partnering with Civilian Universities," *Military Review* 106, no. 4 (July-August 2026): 129–40, https://www.armyupress.army.mil/Journals/Military-Review/English-Edition-Archives/July-August-2026/From-Consumer-to-Custodian/.

37. Nielsen and Liebert, "The Continuing Relevance of Morris Janowitz's *The Professional Soldier* for the Education of Officers," 744.

38. For a good discussion on national historiography, see Terence Ranger, "Nationalist Historiography, Patriotic History and the History of the Nation: The Struggle Over the Past in Zimbabwe," *Journal of Southern African Studies* 30, no. 2 (June 2004): 215–34, https://doi.org/10.1080/0305707042000215338.

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Remembering 9/11

This year marks the twenty-fifth anniversary of the attacks on 11 September 2001. *Military Review* remembers all those killed during the attacks as well as those service members injured or killed in the subsequent Global War on Terrorism.



The Tribute in Light is an art installation of eighty-eight searchlights placed to create two vertical columns of light representing the Twin Towers in remembrance of the 11 September 2001 attacks. (Photo by Jkellgraphics via Wikimedia Commons)

Airman 1st Class Tanja Kambel, 506th Expeditionary Security Forces Squadron, paints names on Kirkuk's memorial wall on 25 November 2009 at the Leroy Webster distinguished visitor helicopter pad on Kirkuk Regional Air Base, Iraq. The memorial wall held the names of all the service members who lost their lives during Operation Iraqi Freedom from the start of the campaign in 2003. (Photo courtesy of the 506th Air Expeditionary Group)

